MING'S PERSONAL STUDY NOTES

STATS 361: Causal Inference

Lectures 1–14

Reviewed English edition

Reading this edition

The author's explanations, intuitions, questions, and marginal supplements are preserved alongside the lecture material. Scan references use one-based PDF page numbers. Reading order follows topics and drawn links, including left-page additions to right-page notes. The English layout is reflowed; reconstructed curves are schematic, not numerical data.

Black, red, and blue reproduce the original pen-color categories. Arrows preserve the relation between an annotation and its target. These colors do not indicate editorial judgments.

Purple: a mistake present in the original handwritten note. The source claim is preserved and the correction is placed beside it.

Orange: a qualification, missing assumption, unclear reading, or unresolved point. This does not necessarily mean the original statement is false.

Recognition and translation mistakes are corrected directly; the audit files record them without adding colored error boxes. Editorial comments are separate from the author's voice. The original scans were unchanged during review and are removed from the current tree after verification at the author's request.

Coverage: 89 scanned pages, merged by textbook. The companion findings and coverage files document the review and primary references.

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Lecture 1: Randomized Controlled Trials

Scan reference: Lecture1-10.pdf, pp. 2, 1, 3; reading order: main text on p. 2, its supplement on p. 1, then p. 3.

Regression in an RCT analysis of the average treatment effect reduces variance; it does not play a role in identifying the ATE.

Regression in an RCT (p. 2) $\longrightarrow$ Variance reduction; ATE identification comes from assignment.

$$\Delta_i = Y_i(1) - Y_i(0), \qquad \tau = E[Y_i(1) - Y_i(0)].$$

Estimate and construct inference for τ using the difference-in-means estimator:

$$\widehat{\tau}_{\text{DM}} = \frac{1}{n_1} \sum_{W_i=1} Y_i - \frac{1}{n_0} \sum_{W_i=0} Y_i, \qquad n_w = |\{i : W_i = w\}|.$$

The note marks this estimator as consistent and unbiased.

Difference in means (p. 2) $\longrightarrow$ $\widehat{\tau}_{\text{DM}}$

Qualification / unresolved point (S1-002): Sampling and allocation conditions

Source: Lecture1-10.pdf, pp. 2

The difference in means is unbiased under the stated random-assignment setup when both groups are nonempty; the population consistency and normal limits also use IID sampling and finite second moments. The balanced-variance formulas below specialize to $P(W = 1) = 1/2$ and centered covariates $E[X] = 0$. The general variance is $\text{Var}(Y(1))/e + \text{Var}(Y(0))/(1 - e)$. Centering also explains the displayed OLS decomposition. Its $N(0, \cdot/n_w)$ labels are asymptotic approximations, not exact finite-sample distributions under merely the stated moment assumptions. See lecture (1.6), (1.8)–(1.13).

Supplement: assumptions.

$$Y_i = Y_i(W_i), \qquad W_i \perp \{Y_i(0), Y_i(1)\}.$$

The first is labeled SUTVA; the second is random treatment assignment. SUTVA means the stable unit treatment value assumption: (1) unit stability, with experimental units independent of one another; (2) a single treatment level/version. These are the author's short descriptions.

$Y_i = Y_i(W_i)$; SUTVA $\longrightarrow$ Stable unit treatment value assumption: the two definitions in the supplement (p. 1).

Original-note correction (S1-001): SUTVA does not mean statistical independence

Source: Lecture1-10.pdf, pp. 1

SUTVA requires no interference between units and no hidden versions of each specified treatment. It does not require statistically independent outcomes, and it permits the two treatment values $0, 1$ used here. Independent sampling is a separate assumption. The handwritten phrase "single

treatment level" is best replaced by "a well-defined version of each treatment." See the discussion and footnote on printed pp. 2–3 of the lecture.

Linear outcome model and variance reduction

$$Y_i(w) = c_{(w)} + X_i\beta_{(w)} + \varepsilon_i(w), \quad E[\varepsilon_i(w) \mid X_i] = 0, \quad \mathrm{Var}[\varepsilon_i(w) \mid X_i] = b^2.$$

The coefficients are fitted separately for the two treatment states. Write $A = \mathrm{Var}[X]$ and $\|v\|_A^2 = v'Av$. The variance in the notes is

$$V_{\mathrm{DM}} = 4b^2 + \|\beta_{(0)} + \beta_{(1)}\|_A^2 + \|\beta_{(0)} - \beta_{(1)}\|_A^2.$$

Also,

$$\tau = c_{(1)} - c_{(0)} + E[X](\beta_{(1)} - \beta_{(0)}), \qquad \widehat{\tau}_{\mathrm{OLS}} = \widehat{c}_{(1)} - \widehat{c}_{(0)} + \overline{X}(\widehat{\beta}_{(1)} - \widehat{\beta}_{(0)}).$$

The error decomposition written in the notebook is

$$\begin{aligned}\widehat{\tau}_{\mathrm{OLS}} - \tau = (\widehat{c}_{(1)} - c_{(1)}) - (\widehat{c}_{(0)} - c_{(0)}) + \overline{X}(\beta_{(1)} - \beta_{(0)}) \\ + \overline{X}(\widehat{\beta}_{(1)} - \widehat{\beta}_{(0)} - \beta_{(1)} + \beta_{(0)}).\end{aligned}$$

The first two terms are annotated $N(0, b^2/n_1)$ and $N(0, b^2/n_0)$; the third is annotated $N(0, \|\beta_{(1)} - \beta_{(0)}\|_A^2/n)$; the last is $O_p(1/n)$. The margin says that uncertainty in the estimated slopes tends to zero. Thus

$$\sqrt{n}(\widehat{\tau}_{\mathrm{OLS}} - \tau) \Rightarrow N(0, V_{\mathrm{OLS}}), \qquad V_{\mathrm{OLS}} = 4b^2 + \|\beta_{(0)} - \beta_{(1)}\|_A^2,$$
$$V_{\mathrm{DM}} = V_{\mathrm{OLS}} + \|\beta_{(0)} + \beta_{(1)}\|_A^2.$$

OLS under a general outcome regression

The next page says that OLS is better than difference in means in this model. Suppose

$$Y_i(w) = \mu_{(w)}(X_i) + \varepsilon_i(w), \quad E[\varepsilon_i(w) \mid X_i] = 0, \quad \mathrm{Var}[\varepsilon_i(w) \mid X_i] = b^2.$$

For difference in means,

$$\sqrt{n}(\widehat{\tau}_{\mathrm{DM}} - \tau) \Rightarrow N(0, V_{\mathrm{DM}}), \quad V_{\mathrm{DM}} = 4b^2 + 2\,\mathrm{Var}[\mu_{(0)}(X_i)] + 2\,\mathrm{Var}[\mu_{(1)}(X_i)].$$

For OLS, the Huber–White calculation starts from the population projection

$$(c^*_{(w)}, \beta^*_{(w)}) = arg\min_{c,\beta} E[(Y_i(w) - X_i\beta - c)^2].$$

The notebook records

$$\sqrt{n_w}\left\{\begin{pmatrix}\widehat{c}_{(w)}\\ \widehat{\beta}_{(w)}\end{pmatrix} - \begin{pmatrix}c^*_{(w)}\\ \beta^*_{(w)}\end{pmatrix}\right\} \Rightarrow N\left(0, \begin{pmatrix}\mathrm{MSE}^*_{(w)} & 0\\ 0 & \cdots\end{pmatrix}\right),$$

with $c^*_{(w)} = E[Y_i(w)]$ and

Original-note correction (S1-003): The misspecified-regression covariance need not be block diagonal

Source: Lecture1-10.pdf, pp. 3

The zero off-diagonal entries do not follow from $E[X] = 0$. With $q = (1, X')'$, $r_w = Y(w) - c^*_w - X\beta^*_w$ and $Q = E[qq']$, the covariance is

$$Q^{-1}E[r_w^2 qq']Q^{-1}.$$

The intercept–slope block is $E[r_w^2 X]A^{-1}$ (with matching row/column orientation), generally nonzero under a misspecified conditional mean. Projection orthogonality gives $E[r_w X] = 0$, not $E[r_w^2 X] = 0$. The intercept variance $E[r_w^2]$ and the subsequent balanced-trial variance-reduction identity remain valid. This is an error in the visible original, also present in lecture (1.16), rather than an OCR error.

$$\mathrm{MSE}^*_{(w)} = E[(Y_i(w) - X_i\beta^*_{(w)} - c^*_{(w)})^2].$$

It concludes

$$\sqrt{n}(\widehat{\tau}_{\mathrm{OLS}} - \tau) \Rightarrow N(0, V_{\mathrm{OLS}}), \qquad V_{\mathrm{OLS}} = V_{\mathrm{DM}} - \|\beta^*_{(0)} + \beta^*_{(1)}\|^2_A.$$

$V_{\mathrm{OLS}} < V_{\mathrm{DM}}$ always holds.

Original-note correction (S1-004): Variance reduction is weak, not always strict

Source: Lecture1-10.pdf, pp. 3

The preceding identity implies $V_{\mathrm{OLS}} \leq V_{\mathrm{DM}}$. Equality occurs whenever $\|\beta^*_0 + \beta^*_1\|^2_A = 0$; for nonsingular A, this means $\beta^*_0 + \beta^*_1 = 0$. Opposite nonzero slopes already give equality. Strict reduction needs a positive squared norm. This follows directly by subtracting the two displayed variances.

Personal observation. OLS can be viewed as

$$\widehat{\tau}_{\mathrm{OLS}} = \frac{1}{n}\sum_{i=1}^{n}\{(\widehat{c}_{(1)} + X_i\widehat{\beta}_{(1)}) - (\widehat{c}_{(0)} + X_i\widehat{\beta}_{(0)})\} = \frac{1}{n}\sum_i[\widehat{\mu}_{(1)}(X_i) - \widehat{\mu}_{(0)}(X_i)].$$

Here $\widehat{\mu}_{(w)}(x)$ is the fitted OLS value at x. It could be replaced with another method.

Lecture 2: Unconfoundedness and the Propensity Score

Scan reference: Lecture1-10.pdf, pp. 5, 4, 7, 6; p. 4 continues the bottom of p. 5, and p. 6 supplements p. 7.

Unconfoundedness: the aim is to remove the influence of confounding factors. Propensity score: the probability of treatment given the covariates.

Aggregating difference-in-means estimators

Simply weight by probability. With discrete, numerically valued X_i, assume

$$\{Y_i(0), Y_i(1)\} \perp W_i \mid X_i = x, \qquad x \in \mathcal{X}.$$

Here x is the grouping variable. Define the group-wise ATE

$$\tau(x) = E[Y_i(1) - Y_i(0) \mid X_i = x].$$

Then

$$\widehat{\tau}_{\mathrm{AGG}} = \sum_{x\in\mathcal{X}} \frac{n_x}{n}\widehat{\tau}(x),$$

$$\widehat{\tau}(x) = \frac{1}{n_{x1}}\sum_{X_i=x, W_i=1} Y_i - \frac{1}{n_{x0}}\sum_{X_i=x, W_i=0} Y_i,$$

$$n_x = |\{i : X_i = x\}|, \qquad n_{xw} = |\{i : X_i = x, W_i = w\}|.$$

Let $e(x) = P(W_i = 1 \mid X_i = x)$. The within-group limit is

$$\sqrt{n_x}(\widehat{\tau}(x) - \tau(x)) \Rightarrow N\left(0, \frac{\mathrm{Var}[Y_i(0) \mid X_i = x]}{1 - e(x)} + \frac{\mathrm{Var}[Y_i(1) \mid X_i = x]}{e(x)}\right)$$
$$= N\left(0, \frac{b^2(x)}{e(x)(1 - e(x))}\right),$$

where the second equality uses the common conditional variance written in the notes. The definition line writes $\widehat{\pi}(x) = P(X_i = x) = n_x/n$.

Original-note correction (S1-005): Population and empirical stratum probabilities

Source: Lecture1-10.pdf, pp. 5

Use $\pi(x) = P(X_i = x)$ for the population mass and $\widehat{\pi}(x) = n_x/n$ for its estimate; they are generally unequal in a finite sample. This distinction is already required by the subsequent terms $\widehat{\pi}(x) - \pi(x)$. The expansion below is read with these distinct definitions. For its fixed-stratum CLT, assume finitely many strata, positive stratum masses and treatment overlap, and finite conditional second moments. See lecture (2.3)–(2.7).

Then

$$\widehat{\tau}_{\mathrm{AGG}} = \sum_x \pi(x)\tau(x) + \sum_x \pi(x)(\widehat{\tau}(x) - \tau(x))$$
$$+ \sum_x (\widehat{\pi}(x) - \pi(x))\tau(x) + \sum_x (\widehat{\pi}(x) - \pi(x))(\widehat{\tau}(x) - \tau(x)).$$

The first term is τ. The second is annotated as asymptotically normal, with variance

$$\sum_x \pi^2(x)\,\mathrm{Var}[\widehat{\tau}(x)].$$

The third has variance $n^{-1}\,\mathrm{Var}[\tau(X_i)]$; the last is $O_p(1/n)$. Following the arrow onto the left page,

Four-term aggregate-estimator expansion (p. 5) $\longrightarrow$ Aggregate variance V_{AGG} (p. 4).

$$\sqrt{n}(\widehat{\tau}_{\mathrm{AGG}} - \tau) \Rightarrow N(0, V_{\mathrm{AGG}}),$$

$$V_{\mathrm{AGG}} = \mathrm{Var}[\tau(X_i)] + \sum_{x \in \mathcal{X}} \pi^2(x) \frac{1}{\pi(x)} \frac{b^2(x)}{e(x)(1 - e(x))}$$
$$= \mathrm{Var}[\tau(X_i)] + E\left[\frac{b^2(X_i)}{e(X_i)(1 - e(X_i))}\right].$$

Continuous covariates and propensity scores

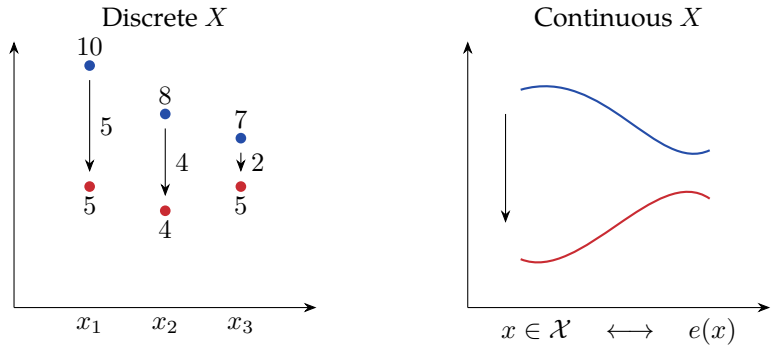
Given X_i, W_i cannot "peek" at Y_i: this is the author's intuition for unconfoundedness. If

$$\{Y_i(0), Y_i(1)\} \perp W_i \mid X_i, \quad e(x) = P(W_i = 1 \mid X_i = x),$$

then

$$\{Y_i(0), Y_i(1)\} \perp W_i \mid e(X_i).$$

Supplementary sketches and calculation.



The proof written beneath the sketches is

$$\begin{aligned} &P(W_i = w \mid \{Y_i(0), Y_i(1)\}, e(X_i)) \\ &\quad = \int_{\mathcal{X}} P(W_i = w \mid \{Y_i(0), Y_i(1)\}, X_i = x) P(X_i = x \mid e(X_i))\, dx \\ &\quad = \int_{\mathcal{X}} P(W_i = w \mid X_i = x) P(X_i = x \mid e(X_i))\, dx \\ &\quad = e(X_i)\mathbf{1}_{\{w=1\}} + (1 - e(X_i))\mathbf{1}_{\{w=0\}}. \end{aligned}$$

The margin labels this "Lebesgue integration."

Original-note correction (S1-006): Conditioning in the propensity-score proof

Source: Lecture1-10.pdf, pp. 6

Let $\mathcal{Y} = (Y(0), Y(1))$. The tower-property integral must initially use $dP(X \mid \mathcal{Y}, e(X))$, not $dP(X \mid e(X))$. Unconfoundedness changes the integrand to $P(W = w \mid X)$, which is constant on each propensity-score level set. Integrating that constant gives the claimed result, regardless of the remaining conditional distribution. For continuous X, use a regular conditional measure rather than the point probability $P(X = x \mid \cdot)\, dx$. Compare the correctly conditioned proof of Proposition 2.1 in the lecture.

Propensity stratification

1. Sort the observations so that $\widehat{e}(X_{i_1}) \le \widehat{e}(X_{i_2}) \le \cdots \le \widehat{e}(X_{i_n})$.
2. Split them into J strata and estimate a difference in means within each stratum. If I_j contains ranks $\lfloor (j-1)n/J \rfloor + 1$ through $\lfloor jn/J \rfloor$, write the abbreviated formula on the page as

$$\widehat{\tau}_j = \frac{\sum_{i \in I_j} W_i Y_i}{\sum_{i \in I_j} W_i} - \frac{\sum_{i \in I_j} (1 - W_i) Y_i}{\sum_{i \in I_j} (1 - W_i)}.$$

   Make treatment and control groups more similar (p. 7) $\longrightarrow$ Sort by $\widehat{e}(X_i)$ and form propensity-score strata.

3. Average the stratum estimates: $\widehat{\tau}_{\text{STRAT}} = J^{-1} \sum_{j=1}^{J} \widehat{\tau}_j$. The notebook calls this consistent.

Qualification / unresolved point (S1-007): Stratification needs shrinking propensity-score strata

Source: Lecture1-10.pdf, pp. 7

A fixed number of broad propensity-score strata generally leaves residual confounding. Consistency requires increasingly fine strata, sufficiently many treated and control observations within them, overlap, and sufficiently accurate propensity estimates. The lecture explicitly states uniform consistency of $\widehat{e}$ and that J grows appropriately with n. Equal weighting is appropriate for the approximately equal-size strata used here.

Inverse-propensity weighting

Scan reference: Lecture1-10.pdf, pp. 9, 8, 10; p. 8 supplies the model used in the variance calculation on p. 9.

$$\widehat{\tau}_{\text{IPW}} = \frac{1}{n}\sum_{i=1}^{n}\left\{\frac{W_i Y_i}{\widehat{e}(X_i)} - \frac{(1-W_i)Y_i}{1-\widehat{e}(X_i)}\right\}.$$

Oracle IPW assumes $e(x)$ is perfectly known:

$$\widehat{\tau}^{*}_{\text{IPW}} = \frac{1}{n}\sum_{i}\left\{\frac{W_i Y_i}{e(X_i)} - \frac{(1-W_i)Y_i}{1-e(X_i)}\right\}.$$

Suppose overlap holds, $\eta \le e(x) \le 1-\eta$ for all $x \in \mathcal{X}$, and $|Y_i| < M$. If

$$\sup_{x\in\mathcal{X}} |e(x) - \widehat{e}(x)| = O_p(a_n) \longrightarrow 0,$$

the notebook gives

$$|\widehat{\tau}_{\text{IPW}} - \widehat{\tau}^{*}_{\text{IPW}}| = O_p(a_n M/\eta).$$

Thus consistency of the oracle estimator transfers to estimated-propensity IPW. Moreover,

$$E[\widehat{\tau}^{*}_{\text{IPW}}] = E[Y_i(1) - Y_i(0)] = \tau.$$

With overlap, oracle IPW converges at the $1/\sqrt{n}$ scale.

Variance calculation: supplementary model. The left page supplies

$$\begin{aligned} Y_i(0) &= c(X_i) - (1-e(X_i))\tau(X_i) + \varepsilon_i(0), \\ Y_i(1) &= c(X_i) + e(X_i)\tau(X_i) + \varepsilon_i(1), \\ E[\varepsilon_i(w) \mid X_i] = 0, &\qquad \text{Var}[\varepsilon_i(w) \mid X_i = x] = b^2(x). \end{aligned}$$

Then the notes state

Supplementary outcome model on p. 8 $\longrightarrow$ Oracle-IPW variance on p. 9, V_{IPW^*}.

$$\begin{aligned} n\,\text{Var}[\widehat{\tau}^{*}_{\text{IPW}}] = V_{\text{IPW}^*} &= E\left[\frac{c^2(X_i)}{e(X_i)(1-e(X_i))}\right] + \text{Var}[\tau(X_i)] \\ &\quad + E\left[\frac{b^2(X_i)}{e(X_i)(1-e(X_i))}\right], \\ \sqrt{n}(\widehat{\tau}^{*}_{\text{IPW}} - \tau) &\Rightarrow N(0, V_{\text{IPW}^*}). \end{aligned}$$

Personal comparison. $c(x)$ describes the relationship between X and Y, rather than the relationship between X and ΔY. Compared with aggregation,

$$V_{\text{IPW}^*} = V_{\text{AGG}} + E\left[\frac{c^2(X_i)}{e(X_i)(1-e(X_i))}\right].$$

$V_{\text{IPW}^*} - V_{\text{AGG}}$ (p. 10) $\longrightarrow$ Unless $c(X_i)$ is zero everywhere, oracle IPW has worse variance than aggregation.

Compared with linear modeling: IPW is based on $\{Y_i(0), Y_i(1)\} \perp W_i \mid X_i$, while the linear model relies on linearity.

Qualification / unresolved point (S1-009): Variance comparison and causal identification

Source: Lecture1-10.pdf, pp. 10

The excess variance is zero exactly when $c(X) = 0$ almost surely; it is positive if $P(c(X) \neq 0) > 0$, under overlap and finite moments. A correctly specified linear outcome model is an estimation restriction, not by itself a causal identification assumption: observational regression still needs suitable exogeneity/unconfoundedness or another identification argument. The original comparison of estimation methods is retained.

Lecture 3: Efficient Treatment Effect Estimation via Augmented IPW

Scan reference: Lecture1-10.pdf, pp. 11–12.

Ordinary IPW has poor large-sample properties. For example, with discrete X, it is worse than the aggregate estimator $\widehat{\tau}$.

Two characterizations of the ATE

1. Using $e(x) = P(W_i = 1 \mid X_i = x)$, $\tau = E[\widehat{\tau}^*_{\text{IPW}}]$, where the oracle estimator is as above.
2. Using $\mu_{(w)}(x) = E[Y_i(w) \mid X_i = x]$,

$$\tau(x) = \mu_{(1)}(x) - \mu_{(0)}(x), \quad \tau = E[\mu_{(1)}(X_i) - \mu_{(0)}(X_i)],$$

which motivates

$$\widehat{\tau}_{\text{REG}} = \frac{1}{n}\sum_i [\widehat{\mu}_{(1)}(X_i) - \widehat{\mu}_{(0)}(X_i)].$$

The augmented IPW estimator

$$\widehat{\tau}_{\text{AIPW}} = \frac{1}{n}\sum_i \left[\widehat{\mu}_{(1)}(X_i) - \widehat{\mu}_{(0)}(X_i) + W_i\frac{Y_i - \widehat{\mu}_{(1)}(X_i)}{\widehat{e}(X_i)} - (1-W_i)\frac{Y_i - \widehat{\mu}_{(0)}(X_i)}{1-\widehat{e}(X_i)}\right].$$

Personal intuition. First use $\mu(x)$ to estimate the outcome, and then use IPW on the regression residuals. IPW makes the covariate composition of the data "fair."

Double robustness

AIPW is consistent if either the outcome regressions $\widehat{\mu}_{(w)}(x)$ are consistent or $\widehat{e}(x)$ is consistent.

Qualification / unresolved point (S1-010): Double robustness and efficiency require regularity

Source: Lecture1-10.pdf, pp. 11, 12, 13

Double robustness is a statement under overlap, suitable integrability and control of fitted-function evaluation (for example cross-fitting); the other nuisance estimate must remain well behaved. Conditioning on X_i, W_i alone does not freeze a function trained on the same observation. In the cross-fitted argument, condition also on the training fold. The efficiency bound below is for regular estimators in the nonparametric unconfoundedness model, not literally every conceivable estimator. The product-risk condition and nuisance consistency listed on the next page are assumptions to verify for a chosen learning method, not automatic properties of all machine-learning algorithms. See lecture (3.8) and the regularity discussion after (3.16).

When $\widehat{\mu}_{(w)}(x) \approx \mu_{(w)}(x)$, the regression average is consistent and the weighted residual terms have mean approximately zero because

$$E[Y_i - \widehat{\mu}_{(W_i)}(X_i) \mid X_i, W_i] \approx 0.$$

The propensity term is held fixed by this conditioning. When instead $\widehat{e}(x) \approx e(x)$, rearrange:

$$\begin{aligned}\widehat{\tau}_{\text{AIPW}} = \frac{1}{n}\sum_i \left[\frac{W_i Y_i}{\widehat{e}(X_i)} - \frac{(1-W_i)Y_i}{1-\widehat{e}(X_i)}\right] \\ + \frac{1}{n}\sum_i \left[\widehat{\mu}_{(1)}(X_i)\left(1 - \frac{W_i}{\widehat{e}(X_i)}\right) - \widehat{\mu}_{(0)}(X_i)\left(1 - \frac{1-W_i}{1-\widehat{e}(X_i)}\right)\right].\end{aligned}$$

The first part is IPW. The second has mean approximately zero since $E[1 - W_i/\widehat{e}(X_i) \mid X_i] \approx 0$; conditioning holds the fitted outcome regression fixed.

Semiparametric efficiency

The notes call AIPW asymptotically optimal among nonparametric estimators. When the nuisance functions are estimated sufficiently accurately,

$$\sqrt{n}(\widehat{\tau}_{\text{AIPW}} - \widehat{\tau}^*_{\text{AIPW}}) \longrightarrow_p 0,$$

where the starred estimator is oracle AIPW. This oracle estimator is just an IID average, so the central limit theorem also applies to the estimated version:

$\sqrt{n}(\widehat{\tau}_{\text{AIPW}} - \widehat{\tau}^*_{\text{AIPW}}) \to_p 0$ (p. 12) $\longrightarrow$ The oracle IID central limit theorem transfers to estimated AIPW.

$$\sqrt{n}(\widehat{\tau}_{\text{AIPW}} - \tau) \Rightarrow N(0, V^*),$$

$$V^* = \text{Var}[\tau(X_i)] + E\left[\frac{b_0^2(X_i)}{1-e(X_i)}\right] + E\left[\frac{b_1^2(X_i)}{e(X_i)}\right].$$

The margin describes this as a Cramer–Rao-type constraint: this is the best attainable effect/variance.

AIPW and cross-fitting

Scan reference: Lecture1-10.pdf, pp. 13–14.

Randomly split the data into two halves, I_1 and I_2. The annotation says to split the data and choose the halves at random. Set

$$\widehat{\tau}_{\text{AIPW}} = \frac{|I_1|}{n}\widehat{\tau}^{I_1} + \frac{|I_2|}{n}\widehat{\tau}^{I_2},$$

$$\widehat{\tau}^{I_1} = \frac{1}{|I_1|}\sum_{i\in I_1}\left[\widehat{\mu}^{I_2}_{(1)}(X_i) - \widehat{\mu}^{I_2}_{(0)}(X_i) + W_i\frac{Y_i - \widehat{\mu}^{I_2}_{(1)}(X_i)}{\widehat{e}^{I_2}(X_i)} - (1-W_i)\frac{Y_i - \widehat{\mu}^{I_2}_{(0)}(X_i)}{1-\widehat{e}^{I_2}(X_i)}\right].$$

Asymptotically, we can ignore the specific properties of the chosen machine-learning adjustment method and instead rely on:

1. Overlap: $\eta < e(x) < 1-\eta$ for all $x \in \mathcal{X}$, with $\eta > 0$.
2. Consistency of the nuisance estimators:

$$\sup_x |\widehat{\mu}^{I_2}_{(w)}(x) - \mu_{(w)}(x)| \longrightarrow_p 0, \quad \sup_x |\widehat{e}^{I_2}(x) - e(x)| \longrightarrow_p 0.$$

   The blue sentence says that the machine-learning adjustment methods have large-sample consistency.
3. Risk decay for the outcome and propensity models:

$$E[(\widehat{\mu}^{I_2}_{(w)}(X_i) - \mu_{(w)}(X_i))^2]\, E[(\widehat{e}^{I_2}(X_i) - e(X_i))^2] = o(1/n).$$

   If both are $\sqrt{n}$-consistent, the product is of order $1/n^2$. A simple sufficient approach is for both regression adjustments to converge at $o(n^{-1/4})$ in RMSE.

Proof recorded on the continuation page. Write the oracle estimator as $\widehat{\tau}^* = |I_1|\widehat{\tau}^{I_1,*}/n + |I_2|\widehat{\tau}^{I_2,*}/n$. To establish $\sqrt{n}(\widehat{\tau}_{\text{AIPW}} - \widehat{\tau}^*) \to_p 0$, it suffices to show the corresponding result for the treated mean; the other components are analogous. Define

$$\widehat{\tau}^{I_1} = \widehat{\mu}^{I_1}_{(1)} - \widehat{\mu}^{I_1}_{(0)}, \qquad \widehat{\mu}^{I_1}_{(1)} = \frac{1}{|I_1|}\sum_{i\in I_1}\left[\widehat{\mu}^{I_2}_{(1)}(X_i) + W_i\frac{Y_i - \widehat{\mu}^{I_2}_{(1)}(X_i)}{\widehat{e}^{I_2}(X_i)}\right].$$

Subtracting its oracle version gives

$$\begin{aligned}\widehat{\mu}^{I_1}_{(1)} - \widehat{\mu}^{I_1,*}_{(1)} = &\frac{1}{|I_1|}\sum_{i\in I_1}(\widehat{\mu}^{I_2}_{(1)}(X_i) - \mu_{(1)}(X_i))\left(1 - \frac{W_i}{e(X_i)}\right)\\ &+ \frac{1}{|I_1|}\sum_{i\in I_1} W_i(Y_i - \mu_{(1)}(X_i))\left(\frac{1}{\widehat{e}^{I_2}(X_i)} - \frac{1}{e(X_i)}\right)\\ &- \frac{1}{|I_1|}\sum_{i\in I_1} W_i(\widehat{\mu}^{I_2}_{(1)}(X_i) - \mu_{(1)}(X_i))\left(\frac{1}{\widehat{e}^{I_2}(X_i)} - \frac{1}{e(X_i)}\right).\end{aligned}$$

The blue annotations bound the first contribution by an expression written as $o_p(1)/n$ and the last by $o_p(1/\sqrt{n})$.

Original-note correction (S1-011): A variance rate was attached to the remainder itself

Source: Lecture1-10.pdf, pp. 14

If T_1 denotes the first average and $\mathcal{D}_2$ its training data, cross-fitting gives $E[T_1 \mid \mathcal{D}_2] = 0$ and

$$E[T_1^2 \mid \mathcal{D}_2] \leq \frac{E[(\widehat{\mu}_1(X) - \mu_1(X))^2 \mid \mathcal{D}_2]}{\eta |I_1|} = o_p(n^{-1}).$$

Thus $T_1 = o_p(n^{-1/2})$, not generally $o_p(n^{-1})$. The smaller rate belongs to its conditional variance/second moment. The second average is handled similarly, and the product term uses the displayed product-risk assumption. This identifies the likely omitted variance operation without changing the original annotation.

For K folds, use nuisance estimates trained without the fold $k(i)$ containing observation i:

$$\widehat{\tau}_{\text{AIPW}} = \frac{1}{n} \sum_i \left[\widehat{\mu}_{(1)}^{(-k(i))}(X_i) - \widehat{\mu}_{(0)}^{(-k(i))}(X_i) + W_i \frac{Y_i - \widehat{\mu}_{(1)}^{(-k(i))}(X_i)}{\widehat{e}^{(-k(i))}(X_i)} \right.$$
$$\left. - (1 - W_i) \frac{Y_i - \widehat{\mu}_{(0)}^{(-k(i))}(X_i)}{1 - \widehat{e}^{(-k(i))}(X_i)} \right].$$

Confidence intervals

Scan reference: Lecture1-10.pdf, pp. 16, 15; p. 15 supplies definitions used on p. 16.

Supplement: terminology. The notebook labels the likelihood derivative an efficient score:

$$S(\theta) = \frac{d \log L(\theta)}{d\theta},$$

where $L(\theta)$ is the likelihood. The empirical variance of n sample scores is

$$\frac{1}{n-1} \sum_i (S(\theta_i) - \overline{S(\theta)})^2.$$

Efficient variance is the smallest attainable variance for a given problem, with a Cramer–Rao lower-bound interpretation.

Original-note correction (S1-012): Likelihood score and efficient influence function are different objects

Source: Lecture1-10.pdf, pp. 15, 16

The displayed derivative is the ordinary likelihood score. With nuisance parameters, an efficient score additionally removes the nuisance-score component. For ATE estimation, the centered AIPW summand $\psi_i^* - \tau$ below is the efficient influence function; its variance is the efficiency bound. It is not the same as a raw likelihood score, whose variance is Fisher information. For observations O_i, empirical score variance would use $S_\theta(O_i)$ at a common θ, rather than different parameter values θ_i. The empirical-variance formula for AIPW on p. 16 is correct under the stated consistency and moment conditions.

Variance estimation. Define the oracle summand

$$\psi_i^* = \mu_{(1)}(X_i) - \mu_{(0)}(X_i) + \frac{W_i(Y_i - \mu_{(1)}(X_i))}{e(X_i)} - \frac{(1-W_i)(Y_i - \mu_{(0)}(X_i))}{1-e(X_i)}.$$

Its empirical variance converges to the efficient variance:

$$\frac{1}{n-1}\sum_i (\psi_i^* - \widehat{\tau}^*)^2 \longrightarrow_p V^*.$$

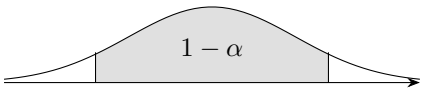
The margin calls $\widehat{\tau}^*$ the non-cross-fitted oracle AIPW estimator.

The non-cross-fitted oracle AIPW estimate (p. 16) $\longrightarrow$ $\widehat{\tau}^*$ in $\sum_i(\psi_i^* - \widehat{\tau}^*)^2/(n-1)$.

Replacing each nuisance function by the estimate trained outside $k(i)$ gives the cross-fitted summand $\widehat{\psi}_i$ and

$$\widehat{V}_{\text{AIPW}} = \frac{1}{n-1}\sum_i (\widehat{\psi}_i - \widehat{\tau}_{\text{AIPW}})^2 \longrightarrow_p V^*.$$

The confidence interval drawn under a central normal probability region is



Qualification / unresolved point (S1-013): Confidence level convention

Source: Lecture1-10.pdf, pp. 16

The formula below has asymptotic coverage $1-\alpha$; α is the error probability. The handwritten "α-level" can be read as a significance-level convention. No formula correction is needed.

$$\widehat{\tau}_{\text{AIPW}} \pm \frac{1}{\sqrt{n}}\Phi^{-1}(1-\alpha/2)\sqrt{\widehat{V}_{\text{AIPW}}}.$$

Lecture 4: Estimating Treatment Heterogeneity

Scan reference: Lecture1-10.pdf, pp. 17–18.

For observations (X_i, Y_i, W_i) with $X_i \in \mathcal{X}$, we previously pursued the ATE $E[Y_i(1) - Y_i(0)]$. Heterogeneity analysis instead considers the conditional average treatment effect (CATE):

$$\tau(x) = E[Y_i(1) - Y_i(0) \mid X_i = x].$$

Average for a category of X_i (p. 17) $\longrightarrow$ $\tau(x) = E[Y_i(1) - Y_i(0) \mid X_i = x]$.

Regularization bias

Under $\{Y_i(0), Y_i(1)\} \perp W_i \mid X_i$,

$$\tau(x) = \mu_{(1)}(x) - \mu_{(0)}(x), \qquad \mu_{(w)}(x) = E[Y_i \mid X_i = x, W_i = w].$$

A direct CATE estimator is $\widehat{\tau}_T(x) = \widehat{\mu}_{(1)}(x) - \widehat{\mu}_{(0)}(x)$.

Personal concern. If there are too few observations for $\widehat{\mu}_{(1)}$, it may be heavily regularized (toward a straight line). If there are many observations for $\widehat{\mu}_{(0)}$, that fit may be accurate; then the difference $\widehat{\tau}_T(x)$ goes wrong because of the first fit.

Semiparametric modeling

$$Y_i(w) = f(X_i) + w\tau(X_i) + \varepsilon_i(w), \qquad P(W_i = 1 \mid X_i) = e(X_i), \qquad \tau(x) = \psi(x)\beta.$$

f is a function of X_i, and $\psi(x)$ is a nonparametric representation of X; β is the treatment-effect coefficient. Under unconfoundedness, the Robinson transformation is

$Y_i(w) = f(X_i) + w\tau(X_i) + \varepsilon_i(w)$ (p. 17) $\longrightarrow$ Robinson: subtract $m(X_i)$ and residualize W_i by $e(X_i)$.

$$Y_i - m(X_i) = (W_i - e(X_i))\psi(X_i)\beta + \varepsilon_i, \quad m(x) = E[Y_i \mid X_i = x] = f(x) + e(x)\tau(x),$$

with $\varepsilon_i = \varepsilon_i(W_i)$.

$\varepsilon_i = \varepsilon_i(W_i)$ (p. 17) $\longrightarrow$ The error in $Y_i - m(X_i) = (W_i - e(X_i))\psi(X_i)\beta + \varepsilon_i$.

Design a residual-on-residual OLS regression using

$$\widetilde{Y}_i^* = Y_i - m(X_i), \qquad \widetilde{Z}_i^* = (W_i - e(X_i))\psi(X_i).$$

Regress $\widetilde{Y}_i^*$ on $\widetilde{Z}_i^*$ to obtain $\widehat{\beta}_R^*$ (the notebook uses a coefficient symbol resembling ζ_R). These transformations use only the observed data and the nuisance functions; the margin calls this data preprocessing. The notebook gives

$$\sqrt{n}(\widehat{\beta}_R^* - \beta) \Rightarrow N(0, V_R),$$
$$V_R = \mathrm{Var}[\widetilde{Z}_i^*]^{-1}\,\mathrm{Var}[\widetilde{Z}_i^*\widetilde{Y}_i^*]\,\mathrm{Var}[\widetilde{Z}_i^*]^{-1}.$$

Under homoskedasticity, $\mathrm{Var}(\varepsilon_i \mid X_i, W_i) = b^2$ is constant and V_R is the semiparametrically efficient variance for β. Under heteroskedasticity, the notes say that V_R still applies but is not the efficient variance.

Original-note correction (S1-014): Use the regression residual in the Robinson sandwich

Source: Lecture1-10.pdf, pp. 18

Let $Z = \widetilde{Z}_i^*$ be a column vector, $Q = E[ZZ']$ and $\varepsilon = \widetilde{Y}_i^* - Z'\beta$. The normal equations give influence function $Q^{-1}Z\varepsilon$, hence

$$V_R = Q^{-1}E[ZZ'\varepsilon^2]Q^{-1}.$$

The middle term is $\mathrm{Var}(Z\varepsilon)$, not $\mathrm{Var}(Z\widetilde{Y}^*)$. The latter includes random variation of $ZZ'\beta$ that the estimated denominator cancels. Here $E[Z] = 0$, so $Q = \mathrm{Var}(Z)$. Under conditional homoskedasticity the corrected variance is b^2Q^{-1}, with Q nonsingular and finite required moments. The source's efficiency discussion refers to this corrected sandwich.

Feasible cross-fitted version. The preceding regression is idealized because $m(X_i)$ and $e(X_i)$ are unknown. Fit nonparametric regressions $Y \sim X$ and $W \sim X$; omit the fold containing observation i when constructing

$$\widetilde{Y}_i = Y_i - \widehat{m}^{(-k(i))}(X_i), \qquad \widetilde{Z}_i = (W_i - \widehat{e}^{(-k(i))}(X_i))\psi(X_i).$$

Regress $\widetilde{Y}_i$ on $\widetilde{Z}_i$ to obtain $\widehat{\beta}_R$. If both nuisance RMSEs are $o_p(n^{-1/4})$, the notes conclude

$$\sqrt{n}(\widehat{\beta}_R^* - \widehat{\beta}_R) \longrightarrow_p 0.$$

A loss function for treatment heterogeneity

Scan reference: Lecture1-10.pdf, pp. 20, 19; p. 19 supplements the residual regression and the R-loss on p. 20.

The previous estimator assumes $\tau(x) = \psi(x)\beta$. Often the relationship is not linear, so use machine learning. With

$$\mu_{(w)}(x) = E[Y(w) \mid X = x], \quad \varepsilon_i(w) = Y_i(w) - [\mu_{(0)}(X_i) + w\tau(X_i)],$$

unconfoundedness yields $E[\varepsilon_i(w) \mid X_i, W_i] = 0$. Here $\mu_{(0)}$ plays the role of the previous f. Again,

$$Y_i - m(X_i) = (W_i - e(X_i))\tau(X_i) + \varepsilon_i, \quad m(x) = \mu_{(0)}(x) + e(x)\tau(x), \quad \varepsilon_i = \varepsilon_i(W_i).$$

Equivalently,

$$\tau(\cdot) = \arg\min_t E[((Y_i - m(X_i)) - (W_i - e(X_i))t(X_i))^2].$$

If m and e are known, minimizing this gives $\tau(\cdot)$.

Qualification / unresolved point (S1-015): R-loss identification and the RCT parenthesis

Source: Lecture1-10.pdf, pp. 20

The excess population R-loss is

$$L(t) - L(\tau) = E[e(X)(1 - e(X))(t(X) - \tau(X))^2].$$

Therefore the CATE is uniquely determined only where $0 < e(X) < 1$ (up to null sets). In an RCT, e is known by design; the outcome mean $m(x) = E[Y \mid X = x]$ is generally still unknown and must be estimated. Thus the parenthetical RCT exception does not make both nuisance functions known.

The oracle penalized estimator is

$$\widehat{\tau}_R^*(\cdot) = \arg\min_t \left\{ \frac{1}{n} \sum_i [(Y_i - m(X_i)) - (W_i - e(X_i))t(X_i)]^2 + \Lambda_n(t) \right\}.$$

Regularize the complexity of $\tau(\cdot)$ (p. 20) $\longrightarrow$ $\Lambda_n(\tau)$: linear examples are lasso/ridge; the margin also lists CNNs.

Usually m and e are unknown (the notes parenthetically mention RCTs), so cross-fit them:

$$\widehat{L}_n(t) = \frac{1}{n} \sum_i [(Y_i - \widehat{m}^{(-k(i))}(X_i)) - (W_i - \widehat{e}^{(-k(i))}(X_i))t(X_i)]^2,$$
$$\widehat{\tau}_R = \arg\min_t \{\widehat{L}_n(t) + \Lambda_n(t)\}.$$

For lasso, $\widehat{\tau}(x) = x\widehat{\beta}$, with

$$\widehat{\beta} = \arg\min_\beta \left\{ \frac{1}{n} \sum_i [(Y_i - \widehat{m}^{(-k(i))}(X_i)) - (W_i - \widehat{e}^{(-k(i))}(X_i))X_i\beta]^2 + \lambda\|\beta\|_1 \right\}.$$

Personal supplementary observation. With a constant treatment effect, $Y_i(w) = f(X_i) + \tau w + \varepsilon_i$ and $E[\varepsilon_i \mid X_i, W_i] = 0$, the residual estimator is

$$\widehat{\tau}_R = \frac{\sum_i (Y_i - \widehat{m}^{(-k(i))}(X_i))(W_i - \widehat{e}^{(-k(i))}(X_i))}{\sum_i (W_i - \widehat{e}^{(-k(i))}(X_i))^2}. \qquad \text{(4.5 in the notes)}$$

The premise is that the nuisance components converge fast enough. Even if τ is not constant, this estimator converges to a nonnegatively weighted average of $\tau(x)$ (the annotation also writes a treated-minus-control mean comparison).

$\overline{Y}_{(1)} - \overline{Y}_{(0)}$ (p. 19 marginal comparison)	$\longrightarrow$	The nonnegative weighted average of $\tau(x)$ in the preceding observation.

When heterogeneity is small and overlap presents difficulties, use (4.5) as a replacement.

Constant-coefficient residual regression and its weighted-effect interpretation (p. 19)	$\longrightarrow$	R-loss for a flexible treatment-effect function (p. 20).

Qualification / unresolved point (S1-016): The constant-coefficient residual regression changes the target

Source: Lecture1-10.pdf, pp. 19

With heterogeneous effects, its population limit is

$$\frac{E[e(X)(1 - e(X))\tau(X)]}{E[e(X)(1 - e(X))]},$$

provided the denominator is positive and nuisances converge sufficiently fast. It emphasizes regions of overlap. It equals the population ATE only with additional conditions, for example constant effects or zero covariance between $e(X)(1 - e(X))$ and $\tau(X)$. It does not identify effects where treatment is deterministic. This makes precise the author's nonnegative-weight intuition.

Lecture 5: Regression Discontinuity Designs

Scan reference: Lecture1-10.pdf, pp. 22, 21; p. 21 explains why the earlier IPW approach does not apply.

Introduce a cutoff using a running variable $Z_i \in \mathbb{R}$:

$$W_i = \mathbf{1}_{\{Z_i \geq c\}}, \qquad c \text{ is the threshold.}$$

As $Z_i \to c$, observations near c have similar characteristics and differ in whether they receive the intervention. Let $\mu_{(w)}(z) = E[Y_i(w) \mid Z_i = z]$. If the two conditional means are continuous, the CATE at $z = c$ is

$$\tau_c = \mu_{(1)}(c) - \mu_{(0)}(c) = \lim_{z \to c^+} E[Y_i \mid Z_i = z] - \lim_{z \to c^-} E[Y_i \mid Z_i = z].$$

Personal supplement: why does ordinary IPW not apply? Earlier we assumed unconfoundedness $\{Y_i(0), Y_i(1)\} \perp W_i \mid Z_i$ and overlap $\eta \leq P(W_i = 1 \mid Z_i) \leq 1 - \eta$. In RDD, unconfoundedness holds, but overlap fails because the treatment probability given $Z_i = z$ is either zero or one. The blue comment says this fails regardless of the nondegenerate choice of η. We must compare observations near c, using similarity rather than overlap. Failure of overlap means τ_c cannot be estimated at the $\sqrt{n}$ rate. Its minimax error converges at a nonparametric rate depending on the smoothness of $\mu_{(w)}(z)$. For example, a uniformly bounded second derivative at c allows an $n^{-2/5}$ estimation-error rate.

Qualification / unresolved point (S1-017): Scope of the RDD identification and rate claims

Source: Lecture1-10.pdf, pp. 21, 22, 24

The no-$\sqrt{n}$ statement concerns the nonparametric smoothness class used here, not every possible parametric restriction. The $n^{-2/5}$ rate uses a second-derivative bound on a neighborhood of the cutoff, positive continuous design density on both sides, independent observations, and bounded conditional variance. Continuity identifies the boundary contrast only when the cutoff is approached from both observed sides. The arithmetic balancing squared bias h^4 against variance $(nh)^{-1}$ is correct; under order-k smoothness and suitable local polynomial fitting it gives error $n^{-k/(2k+1)}$.

Estimation via local linear regression

Choose a small bandwidth $h_n \to 0$ and a symmetric weighting function K. Fit the two conditional means locally. The coefficient on W_i in

$$\arg \min_{a,\tau,\beta_{(0)},\beta_{(1)}} \sum_i K\left(\frac{|Z_i - c|}{h_n}\right) [Y_i - a - \tau W_i - \beta_{(0)}(Z_i - c)_- - \beta_{(1)}(Z_i - c)_+]^2$$

is $\widehat{\tau}_c$.

Weight; negative side; positive side (p. 22) $\longrightarrow$ $K(|Z_i - c|/h_n)$; $(Z_i - c)_-$; $(Z_i - c)_+$.

The intercept a and slopes $\beta_{(w)}$ are nuisance parameters; the handwritten labels identify negative-side and positive-side parts. Common kernels are the window function $K(x) = \mathbf{1}_{\{|x| \le 1\}}$ and triangular kernel $K(x) = (1 - |x|)_+$.

Estimator properties and smoothness

Scan reference: Lecture1-10.pdf, pp. 23–24.

Quantify continuity by assuming $\mu_{(w)}$ is twice differentiable with a uniformly bounded second derivative:

$$\left|\frac{d^2}{dz^2}\mu_{(w)}(z)\right| \le B, \qquad z \in \mathbb{R}, \quad w \in \{0, 1\}.$$

If smoothness is weaker, with only Lipschitz continuity (the note says bounded but not differentiable), local averaging may work better because there is no clear linear trend locally. If smoothness is stronger, with bounded derivatives of order $k \ge 3$, higher-order local regression may improve the convergence rate.

Original-note correction (S1-018): Lipschitz continuity is not bounded nondifferentiability

Source: Lecture1-10.pdf, pp. 23

Lipschitz continuity means $|f(x) - f(y)| \le L|x - y|$ for a finite constant L. It permits both differentiable functions and functions with corners, such as $|x|$. Boundedness alone does not imply Lipschitz continuity, and a Lipschitz function need not be bounded on an unbounded domain. The author's intended weaker-smoothness comparison remains valid.

Taylor-expand at c:

$$\mu_{(w)}(z) = a_{(w)} + \beta_{(w)}(z-c) + \tfrac{1}{2}\rho_{(w)}(z-c), \qquad |\rho_{(w)}(u)| \le Bu^2.$$

The last term is the remainder, and $\tau_c = a_{(1)} - a_{(0)}$. Fit each side separately:

$$(\widehat{a}_{(1)}, \widehat{\beta}_{(1)}) = arg\min_{a,\beta} \sum_{Z_i \ge c} K\left(\frac{|Z_i - c|}{h_n}\right)[Y_i - a - \beta(Z_i - c)]^2,$$

and analogously on the control side; then $\widehat{\tau}_c = \widehat{a}_{(1)} - \widehat{a}_{(0)}$.

Closed-form weights for the window kernel. On $c \le Z_i \le c + h_n$, the scan writes

$$\widehat{a}_{(1)} = \sum_i \gamma_i Y_i, \qquad \gamma_i = \frac{\widehat{E}_{(1)}[(Z_i - c)^2] - \widehat{E}_{(1)}[Z_i - c](Z_i - c)}{\widehat{E}_{(1)}[(Z_i - c)^2] - \widehat{E}_{(1)}[Z_i - c]^2},$$

where $\widehat{E}_{(1)}[Z_i - c]$ is the average of $Z_i - c$ within the bandwidth. It also records

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Original-note correction (S1-019): The local-linear weights need a sample-size denominator

Source: Lecture1-10.pdf, pp. 24

Let $N_+ = |\{i : c \le Z_i \le c + h_n\}|$, $u_i = Z_i - c$, and $m_j = N_+^{-1} \sum_i u_i^j$ within this band. The normalized intercept weights are

$$\gamma_i = \frac{m_2 - m_1 u_i}{N_+(m_2 - m_1^2)}.$$

Assume $m_2 - m_1^2 > 0$. Then $\sum_i \gamma_i = 1$ and $\sum_i \gamma_i u_i = 0$, as required. The source formula without N_+ sums to N_+. The later bias/noise calculations require the normalized weights.

$$\sum_i \gamma_i = 1, \qquad \sum_i \gamma_i (Z_i - c) = 0.$$

The decomposition on the page is

$$\widehat{a}_{(1)} = a_{(1)} + \underbrace{\sum_i \gamma_i \rho_{(1)}(Z_i - c)}_{\text{curvature bias}} + \underbrace{\sum_i \gamma_i (Y_i - \mu_{(1)}(Z_i))}_{\text{sampling noise}}.$$

The control intercept has an analogous decomposition.

Original-note correction (S1-020): Keep the Taylor remainder normalization consistent

Source: Lecture1-10.pdf, pp. 23, 24

With the earlier definition, the curvature term is $\frac{1}{2}\sum_i \gamma_i \rho_{(1)}(Z_i - c)$. Equivalently redefine $r_w = \rho_w/2$, with $|r_w(u)| \le Bu^2/2$, and write $\sum_i \gamma_i r_w(u_i)$. The omission affects constants, not the stated rate. Bounds on curvature bias use $\sum_i |\gamma_i| u_i^2$ because local-linear weights may have either sign.

Curvature bias and sampling noise (p. 24) ⟶ Both terms need bounds.

Curvature bias is bounded at order Bh_n^2, where B bounds the second derivative and h_n is the bandwidth. Sampling noise has mean zero; if $\mathrm{Var}[Y_i \mid Z_i] \le b^2$, its variance is bounded by $b^2 \sum_i \gamma_i^2$. If Z_i has a continuous nonzero density f near c,

$$b^2 \sum_i \gamma_i^2 \approx \frac{4b^2}{|\{i : c \le Z_i \le c + h_n\}|} \approx \frac{4b^2}{f(c) n h_n}.$$

Thus squared bias is of order h_n^4 and variance is of order $(nh_n)^{-1}$, giving

$$\widehat{\tau}_c = \tau_c + O_p(n^{-2/5}), \qquad h_n \asymp n^{-1/5}.$$

The red note identifies $n^{-2/5}$ as the estimation error and adds that with k-th order smoothness, $h_n \asymp n^{-1/(2k+1)}$.

Noise in the running variable

Scan reference: Lecture1-10.pdf, p. 25.

May contain noise (p. 25) ⟶ The running variable Z_i.

Introduce a latent $U_i \sim G$ and two conditions:

1. A noisy running variable: the scan writes $Z_i \mid U_i \sim N(0, \nu^2)$, with $\nu > 0$.
2. Unconfoundedness of the running variable: $\{Y_i(0), Y_i(1)\} \perp Z_i \mid U_i$.

With $Y_i = Y_i(W_i)$ and $\{Y_i(0), Y_i(1)\} \perp W_i \mid U_i$,

$$e(u) = P(W_i = 1 \mid U_i = u) = P(Z_i \ge c \mid U_i = u) = 1 - \Phi\left(\frac{c - u}{\nu}\right),$$

$$\alpha_{(w)}(u) = E[Y_i(w) \mid U_i = u], \qquad \tau(u) = E[Y_i(1) - Y_i(0) \mid U_i = u].$$

The annotation defines the standard normal CDF,

Original-note correction (S1-021): The noisy running variable must be centered at the latent variable

Source: Lecture1-10.pdf, pp. 25

The subsequent propensity and convolution formulas require $Z_i \mid U_i = u \sim N(u, \nu^2)$, equivalently $Z_i - U_i \mid U_i \sim N(0, \nu^2)$. If the written mean were really zero for every u, the propensity would be $1 - \Phi(c/\nu)$, independent of u. The source therefore omits u in the conditional mean. The later decomposition is correct for the latent-location model; its weight normalization alone does not remove confounding bias unless the induced h_+ and h_- are suitably balanced.

$\Phi(x) = (2\pi)^{-1/2} \int_{-\infty}^{x} e^{-t^2/2}\, dt.$

U_i is unobserved, so we cannot directly use its propensity score. Consider a deconvolution-type estimator:

Unobserved U_i: cannot directly use its propensity score (p. 25) $\longrightarrow$ Deconvolution weights $\gamma_+(z), \gamma_-(z)$.

$$\widehat{\tau}_\gamma = \frac{1}{n}\sum_{Z_i \geq c} \gamma_+(Z_i) Y_i - \frac{1}{n}\sum_{Z_i < c} \gamma_-(Z_i) Y_i.$$

Choose the weights to satisfy

$$E[\mathbf{1}_{\{Z_i \geq c\}} \gamma_+(Z_i)] = E[\mathbf{1}_{\{Z_i < c\}} \gamma_-(Z_i)] = 1.$$

Then

$$E[\widehat{\tau}_\gamma] = \underbrace{\int h_+(u)\tau(u)\, dG(u)}_{\text{weighted treatment effect}} + \underbrace{\int (h_+(u) - h_-(u))\alpha_{(0)}(u)\, dG(u)}_{\text{confounding bias}},$$

where the weighting functions are

$$h_+(u) = \int_c^\infty \gamma_+(z)\varphi_\nu(z-u)\, dz, \qquad h_-(u) = \int_{-\infty}^c \gamma_-(z)\varphi_\nu(z-u)\, dz,$$

$$\varphi_\nu(z-u) = \frac{1}{\sqrt{2\pi\nu^2}} \exp\left[-\frac{1}{2}\left(\frac{z-u}{\nu}\right)^2\right].$$

Lecture 6: Finite-Sample Inference in RDDs

Scan reference: Lecture1-10.pdf, pp. 27, 26, 29, 28; p. 26 continues p. 27, and p. 28 explains the weight constraints on p. 29.

Finite-sample RDDs allow a discontinuous/discrete running variable or a more complex cutoff rule. The local-linear regression solution (6.2) can be written as a linear estimator,

$$\widehat{\tau}_c(\gamma) = \sum_{i=1}^n \gamma_i Y_i. \qquad \text{(6.3 in the notes)}$$

The weights depend only on Z_i.

The local-linear solution (6.2) (p. 27) $\longrightarrow$ The linear estimator (6.3), whose properties follow below.

If

$$Y_i(w) = \mu_{(w)}(Z_i) + \varepsilon_i(w), \qquad \varepsilon_i(w) \mid Z_i \sim N(0, b^2),$$

then, conditional on $Z_1, \ldots, Z_n$,

$$\widehat{\tau}_c(\gamma) \sim N(\widehat{\tau}_c^*(\gamma), b^2\|\gamma\|_2^2), \qquad \widehat{\tau}_c^*(\gamma) = \sum_i \gamma_i \mu_{(W_i)}(Z_i).$$

Here $W_i = \mathbf{1}_{\{Z_i \geq c\}}$ and $\|\gamma\|_2 = (\gamma_1^2 + \cdots + \gamma_n^2)^{1/2}$ is the Euclidean norm.

Minimax linear estimation

Minimax means minimizing the worst-case MSE. For a linear estimator, the conditional variance is directly accessible as $b^2\|\gamma\|_2^2$ under homoskedasticity. Its bias depends on unknown conditional means:

$$\text{Bias}(\widehat{\tau}_c(\gamma) \mid Z_1, \ldots, Z_n) = \sum_i \gamma_i \mu_{(W_i)}(Z_i) - [\mu_{(1)}(c) - \mu_{(0)}(c)].$$

A smoothness restriction $|\mu''_{(w)}(z)| \le B$ bounds the bias by

$$I_B(\gamma) = \sup_{|\mu''_{(w)}| \le B} \left\{ \sum_i \gamma_i \mu_{(W_i)}(Z_i) - [\mu_{(1)}(c) - \mu_{(0)}(c)] \right\}.$$

The blue note says the supremum does not need an absolute value because the second-derivative class is symmetric. Hence

$$|\operatorname{Bias}(\widehat{\tau}_c(\gamma) \mid Z_1, \dots, Z_n)| \le I_B(\gamma),$$
$$\operatorname{MSE}(\widehat{\tau}_c(\gamma) \mid Z_1, \dots, Z_n) \le b^2 \|\gamma\|_2^2 + I_B^2(\gamma).$$

The red annotation emphasizes "MSE = variance + squared bias" and that this bound does not depend on the unknown μ. The left-page continuation defines

$$\gamma^B = \arg\min_\gamma \{b^2 \|\gamma\|_2^2 + I_B^2(\gamma)\}, \qquad \widehat{\tau}_c(\gamma^B) = \sum_i \gamma_i^B Y_i. \qquad \text{(6.8 in the notes)}$$

The blue comparison says that, at a fixed bias bound $I_B(\gamma)$, this yields smaller MSE than local-linear estimation.

Qualification / unresolved point (S1-022): What the finite-sample minimax guarantee says

Source: Lecture1-10.pdf, pp. 26, 27, 29, 30

The exact conditional normal calculation needs independent Gaussian errors across units (or the corresponding joint Gaussian covariance), with known variance and a valid curvature bound B. Marginal normality alone is insufficient. Minimizing $b^2\|\gamma\|_2^2 + I_B(\gamma)^2$ minimizes worst-case conditional MSE over the stated function class; it does not guarantee smaller actual MSE at every particular pair of response functions, nor strict improvement. The bias-adjusted interval is valid conditional on the design, including discrete designs, whenever the optimization is feasible. A plug-in choice of b or B requires further justification.

Deriving the minimax linear weights

Decompose

$$\mu_{(w)}(z) = a_{(w)} + \beta_{(w)}(z - c) + \rho_{(w)}(z),$$

where $\rho_{(w)}(c) = \rho'_{(w)}(c) = 0$ and the second derivative is bounded by B. The intercepts and slopes are unrestricted, so $I_B(\gamma)$ can be infinite unless

$$\sum_i \gamma_i W_i = 1, \qquad \sum_i \gamma_i = 0, \qquad \sum_i \gamma_i (Z_i - c)_+ = 0, \qquad \sum_i \gamma_i (Z_i - c)_- = 0.$$

After these constraints, only the remainders need control. Rewrite (6.8) as

$$(\gamma^B, t) = \arg\min_{\gamma, t} \{b^2 \|\gamma\|_2^2 + B^2 t^2\},$$

subject to

$$\sum_i \gamma_i W_i \rho_{(1)}(Z_i) + \sum_i \gamma_i (1 - W_i) \rho_{(0)}(Z_i) \le t$$

for every $\rho_{(w)}$ with $\rho_{(w)}(c) = \rho'_{(w)}(c) = 0$ and $|\rho''_{(w)}(z)| \le 1$, together with

$$\sum_i \gamma_i W_i = 1, \quad \sum_i \gamma_i = 0, \quad \sum_i \gamma_i W_i (Z_i - c) = 0, \quad \sum_i \gamma_i (1 - W_i)(Z_i - c) = 0. \quad \text{(6.10 in the notes)}$$

The blue instruction is to solve the corresponding dual problem.

Primal minimax optimization (6.10), p. 29 ⟶ Transform to the dual problem to solve it.

Personal supplement: why do the weights sum to zero? The final estimate is a difference. Split its sum into the treated and control parts:

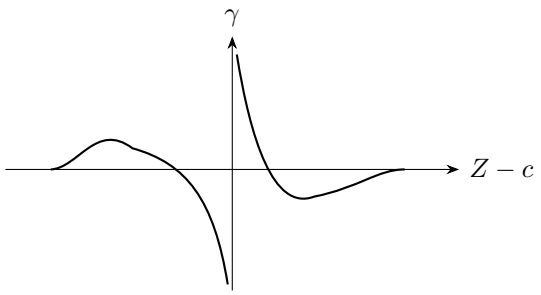
$$\widehat{\tau}_c(\gamma) = \sum_i \gamma_i^{(1)} Y_i + \sum_i \gamma_i^{(2)} Y_i.$$

The treated weights sum to 1 and the control weights sum to -1, so

$Y = 1$ group; $Y = 0$ group (source labels, p. 28) ⟶ Weights sum to 1 and -1, respectively; total sum 0.

$$\sum_i \gamma_i = 1 + (-1) = 0, \quad \sum_i \gamma_i^{(1)} = \sum_i \gamma_i W_i = 1, \quad \sum_i \gamma_i^{(2)} = \sum_i \gamma_i (1 - W_i) = -1.$$

The sketch shows a weight profile with a negative spike just to the left of the threshold and a positive spike just to the right; farther away, the weights change sign and then approach zero.



The red labels on the original say $Y = 1$ and $Y = 0$, while the equations below them use treatment indicators W.

Original-note correction (S1-023): The red group labels refer to treatment, not outcome

Source: Lecture1-10.pdf, pp. 28

These labels should be $W = 1$ and $W = 0$. The outcome Y need not be binary. The sum constraints themselves are correct: treated weights sum to 1 and control weights to -1.

Inference with linear estimators

Scan reference: Lecture1-10.pdf, pp. 30–31.

Construct a confidence interval for $\widehat{\tau}_c(\gamma^B)$. Let

$$\text{err} = \widehat{\tau}_c - \tau_c, \qquad \text{err} \mid Z_1, \ldots, Z_n \sim N(\text{bias}, b^2 \|\gamma^B\|_2^2),$$

where $|\text{bias}| \le Bt$ from (6.10). By unimodality of the Gaussian distribution, for $S \sim N(0,1)$,

$$P(|\text{err}| \ge \zeta) \le P(|Bt + b\|\gamma^B\|_2 S| \ge \zeta).$$

A confidence interval with conditional coverage at least $1 - \alpha$ is

$$I_\alpha = (\widehat{\tau}_c(\gamma^B) - \zeta_\alpha^B, \widehat{\tau}_c(\gamma^B) + \zeta_\alpha^B),$$

$$\zeta_\alpha^B = \inf\{\zeta : P(|Bt + b\|\gamma^B\|_2 S| > \zeta) \le \alpha\}. \qquad \text{(6.13 in the notes)}$$

The blue comment stresses that conditioning on Z_i makes this work for running variables with any distribution, including nonstandard RDD settings.

Discrete running variables. The restriction $|\mu''_{(w)}(z)| \leq B$ alone need not identify τ_c, but (6.13) remains valid. For continuous Z_i, the interval shrinks at rate $n^{-2/5}$; for discrete Z_i, its length may not tend to zero. When c is outside the support of the running variable, one needs an interpolating function representing $\mu_{(w)}(z)$ and satisfying the second-derivative bound.

Multivariate running variables. Examples include two- or three-dimensional regions and geographic boundaries. A scalar-cutoff RDD cannot be used directly; some reduce the problem to one-dimensional distance, which the note calls less stable. The minimax method can instead be extended directly to multiple dimensions.

Beyond homoskedasticity

When the errors $\varepsilon_i = Y_i - \mu_{(W_i)}(Z_i)$ are no longer homoskedastic Gaussian with variance b^2, the exact normal statement (6.11) fails. Use a central limit theorem to justify

$$\widehat{\tau}_c(\gamma) \mid Z_1, \ldots, Z_n \;\approx\; N\left(\widehat{\tau}^*_c(\gamma), \sum_i \gamma_i^2 \operatorname{Var}[Y_i \mid Z_i, W_i]\right). \qquad \text{(6.14 in the notes)}$$

With a valid Gaussian approximation, confidence intervals can still be obtained. Estimate the conditional variance by

$$\widehat{V}_n = \sum_i \gamma_i^2 (Y_i - \widehat{\mu}_{(W_i)}(Z_i))^2,$$

where the fitted mean comes from local-linear regression.

Local-linear fitted mean (p. 31) $\longrightarrow$ $\widehat{\mu}_{(W_i)}(Z_i)$ in the residual-variance estimator.

If the fitted mean is wrong (p. 31) $\longrightarrow$ Residuals inflate; the result is too conservative.

The minimax criterion (6.8) was for homoskedasticity; under heteroskedasticity, introduce individual variances b_i^2 into (6.10), the problem used to construct the dual.

Qualification / unresolved point (S1-024): Residual variance estimates are not automatically conservative

Source: Lecture1-10.pdf, pp. 31

For an independent evaluation observation and a fixed fitted mean, expected squared residual equals conditional variance plus squared mean error. With an in-sample fitted regression, overfitting and leverage can instead shrink residuals. Conservativeness therefore needs suitable leave-one-out/cross-fitted variance estimation or a theorem controlling the in-sample fit. The conditional Gaussian approximation also needs a weighted Lindeberg condition (no dominant observation), and the variance estimate must be consistent or conservatively valid. These requirements are not supplied merely by changing b^2 to b_i^2.

Lecture 7: Balancing Estimators

Scan reference: Lecture1-10.pdf, pp. 32–33.

The heading links balance to the principle behind estimating $\widehat{e}(x)$.

Balance / the design principle for $\widehat{e}(x)$ (p. 32) $\longrightarrow$ Improve propensity and regression functions; high-dimensional ATE estimation.

Two aims are noted: improving the propensity and regression functions, and estimating ATEs with high-dimensional covariates. IPW and AIPW use propensity scores; this lecture uses convex optimization to derive how to estimate $e(x)$.

Review of IPW

For IID observations $(X_i, Y_i, W_i) \in \mathcal{X} \times \mathbb{R} \times \{0, 1\}$, the goal is $\tau = \mu_{(1)} - \mu_{(0)}$, where $\mu_{(w)} = E[Y_i(w)]$. Assume

$$\{Y_i(0), Y_i(1)\} \perp W_i \mid X_i, \qquad 0 < \eta \leq e(X_i) \leq 1 - \eta < 1,$$

with $e(x) = P(W_i = 1 \mid X_i = x)$ and positive constant η. Estimate the treated mean using

$$\widehat{\mu}_{\text{IPW}}(1) = \frac{1}{n} \sum_i \frac{W_i Y_i}{\widehat{e}(X_i)}. \qquad \text{(7.3 in the notes)}$$

The notebook labels the following consistency argument "brief, not important" but records three steps:

1. Population balance: under unconfoundedness, $\mu_{(1)} = E[W_i Y_i / e(X_i)]$.
2. Oracle estimator: with the true $e(x)$ it is unbiased; overlap gives finite variance.
3. Feasible approximation: if the estimated propensities also satisfy overlap, the Cauchy–Schwarz inequality gives

$$\begin{aligned} |\widehat{\mu}_{\text{IPW}}(1) - \widehat{\mu}^*_{\text{IPW}}(1)| &\leq \left[\frac{1}{n}\sum_i \left(\frac{1}{\widehat{e}(X_i)} - \frac{1}{e(X_i)}\right)^2\right]^{1/2} \left[\frac{1}{n}\sum_i (W_i Y_i)^2\right]^{1/2} \\ &\leq \frac{1}{\eta^2}\left[\frac{1}{n}\sum_i (\widehat{e}(X_i) - e(X_i))^2\right]^{1/2} \left[\frac{1}{n}\sum_i (W_i Y_i)^2\right]^{1/2}. \end{aligned}$$

The margin recalls $|a \cdot b|^2 \leq \|a\|^2 \|b\|^2$. It adds that the analogous AIPW error is negligible relative to $1/\sqrt{n}$ as $n \to \infty$.

Population versus sample balance

Expand the conditional mean in a predefined basis:

$$\mu_{(w)}(x) = \sum_{j=1}^{\infty} \beta_{j(w)} \psi_j(x).$$

Polynomial or Fourier basis (p. 33) $\longrightarrow$ $\psi_j(x)$ in the conditional-mean expansion.

Under reasonable regularity conditions,

$$\mu_{(w)} = \sum_{j=1}^{\infty} \beta_{j(w)} E[\psi_j(X_i)].$$

Take expectations on both sides (p. 33) $\longrightarrow$ $\mu_{(w)} = \sum_j \beta_{j(w)} E[\psi_j(X)]$.

Could these regularity assumptions have a problem?

Qualification / unresolved point (S1-025): Answer to the basis-expansion regularity question

Source: Lecture1-10.pdf, pp. 33

The question is justified: pointwise convergence of a series alone does not justify exchanging its sum and expectation. A sufficient condition is $\sum_j |\beta_{j(w)}| E|\psi_j(X)| < \infty$; alternatively, convergence of partial sums in L^1 suffices. L^2 convergence under a probability distribution also implies L^1 convergence. Under overlap, the same reasoning controls the weighted expectation. The population identity can also be proved directly by conditioning on X, without an infinite basis expansion.

With unconfoundedness, $Y_i = \mu_{(W_i)}(X_i) + \varepsilon_i$ and $E[\varepsilon_i \mid X_i, W_i] = 0$. The population IPW argument is

$$E\left[\frac{W_i Y_i}{e(X_i)}\right] = E\left[\frac{W_i}{e(X_i)} \sum_j \beta_{j(1)} \psi_j(X_i)\right]$$
$$= \sum_j \beta_{j(1)} E\left[\frac{W_i \psi_j(X_i)}{e(X_i)}\right] = \sum_j \beta_{j(1)} E[\psi_j(X_i)] = \mu_{(1)}.$$

The core: $E[W_i \mid X_i] = e(X_i)$ (p. 33) $\longrightarrow$ $E[W_i \psi_j(X_i)/e(X_i)] = E[\psi_j(X_i)]$.

Thus choose weights so that, in the sample,

$$\frac{1}{n} \sum_i \frac{W_i \psi_j(X_i)}{\widehat{e}(X_i)} \approx \frac{1}{n} \sum_i \psi_j(X_i), \qquad j = 1, 2, \ldots. \tag{7.8 in the notes}$$

A balancing loss for propensity estimation

Scan reference: Lecture1-10.pdf, pp. 35, 34; p. 34 supplies the norm reminders used in the high-dimensional bounds.

Suppose the outcome model is linear, $\mu_{(w)}(x) = x\beta_{(w)}$, and the propensity model is logistic,

$$e(x) = \frac{1}{1 + \exp(-x\theta)}.$$

The blue annotation says that sample balance then involves only the covariates X_i, and the targeted balance concerns means. Equation (7.8) becomes

$$\frac{1}{n} \sum_i (1 + e^{-X_i \widehat{\theta}}) W_i X_i = \frac{1}{n} \sum_i X_i. \tag{7.9 in the notes}$$

This is the KKT condition of the convex problem

$$\widehat{\theta} = \arg\min_\theta \frac{1}{n} \sum_i \ell_\theta(X_i, Y_i, W_i), \qquad \ell_\theta(X_i, Y_i, W_i) = W_i e^{-X_i \theta} + (1 - W_i) X_i \theta. \tag{7.10 in the notes}$$

It can therefore be solved by Newton descent.

Balance equation (7.9) is the convex loss KKT condition (p. 35)	$\longrightarrow$	Newton descent; the source claims a unique solution.

The handwritten text describes a unique solution and says this serves as the loss function.

Original-note correction (S1-026): Convexity alone does not guarantee a unique finite solution

Source: Lecture1-10.pdf, pp. 35

The Hessian is $n^{-1}\sum_i W_i e^{-X_i\theta} X_i' X_i$. It is positive semidefinite, and positive definite only with suitable span/rank conditions on treated covariates. A finite minimizer also needs existence; exact balance can fail when the target covariate mean lies outside the attainable treated convex support. Thus convexity alone establishes neither uniqueness nor attainment, and Newton iterations require the usual safeguards. The displayed first-order balance equation is correct.

The margin notes that estimating both $\mu_{(0)}$ and $\mu_{(1)}$ requires two propensity models.

Qualification / unresolved point (S1-027): Two calibration fits versus one true propensity

Source: Lecture1-10.pdf, pp. 35

The treatment mechanism has one propensity $e(x)$; control probabilities are $1-e(x)$. Two separate calibration fits are one way to enforce arm-specific balance in this construction, but two distinct true propensity models are not mathematically required for estimating both means.

ATE estimation with high-dimensional confounders

Suppose $X_i \in \mathbb{R}^p$, possibly $p \gg n$, and $\mu_{(w)}(x) = x\beta_{(w)}$. In high dimensions, it is difficult to estimate $e(x)$ adequately; exact weights are not attainable, so seek approximate balance. By Holder's inequality,

$$\begin{aligned}
&\frac{1}{n}\sum_i \frac{W_i \mu_{(1)}(X_i)}{\widehat{e}(X_i)} - \frac{1}{n}\sum_i \mu_{(1)}(X_i) \\
&\quad = \left(\frac{1}{n}\sum_i \frac{W_i X_i}{\widehat{e}(X_i)} - \frac{1}{n}\sum_i X_i\right)\beta_{(1)} \\
&\quad \le \left\| \frac{1}{n}\sum_i \frac{W_i X_i}{\widehat{e}(X_i)} - \frac{1}{n}\sum_i X_i \right\|_\infty \|\beta_{(1)}\|_1 .
\end{aligned}$$

The first norm is labeled imbalance. The handwritten reminder states Holder's bound $\sum_i |a_i b_i| \le (\sum_i |a_i|^p)^{1/p} (\sum_i |b_i|^q)^{1/q}$.

Personal interpretation and questions. Good inverse-probability weights make the worst case of imbalance as good as possible. Two problems remain: (1) how to parameterize $\widehat{e}(X_i)$ to obtain sup-norm approximate balance; (2) the bound is meaningful only if $\|\beta_{(1)}\|_1$ is bounded. High-dimensional statistics usually assumes a sparse coefficient vector rather than constraining its 1-norm directly.

Supplement: norm reminders. The left page lists the "0-norm" as $\sum_i X_i^0$, the 1-norm as $\sum_i |X_i|$, the 2-norm as $(\sum_i X_i^2)^{1/2}$, and the infinity norm as $\max_i |X_i|$. It gives the general expression $\|x\|_p = (\sum_i x_i^p)^{1/p}$.

Original-note correction (S1-028): Correct the norm shorthand and specify Holder exponents

Source: Lecture1-10.pdf, pp. 34, 35

Use $\|x\|_0 = \sum_i \mathbf{1}\{x_i \neq 0\}$ (a count, not a norm) and $\|x\|_p = (\sum_i |x_i|^p)^{1/p}$ for $p \geq 1$. x_i^0 does not properly handle zero entries, and odd/noninteger powers do not define the intended real norm for negative entries. Holder requires $p, q \geq 1$ with $1/p + 1/q = 1$, including the $(1, \infty)$ endpoint. The displayed imbalance bound follows by this endpoint inequality; its absolute-value version bounds the magnitude of the error.

Approximate balancing weights and regression augmentation

Scan reference: Lecture1-10.pdf, pp. 36–37.

For a well-chosen tuning parameter ζ, solve

$$\widehat{\gamma} = \arg\min_{\gamma_i \geq 1} \left\{ \frac{1}{n^2}\|\gamma\|_2^2 + \zeta \left\| \frac{1}{n}\sum_i (\gamma_i W_i - 1) X_i \right\|_\infty^2 \right\}. \qquad \text{(7.12 in the notes)}$$

Squared L_2 size; imbalance; tuning weight (p. 36) $\longrightarrow$ $n^{-2}\|\gamma\|_2^2$; the squared infinity norm; ζ.

Then use $1/\widehat{e}(X_i) = \widehat{\gamma}_i$ as inverse-propensity weights. Sub-Gaussian concentration inequalities verify, for oracle weights, bounds of the form

$$\frac{1}{n^2}\|1/e(X_i)\|_2^2 \leq \frac{\eta^{-2}}{n}, \qquad \left\| \frac{1}{n}\sum_i (\gamma_i W_i - 1) X_i \right\|_\infty^2 = O_p\left(\frac{\eta^{-2}\log p}{n} \right).$$

The notes say $\widehat{e}(X_i)$ need not estimate $e(X_i)$ well; with a suitable ζ, these inverse weights nonetheless satisfy a bias bound. Next augment the weighted estimator with high-dimensional regression to remove dependence on $\|\beta_{(1)}\|_1$. The augmented balancing estimator is

$$\widehat{\mu}_{\mathrm{AB}}(1) = \frac{1}{n}\sum_i \left[\widehat{\mu}_{(1)}(X_i) + \widehat{\gamma}_i W_i (Y_i - \widehat{\mu}_{(1)}(X_i)) \right], \quad \widehat{\mu}_{(1)}(X_i) = X_i \widehat{\beta}_{(1)}. \qquad \text{(7.14 in the notes)}$$

Its decomposition is

$$\widehat{\mu}_{\mathrm{AB}}(1) = \underbrace{\frac{1}{n}\sum_i X_i \beta_{(1)}}_{\text{mean of } \mu_{(1)}(X_i)} + \underbrace{\frac{1}{n}\sum_i \widehat{\gamma}_i W_i (Y_i - X_i \beta_{(1)})}_{\text{mean-zero noise}} + \left[\frac{1}{n}\sum_i (1 - \widehat{\gamma}_i W_i) X_i \right] (\widehat{\beta}_{(1)} - \beta_{(1)}).$$

The last term is bounded in magnitude by

$$\left\| \frac{1}{n}\sum_i (1 - \widehat{\gamma}_i W_i) X_i \right\|_\infty \|\widehat{\beta}_{(1)} - \beta_{(1)}\|_1.$$

Weights need only correct $\beta_{(1)} - \widehat{\beta}_{(1)}$, not the whole coefficient vector (p. 36) $\longrightarrow$ The imbalance times regression-error remainder.

High-dimensional regression supplies a bound on this error: under sparsity $\|\beta_{(1)}\|_0 \leq k$ and a restricted-eigenvalue condition, lasso achieves

$$\|\widehat{\beta}_{(1)} - \beta_{(1)}\|_1 = O_p\left(k\sqrt{\frac{\log p}{n}}\right).$$

Putting the steps together. Suppose the design satisfies the conditions for this bound, $\beta_{(1)}$ is k-sparse, and $k \ll \sqrt{n}/\log p$.

1. Fit lasso on the treated units; the coefficient 1-norm error is $o_p(1/\sqrt{\log p})$.
2. Fit $\widehat{\gamma}$ using (7.12), giving infinity-norm imbalance $O_p(\sqrt{\log(p)/n})$.
3. Use (7.14), and insert these bounds into the decomposition:

$$\widehat{\mu}_{\text{AB}}(1) - \frac{1}{n}\sum_i X_i\beta_{(1)} = \frac{1}{n}\sum_i \widehat{\gamma}_i W_i(Y_i - X_i\beta_{(1)}) + o_p(n^{-1/2}). \qquad \text{(7.17 in the notes)}$$

The noise term is $O_p(n^{-1/2})$.

Qualification / unresolved point (S1-029): Conditions behind high-dimensional balancing inference

Source: Lecture1-10.pdf, pp. 32, 36, 37

Overlap does not by itself give finite variance: require $E[Y(w)^2] < \infty$. The concentration bounds also use controlled sub-Gaussian covariates, and the lasso rate needs an appropriate tuning level, restricted eigenvalues and noise assumptions. The stated sparsity condition makes the remainder $o_p(n^{-1/2})$ by multiplying the two rates. Conditional on (X, W), weights must depend on these variables rather than outcome noise. A CLT additionally needs no dominant weighted residual; the displayed studentization uses the unknown true β_1, so feasible inference must justify replacing it by an estimate.

A central limit theorem also holds:

$$\frac{\sum_i \widehat{\gamma}_i W_i(Y_i - X_i\beta_{(1)})}{\sqrt{\sum_i \widehat{\gamma}_i^2 W_i^2 (Y_i - X_i\beta_{(1)})^2}} \Rightarrow N(0,1).$$

Personal caveat. This is weaker than expected: it only gives a confidence interval for the sample average $\overline{\mu}_{(1)} = n^{-1}\sum_i X_i\beta_{(1)}$, not for $\mu_{(1)}$ itself. To obtain a result for the population mean, one needs to establish convergence of the estimated weights $\widehat{\gamma}_i$.

Qualification / unresolved point (S1-030): Sample-target and population-target uncertainty

Source: Lecture1-10.pdf, pp. 37

For the population target, also include the sampling fluctuation $n^{-1}\sum_i \mu_1(X_i) - E[\mu_1(X)]$. Convergence of the fitted weights to a stable function is a useful sufficient route to an influence-function result; it is not a logically necessary condition in every construction. One must establish the joint limit and its variance. The author's warning that the displayed conditional interval only covers the sample mean is correct.

Lecture 8: Methods for Panel Data

Scan reference: Lecture1-10.pdf, pp. 39, 38; p. 38 supplies the panel layout and within-unit variance supplement.

Panel (longitudinal) data observe units $i = 1, \dots, n$ at times $t = 1, \dots, T$. The left-page sketch is a matrix with units down the rows and time points $t_1, t_2, \dots, T$ across the columns.

	$t = t_1$	$t = t_2$	$\cdots$	$t = T$
$i = 1$				
$i = 2$				
$\vdots$				
$i = n$				

A constant treatment-effect model

Outcomes are $Y_{it} \in \mathbb{R}$ and treatment assignments are $W_{it} \in \{0, 1\}$. The basic model is

$$Y_{it} = Y_{it}(0) + W_{it}\tau, \qquad i = 1, \dots, n, \quad t = 1, \dots, T. \tag{8.1 in the notes}$$

This very basic model leaves out treatment heterogeneity and treatment dynamics. The notes explain that the treatment effect is the same for all units and all times; a unit's outcome at t depends only on its treatment at t, not on previous treatment. τ does not depend on i, t; past times do not enter.

The two-way model

$$Y_{it} = \alpha_i + \beta_t + W_{it}\tau + \varepsilon_{it}, \qquad E[\varepsilon \mid \alpha, \beta, W] = 0. \tag{8.2 in the notes}$$

α_i and β_t are unit and time shifts. The red comment calls the condition strict exogeneity of treatment, stronger than unconfoundedness.

Qualification / unresolved point (S1-031): Strict exogeneity and unconfoundedness have different scopes

Source: Lecture1-10.pdf, pp. 39

Here strict exogeneity requires mean-zero errors conditional on the entire treatment path and the fixed effects; treatment decisions cannot respond to the idiosyncratic errors in any period. It is stronger than a corresponding contemporaneous mean-exogeneity condition, but it is not literally a universal implication ordering with every distributional unconfoundedness assumption. State the conditioning sets and whether mean or full independence is intended.

For example, with two periods, let one group have treatment path $(0, 0)$ and another $(0, 1)$. The closed-form difference-in-differences estimate is

DID (p. 39) $\longrightarrow$ Difference of the two groups' before–after differences.

$$\widehat{\tau} = \frac{1}{|\{i : W_{i2} = 1\}|} \sum_{W_{i2}=1} (Y_{i2} - Y_{i1}) - \frac{1}{|\{i : W_{i2} = 0\}|} \sum_{W_{i2}=0} (Y_{i2} - Y_{i1}).$$

Comments in the notes. A limitation is that with large n, T and general treatment-assignment distributions, model (8.2) may fail. Analogously, OLS with more than two periods is sometimes also called DID, although the note says there is no closed-form OLS solution. The listed advantage is the intuitive parallel-trends requirement (units differ only in α_i); without parallel trends, it cannot be used.

Original-note correction (S1-032): General two-way OLS still has a closed form

Source: Lecture1-10.pdf, pp. 39

For a balanced panel, residualize Y and W on unit and time indicators: $\widetilde{W}_{it} = W_{it} - \overline{W}_{i\cdot} - \overline{W}_{\cdot t} + \overline{W}_{\cdot\cdot}$, and similarly for Y. When $\sum_{it} \widetilde{W}_{it}^2 > 0$,

$$\widehat{\tau} = \frac{\sum_{it} \widetilde{W}_{it} \widetilde{Y}_{it}}{\sum_{it} \widetilde{W}_{it}^2}.$$

Thus the simple two-group before/after expression need not apply, but a closed-form OLS coefficient exists. The fixed-effect model implies parallel conditional mean trends, not exactly parallel noisy realized paths.

Supplement: modeling the noise and estimating variance. Treat the row vectors $\varepsilon_i = (\varepsilon_{i1}, \ldots, \varepsilon_{iT})$ as IID across units, with $\mathrm{Var}[\varepsilon_i] = \Sigma$. Treating every ε_{it} as independent is hard to justify and unreasonable. For $T = 2$, estimate the variance of the differences directly:

$$\widehat{\mathrm{Var}}[\widehat{\tau}] = \frac{\widehat{\mathrm{Var}}[Y_{i2} - Y_{i1} \mid W_{i2} = 1]}{|\{i : W_{i2} = 1\}|} + \frac{\widehat{\mathrm{Var}}[Y_{i2} - Y_{i1} \mid W_{i2} = 0]}{|\{i : W_{i2} = 0\}|}. \qquad \text{(8.4 in the notes)}$$

This remains robust when errors within a row/unit are correlated.

Interactive panel models

Scan reference: Lecture1-10.pdf, page 41, with the matrix intuition on page 40

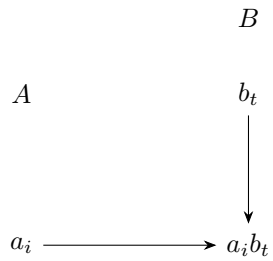
Consider the interactive panel model

$$Y_{it} = A_{i\cdot} B_{t\cdot}^{\mathsf{T}} + W_{it}\tau + \varepsilon_{it}, \qquad \mathbb{E}[\varepsilon \mid A, B, W] = 0, \tag{8.5}$$

where $A \in \mathbb{R}^{n\times k}$, $B \in \mathbb{R}^{T\times k}$ and k is a rank parameter. Equivalently,

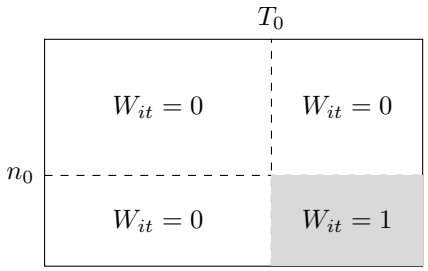
$$Y_{it} = L_{it} + W_{it}\tau + \varepsilon_{it} \qquad \text{for a rank-}k\text{ matrix } L.$$

My left-page matrix intuition. Place the A values down the rows and the B values across the columns. For rank one, their intersection has entry $a_i b_t$; for $k \neq 1$, the entry is a sum $a_{i1}b_{t1} + a_{i2}b_{t2} + \cdots$.



For $k > 1$: $\sum_{j=1}^{k} a_{ij} b_{tj}$

To handle this model, consider synthetic control and synthetic differences-in-differences. Suppose $W_{it} = 1$ exactly when $i > n_0$ and $t > T_0$. The diagram partitions the outcome matrix at row n_0 and column T_0, with only the bottom-right block treated.



Fit the untreated units to the pretreatment trajectory of the treated group:

$$\sum_{i=1}^{n_0} \gamma_i Y_{it} \approx \alpha + \frac{1}{n - n_0} \sum_{i=n_0+1}^{n} Y_{it}, \qquad t = 1, \ldots, T_0.$$

The first group is never treated; the second is treated after T_0. The intercept α plays a role similar to a fixed effect. Choose weights by

$$\begin{aligned} (\widehat{\gamma}, \widehat{\alpha}) \in \underset{\gamma', \alpha}{\arg\min} \quad & \left\| \sum_{i=1}^{n_0} \gamma_i' Y_{i,1:T_0} - \frac{1}{n - n_0} \sum_{i=n_0+1}^{n} Y_{i,1:T_0} - \alpha \mathbf{1} \right\|_2^2, \\ \text{subject to} \quad & \sum_{i=1}^{n_0} \gamma_i' = 1, \qquad \gamma_i' \geq 0. \end{aligned} \tag{8.7}$$

The purpose of γ_i is to create parallel trends, so that the latent A_i terms can be balanced.

Finally, estimate the treatment coefficient in the weighted two-way model:

$$(\widehat{\tau}, \widehat{\alpha}, \widehat{\beta}) \in \arg\min_{\tau', \alpha', \beta'} \sum_{i,t} \gamma_i (Y_{it} - \alpha_i' - \beta_t' - W_{it}\tau')^2, \tag{8.8}$$

where $\gamma_i = 1/(n - n_0)$ for $i > n_0$.

Treated-unit weights in the weighted regression (p. 41) $\longrightarrow$ $\gamma_i = 1/(n - n_0)$ for $i > n_0$.

The closed-form treatment coefficient is

$$\widehat{\tau} = \frac{1}{n - n_0} \sum_{i=n_0+1}^{n} \left\{ \frac{1}{T - T_0} \sum_{t=T_0+1}^{T} Y_{it} - \frac{1}{T_0} \sum_{t=1}^{T_0} Y_{it} \right\} - \sum_{i=1}^{n_0} \gamma_i \left\{ \frac{1}{T - T_0} \sum_{t=T_0+1}^{T} Y_{it} - \frac{1}{T_0} \sum_{t=1}^{T_0} Y_{it} \right\}.$$

The red margin says this depends on T_0 and n. Under an appropriate large-panel asymptotic regime, $n_0, n_1, T \to \infty$, the estimator is consistent. A further handwritten qualification below this statement is clipped by the bottom edge of the scan; the visible fragments refer to the specification in (8.5) and asymptotic normal inference, but a complete sentence cannot be recovered.

Qualification / unresolved point (S1-033): Latent-factor balance and panel asymptotics

Source: Lecture1-10.pdf, pp. 40, 41

A close pretreatment fit alone does not prove that posttreatment counterfactual trends are balanced: preperiod factors must reveal the directions relevant after treatment, and noise/weight concentration must be controlled. The factorization has rank at most k; exact rank k needs full-rank factors. The consistency claim is conditional on a suitable asymptotic regime and assumptions, not on $n, T \to \infty$ alone. The full synthetic-DID method may also use time weights and regularization. See Arkhangelsky et al. (2021) and lecture (8.7)–(8.9); the clipped handwriting is not reconstructed from that reference.

Left-page postscript. Synthetic control is not the only method for interactive fixed effects. For example, apply k-means clustering to the rows of the outcome matrix Y, then fit a time-varying baseline model separately for each group. A short reference is written as "p66, literature" [page number/reference unclear].

Identification through exchangeability

Scan reference: Lecture1-10.pdf, page 43, with the extension on page 42

Recall $Y_{it} = Y_{it}(0) + W_{it}\tau$. A design-based assumption is

$$Y_{i\cdot}(0) \perp W_{i\cdot} \mid S_i, \qquad S_i = \sum_{t=1}^{T} W_{it}. \tag{8.10}$$

One implication: consider a linear estimator

$$\widehat{\tau} = \sum_{i,t} \gamma_{it} Y_{it}, \tag{8.11}$$

whose weight matrix depends only on W. Under (8.10), the untreated potential outcomes are exchangeable within a value of S_i, and

$$\mathbb{E}[\widehat{\tau} \mid W, \gamma] = \sum_{i,t} \gamma_{it} \mathbb{E}[Y_{it}(0) \mid S_i] + \tau \sum_{i,t} \gamma_{it} W_{it}.$$

Thus unbiasedness follows if

$$\sum_{i,t} \gamma_{it} W_{it} = 1, \qquad \sum_{i:S_i=s} \gamma_{it} = 0 \quad \text{for every } t \text{ and } s \in \mathcal{S},$$

where $\mathcal{S}$ is the support of S_i.

Within a stratum, the conditional expectation is constant (p. 43) $\longrightarrow$ $\sum_{i:S_i=s} \gamma_{it} = 0$ removes $\sum_i \gamma_{it} E[Y_{it}(0) \mid S_i]$.

Qualification / unresolved point (S1-034): Exchangeability across units and the variance criterion

Source: Lecture1-10.pdf, pp. 42, 43

For the cancellation, require a common conditional mean $E[Y_{it}(0) \mid S_i = s] = m_t(s)$ across units, and conditions such as IID rows ensuring that conditioning on the full assignment matrix adds no information about a row beyond its own path. Within-row independence of $Y_i(0)$ and W_i given S_i alone does not impose cross-unit exchangeability. The combined restrictions give the claimed double robustness when feasible. Minimizing $\sum_{it} \gamma_{it}^2$ minimizes variance for conditionally uncorrelated homoskedastic cells; with within-unit covariance Σ_i, the variance criterion is $\sum_i \gamma_i' \Sigma_i \gamma_i$. Unbiasedness of feasible weights does not require that working variance model to be correct.

Choose γ by minimizing variance under these unbiasedness constraints:

$$\begin{aligned} \gamma \in \arg\min_{\gamma'} \quad & \sum_{i,t} (\gamma'_{it})^2, \\ \text{subject to} \quad & \sum_{i,t} \gamma'_{it} W_{it} = 1, \quad \sum_{i:S_i=s} \gamma'_{it} = 0 \quad \text{for all } s, t. \end{aligned} \tag{8.14}$$

Roughly, we need variation in W_{it} among units with the same S_i.

Left-page extension: two sets of identifying restrictions. Recall the two-way model

$$Y_{it} = \alpha_i + \beta_t + W_{it}\tau + \varepsilon_{it}, \qquad \mathbb{E}[\varepsilon \mid \alpha, \beta, W] = 0. \tag{8.2, recalled}$$

The estimator (8.11) is also unbiased under this model if

$$\sum_t \gamma_{it} = 0 \text{ for every } i, \qquad \sum_i \gamma_{it} = 0 \text{ for every } t, \qquad \sum_{i,t} \gamma_{it} W_{it} = 1.$$

The first two equalities cancel the fixed effects α_i and β_t, and the third normalizes the treatment coefficient.

Combine the restrictions to obtain a doubly robust estimator: minimize $\sum_{i,t} \gamma_{it}^2$ subject to

$$\sum_{i,t} \gamma_{it} W_{it} = 1, \qquad \sum_{i:S_i=s} \gamma_{it} = 0 \quad \text{for all } s, t,$$

$$\sum_t \gamma_{it} = 0 \quad \text{for all } i, \qquad \sum_i \gamma_{it} = 0 \quad \text{for all } t.$$

It is unbiased if either (8.2) or (8.10) holds. An abbreviated reference near the top of the supplement reads "p67, [2019]" [the citation is not fully identifiable from the scan].

Lecture 9: Instrumental variables regression

Scan reference: Lecture1-10.pdf, page 45, with the three IV conditions on page 44

Instrumental variables regression addresses endogenous treatment effects, where $Y_i(0)$ and W_i are dependent.

A structural model

Start with a constant-effect model:

$$Y_i(w) = Y_i(0) + w\tau, \qquad Y_i = Y_i(W_i). \tag{9.1}$$

Unconfoundedness ordinarily means $\{Y_i(w)\} \perp W_i$; in (9.1), this is $Y_i(0) \perp W_i$. Rewrite the model as

Constant-effect potential outcomes and unconfoundedness (p. 45) $\longrightarrow$ $Y_i = \alpha + W_i\tau + \varepsilon_i$.

$$Y_i = \alpha + W_i\tau + \varepsilon_i, \tag{9.2}$$

where $\alpha = \mathbb{E}[Y_i(0)]$ and $\varepsilon_i = Y_i(0) - \mathbb{E}[Y_i(0)]$. Under unconfoundedness, $\mathbb{E}[\varepsilon_i \mid W_i] = 0$, so OLS of Y_i on W_i estimates τ.

With confounding, equation (9.2) still holds, but $\mathbb{E}[\varepsilon_i \mid W_i] \neq 0$. OLS then targets

$$\begin{aligned}\tau_{\text{OLS}} &= \frac{\text{Cov}(Y_i, W_i)}{\text{Var}(W_i)} \\ &= \frac{\text{Cov}(\tau W_i + \varepsilon_i, W_i)}{\text{Var}(W_i)} = \tau + \frac{\text{Cov}(\varepsilon_i, W_i)}{\text{Var}(W_i)} \neq \tau.\end{aligned}$$

The ratio is underlined as the OLS coefficient.

Qualification / unresolved point (S1-035): Dependence need not create a nonzero linear bias

Source: Lecture1-10.pdf, pp. 45

The exact identity $\tau_{\text{OLS}} = \tau + \text{Cov}(\varepsilon, W)/\text{Var}(W)$ is correct. However, dependence of $Y(0)$ and W does not necessarily imply nonzero conditional mean or nonzero covariance. For example, $Y(0)$ can have treatment-dependent variance but the same conditional mean. The displayed $\neq \tau$ describes the confounded case with nonzero covariance, not a consequence of every dependence. Identification by the subsequent IV ratio requires its nonzero denominator.

Identification with an instrument

Introduce Z_i:

$$Y_i = \alpha + W_i\tau + \varepsilon_i, \qquad \varepsilon_i \perp Z_i, \qquad W_i = Z_i\gamma + \eta_i. \tag{9.4}$$

Since Z_i is uncorrelated with ε_i,

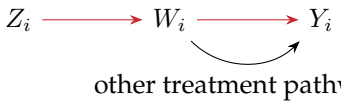
$$\text{Cov}(Y_i, Z_i) = \text{Cov}(\tau W_i + \varepsilon_i, Z_i) = \tau\,\text{Cov}(W_i, Z_i).$$

Hence

$$\tau = \frac{\text{Cov}(Y_i, Z_i)}{\text{Cov}(W_i, Z_i)}, \qquad \widehat{\tau} = \frac{\widehat{\text{Cov}}(Y_i, Z_i)}{\widehat{\text{Cov}}(W_i, Z_i)}.$$

My left-page checklist for an IV.

1. **Exogenous:** $\varepsilon_i \perp Z_i$.
2. **Relevant:** $\mathrm{Cov}(W_i, Z_i) \neq 0$.
3. **Exclusion restriction:** every effect of Z_i on Y_i must pass through W_i.



Optimal instruments

Scan reference: Lecture1-10.pdf, page 47, with extensions on page 46 and the estimating-equation reminder on page 48

Allow a high-dimensional instrument $Z_i \in \mathcal{Z}$:

$$Y_i = \tau W_i + \varepsilon_i, \qquad \varepsilon_i \perp Z_i, \qquad Y_i, W_i \in \mathbb{R}.$$

For a function $w : \mathcal{Z} \to \mathbb{R}$,

$$\tau = \frac{\mathrm{Cov}(Y_i, w(Z_i))}{\mathrm{Cov}(W_i, w(Z_i))}, \tag{9.8}$$

with nonzero denominator. Which transformation $w(\cdot)$ is optimal? The estimator is

$$\begin{aligned} \widehat{\tau}_w &= \frac{\widehat{\mathrm{Cov}}(Y_i, w(Z_i))}{\widehat{\mathrm{Cov}}(W_i, w(Z_i))} \\ &= \frac{n^{-1}\sum_i (Y_i - \overline{Y})(w(Z_i) - \overline{w(Z)})}{n^{-1}\sum_i (W_i - \overline{W})(w(Z_i) - \overline{w(Z)})}, \end{aligned} \tag{9.9}$$

where $\overline{Y} = n^{-1}\sum_i Y_i$, with analogous definitions for the other averages. It solves the estimating equation

$$\frac{1}{n}\sum_i (w(Z_i) - \overline{w(Z)})\{Y_i - \overline{Y} - \widehat{\tau}_w(W_i - \overline{W})\} = 0. \tag{9.10}$$

The source writes

$$n(\widehat{\tau}_w - \tau) \Rightarrow \mathcal{N}(0, V_w), \qquad V_w = \frac{\mathrm{Var}(\varepsilon_i)\,\mathrm{Var}(w(Z_i))}{\mathrm{Cov}(W_i, w(Z_i))^2}. \tag{9.11, source scaling}$$

Original-note correction (S1-036): The IV normal limit uses square-root-n scaling

Source: Lecture1-10.pdf, pp. 47, 48

The correct limit is $\sqrt{n}(\widehat{\tau}_w - \tau) \Rightarrow N(0, V_w)$. The first-order expansion is

$$\sqrt{n}(\widehat{\tau}_w - \tau) = \frac{n^{-1/2}\sum_i (w(Z_i) - E[w(Z)])\varepsilon_i}{\mathrm{Cov}(W, w(Z))} + o_p(1),$$

with centered errors, IID observations, finite required moments and a nonzero limiting covariance. This gives the displayed V_w under $\varepsilon \perp Z$. Compare official lecture (9.12); its equation numbering differs from the handwritten notes.

Because $Z_i \perp \varepsilon_i$ and $\mathbb{E}[\varepsilon_i] = 0$,

$$\mathrm{Var}\{\varepsilon_i(w(Z_i) - \mathbb{E}[w(Z_i)])\} = \mathrm{Var}(\varepsilon_i)\,\mathrm{Var}(w(Z_i)).$$

The optimization written in the notebook is

Use the reciprocal of the variance-dependent ratio (p. 47) $\longrightarrow$ The source optimization for $w^*(\cdot)$ below.

$$w^*(\cdot) \in \arg\min_{w'} \frac{\mathrm{Cov}(W_i, w'(Z_i))^2}{\mathrm{Var}(w'(Z_i))}.$$

Its stated solution is

$$w^*(z) \propto \mathbb{E}[W_i \mid Z_i = z].$$

$w^*(z) \propto E[W_i \mid Z_i = z]$ (p. 47) $\longrightarrow$ The best prediction of W_i from Z_i.

Original-note correction (S1-037): Maximize the covariance ratio

Source: Lecture1-10.pdf, pp. 47

Minimizing V_w requires

$$\arg\max_{w} \frac{\mathrm{Cov}(W, w(Z))^2}{\mathrm{Var}(w(Z))}.$$

Writing $m(Z) = E[W \mid Z]$, conditional expectation and Cauchy–Schwarz give $\mathrm{Cov}(W, w)^2 = \mathrm{Cov}(m, w)^2 \leq \mathrm{Var}(m)\,\mathrm{Var}(w)$. Equality holds for $w = a + bm$, $b \neq 0$, provided $\mathrm{Var}(m) > 0$. Thus the prediction intuition is correct; the optimization direction is not. The result uses the source's independence/homoskedastic-instrument setup; it is not the general heteroskedastic optimal-instrument formula. See official lecture (9.13)–(9.15).

Left-page extension: conditioning on covariates. The condition can be extended to $\varepsilon_i \perp Z_i \mid X_i$, giving

$$\tau(x) = \frac{\mathrm{Cov}(Y_i, w(Z_i) \mid X_i = x)}{\mathrm{Cov}(W_i, w(Z_i) \mid X_i = x)}.$$

The annotation mentions a doubly robust estimator of the average effect $\tau = \mathbb{E}[\tau(X)]$, and a random-forest estimator of $\tau(\cdot)$.

Qualification / unresolved point (S1-038): What the conditional IV ratio identifies

Source: Lecture1-10.pdf, pp. 46

The ratio equals $\tau(x)$ under a conditional structural model $Y = a(X) + W\tau(X) + \varepsilon$ with $E[\varepsilon \mid Z, X] = 0$ and nonzero conditional first-stage covariance. Under unrestricted individual treatment heterogeneity, conditional exogeneity of the instrument alone does not make this ratio the general CATE; with appropriate monotonicity it may instead identify a conditional local effect. A doubly robust estimator or forest must target the identified effect under its particular assumptions.

Another reminder on the same page: for independent X_1, X_2,

$$\begin{aligned}\mathrm{Var}(X_1X_2) = {} & \mathrm{Var}(X_1)\,\mathrm{Var}(X_2) \\ & + \mathbb{E}[X_2]^2\,\mathrm{Var}(X_1) + \mathbb{E}[X_1]^2\,\mathrm{Var}(X_2).\end{aligned}$$

Starred supplement: estimating equations and the delta method. If functions ψ_i satisfy $\mathbb{E}[\psi_i(\theta)] = 0$, and $\widehat{\theta}$ is estimated by

$$\frac{1}{n}\sum_{i=1}^{n}\psi_i(\widehat{\theta}) = 0,$$

then, in the regular setting described by the note's nonparametric delta-method reminder,

$$\sqrt{n}(\widehat{\theta} - \theta) \Rightarrow \mathcal{N}(0, V), \qquad V = \frac{\mathrm{Var}(\psi_i(\theta))}{\mathbb{E}[\psi_i'(\theta)]^2}.$$

Cross-fitting and feasible estimation

Scan reference: Lecture1-10.pdf, page 49

The optimal instrument is $w^*(z) = \mathbb{E}[W_i \mid Z_i = z]$. The general procedure is:

1. Fit a nonparametric first-stage regression to estimate $\mathbb{E}[W_i \mid Z_i = z]$, obtaining $\widehat{w}(\cdot)$.
2. Use $\widehat{w}(\cdot)$ as the instrument in (9.9), obtaining $\widehat{\tau}_w$.

However, with many instruments this can create overfitting bias. If $\widehat{w}(Z_i)$ is fit using the entire data set, it need not be independent of ε_i: the fitted instrument can inherit the dependence of W_i on ε_i.

My blue example in the margin. Suppose Z_i is noise. The note writes

$$W_i = \alpha + \beta Z_i + \eta_i, \qquad Y_i = \gamma + \delta\widehat{W}_i + \varepsilon_i = \gamma + \delta(\widehat{\alpha} + \widehat{\beta}Z_i) + \varepsilon_i,$$

marks the fitted term as biased, and draws an arrow comparing it to $Y_i = \gamma + \delta W_i + \varepsilon_i$.

Fitted-treatment equation in the blue example (p. 49) $\longrightarrow$ $Y_i = \gamma + \delta W_i + \varepsilon_i$; "equivalent" in the source.

The point of the annotation is that using an overfit first stage does not remove the problematic dependence.

Original-note correction (S1-039): Substituting a fitted treatment changes the structural residual

Source: Lecture1-10.pdf, pp. 49

From $Y = \gamma + \delta W + \varepsilon$ one obtains

$$Y = \gamma + \delta\widehat{W} + \{\varepsilon + \delta(W - \widehat{W})\}.$$

The same ε cannot generally be retained unless $\delta(W - \widehat{W}) = 0$. The author's warning about first-stage overfitting is sound, but the displayed equivalence needs this changed residual. If Z is pure noise, relevance fails; cross-fitting prevents reuse of an observation's noise but does not create a relevant instrument.

Use K-fold cross-fitting instead. With $k(i)$ the fold containing observation i,

$$\widehat{\tau} = \frac{\widehat{\mathrm{Cov}}(Y_i, \widehat{w}^{(-k(i))}(Z_i))}{\widehat{\mathrm{Cov}}(W_i, \widehat{w}^{(-k(i))}(Z_i))}. \tag{9.13}$$

The corresponding fitted instrument is independent of the observation's error under the stated sampling setup. In particular, if $\widehat{w}^{(-k(i))}(z)$ is mean-square consistent for $\mathbb{E}[W_i \mid Z_i = z]$, then (9.13) is asymptotically equivalent to the oracle version of (9.9). The red reminder writes this consistency as

$$\lim_{n\to\infty} \mathbb{E}[(\widehat{w}_n - \mathbb{E}[W \mid Z])^2] = 0,$$

where n is the sample size.

Qualification / unresolved point (S1-040): Cross-fitted IV still needs a nonvanishing first stage

Source: Lecture1-10.pdf, pp. 49

With independent observations, a fixed number of folds, suitable moments, and $\mathrm{Var}(E[W \mid Z]) > 0$ bounded away from zero, mean-square-consistent first-stage prediction can emulate the oracle at first order. Training outside the observation's fold removes own-observation dependence under $\varepsilon \perp Z$. It does not justify normal inference when instruments become weak, nor independence across all fitted observations (their training sets overlap).

Nonparametric instrumental variables regression

Scan reference: Lecture1-10.pdf, page 50

The previous model assumed $Y_i = W_i\tau + \varepsilon_i$. The handwritten parenthesis says "Z_i acts linearly on Y_i."

Original-note correction (S1-041): The linear structural coefficient multiplies treatment

Source: Lecture1-10.pdf, pp. 50

The constant structural effect is linear in W_i, not necessarily in Z_i. The instrument may influence treatment nonlinearly. The subsequent generalization replaces $W_i\tau$ by $g(W_i)$; that formula correctly describes the intended extension.

Generalize it to

$$Y_i = g(W_i) + \varepsilon_i, \qquad Z_i \perp \varepsilon_i, \qquad Y_i, W_i \in \mathbb{R}, \quad Z_i \in \mathcal{Z},$$

where g is a general smooth function.

Blue warning: we cannot simply run a nonparametric regression of Y_i on W_i, because

$$g(w) \neq \mathbb{E}[Y_i \mid W_i = w].$$

Since $Z_i \perp \varepsilon_i$ and $\mathbb{E}[\varepsilon_i] = 0$,

$$\begin{aligned}\mathbb{E}[Y_i \mid Z_i = z] &= \mathbb{E}[g(W_i) + \varepsilon_i \mid Z_i = z] \\ &= \mathbb{E}[g(W_i) \mid Z_i = z] = \int_{\mathbb{R}} g(w) f(w \mid z)\, dw. \end{aligned} \tag{9.15}$$

Here $f(w \mid z)$ is the conditional density of W_i given $Z_i = z$.

A learning procedure:

1. Estimate $\widehat{f}(w \mid z)$ using cross-fitting.
2. Choose a suitable function class $\mathcal{G}$ and minimize the loss to estimate g:

$$\widehat{g} \in \arg\min_{g \in \mathcal{G}} \frac{1}{n} \sum_{i=1}^{n} \left\{ Y_i - \int_{\mathbb{R}} g(w) \widehat{f}^{(-k(i))}(w \mid Z_i)\, dw \right\}^2. \tag{9.16}$$

To solve this in practice, approximate g with a basis expansion,

$$g_K(w) = \sum_{k=1}^{K} \beta_k \psi_k(w),$$

where the ψ_k are predetermined basis functions. As K grows, g_K better approximates g. Then

$$\widehat{\beta} \in \arg\min_{\beta} \frac{1}{n} \sum_{i=1}^{n} \{ Y_i - \widehat{m}^{(-k(i))}(Z_i)^{\mathsf{T}} \beta \}^2, \tag{9.17}$$

with components

$$\widehat{m}_k^{(-k(i))}(Z_i) = \int_{\mathbb{R}} \psi_k(w) \widehat{f}^{(-k(i))}(w \mid Z_i)\, dw.$$

Bottom-margin interpretation: consistency for $g(\cdot)$ needs conditions; (9.15) is complicated, while (9.17) turns the problem into regression by minimizing the loss for each basis coefficient. The blue margin adds that using (9.17) comes at the cost of regularization. The remaining bottom-edge wording is clipped and cannot be fully read. A nearby note says "analogous to 9.13."

Qualification / unresolved point (S1-042): NPIV identification and inverse-problem regularization

Source: Lecture1-10.pdf, pp. 50

Let $(Th)(z) = E[h(W) \mid Z = z]$. The moment equation identifies g only if T is injective on the chosen function class: $E[h(W) \mid Z] = 0$ implies $h(W) = 0$ almost surely (a completeness condition). Even then inversion may be ill posed, so accurate first-stage density estimates alone do not ensure accurate recovery of g. Sieve dimension or other regularization must control this instability and approximation error. Also, $E[Y \mid W = w] = g(w) + E[\varepsilon \mid W = w]$; inequality with $g(w)$ holds when that conditional error mean is nonzero. See Newey and Powell (2003), which characterizes identification through completeness and develops regularized series estimation.

Lecture 10: Local average treatment effects

Scan reference: Lecture1-10.pdf, page 52, then the compliance table and derivation on page 51

Recall the Lecture 9 IV model:

$$Y_i = \alpha + W_i \tau + \varepsilon_i, \quad \varepsilon_i \perp Z_i, \qquad W_i = Z_i \gamma + \eta_i. \tag{10.1}$$

When Z_i is binary,

$$\begin{aligned} \tau &= \frac{\mathrm{Cov}(Y_i, Z_i)}{\mathrm{Cov}(W_i, Z_i)} \\ &= \frac{\mathbb{E}[Y_i \mid Z_i = 1] - \mathbb{E}[Y_i \mid Z_i = 0]}{\mathbb{E}[W_i \mid Z_i = 1] - \mathbb{E}[W_i \mid Z_i = 0]}. \end{aligned} \tag{10.2}$$

Treatment effects under noncompliance

In an RCT with noncompliance, $Y_i \in \mathbb{R}$ is the ordinary outcome; $W_i \in \{0,1\}$ is the treatment actually received and is not randomized; $Z_i \in \{0,1\}$ is randomized treatment assignment.

The constant τ in (10.1) may not be appropriate, and treatment receipt may be related to the treatment effect. Define potential treatment receipt $W_i(0), W_i(1)$, with $W_i = W_i(Z_i)$ and $Y_i = Y_i(W_i(Z_i))$.

The IV validity assumptions are:

- **Exclusion restriction:** $Y_i(w,z) = Y_i(w)$. Assignment affects the outcome only through treatment receipt.
- **Exogeneity:** $\{Y_i(0), Y_i(1), W_i(0), W_i(1)\} \perp Z_i$. Assignment is randomized.
- **Relevance:** $\mathbb{E}[W_i(1) - W_i(0)] \neq 0$. Assignment affects treatment receipt.

There are four compliance types:

	$W_i(1) = 0$	$W_i(1) = 1$
$W_i(0) = 0$	Never-taker	Complier
$W_i(0) = 1$	Defier	Always-taker

Assume there are no defiers, that is, **monotonicity**:

$$W_i(1) \geq W_i(0) \quad \text{for every } i.$$

Then

$$\begin{aligned}
&\mathbb{E}[Y_i \mid Z_i = 1] - \mathbb{E}[Y_i \mid Z_i = 0] \\
&\quad = \mathbb{E}[Y_i(W_i(1)) \mid Z_i = 1] - \mathbb{E}[Y_i(W_i(0)) \mid Z_i = 0] \\
&\quad = \mathbb{E}[Y_i(W_i(1)) - Y_i(W_i(0))] \\
&\quad = \mathbb{E}[\mathbf{1}\{C_i = \text{complier}\}(Y_i(1) - Y_i(0))].
\end{aligned}$$

If compliers exist, applying Bayes' rule gives

$$\begin{aligned}
\tau_{\text{LATE}} &= \frac{\mathbb{E}[Y_i \mid Z_i = 1] - \mathbb{E}[Y_i \mid Z_i = 0]}{\mathbb{E}[W_i \mid Z_i = 1] - \mathbb{E}[W_i \mid Z_i = 0]} \\
&= \mathbb{E}[Y_i(1) - Y_i(0) \mid C_i = \text{complier}].
\end{aligned} \tag{10.3}$$

This is the complier average treatment effect.

Qualification / unresolved point (S1-043): LATE, ATE, and individual contrasts

Source: Lecture1-10.pdf, pp. 51, 52

The derivation also requires $0 < P(Z = 1) < 1$ and a positive complier share. Under the stated exclusion, exogeneity and monotonicity, the Wald ratio identifies the complier average effect, not the population ATE in general. The blue expression $E[Y(1) - Y(0)]$ is an ATE; an individual effect is $Y_i(1) - Y_i(0)$. The marginal label itself remains ambiguous. The author's point that never-takers reveal no treated outcome is correct.

My blue postscript. When the constant-effect model (10.1) fails and there are never-takers, always-takers, defiers, and so on, the individual contrast $Y_i(1) - Y_i(0)$ is not generally observable. In particular, a never-taker is never observed with $W_i = 1$. The marginal line writes an "IATE" expression as $\mathbb{E}[Y_i(1) - Y_i(0)]$ [the label and intended distinction between individual and average effects are not resolved in the scan].

Supply and demand

Scan reference: Lecture1-10.pdf, page 53

For each market $i = 1, \dots, n$, let $S_i(p, z)$ be its supply curve and $Q_i(p, z)$ its demand curve, where price $p \in \mathbb{R}$ and the instrument $z \in \{0, 1\}$. Assume supply is continuous and increasing in price, while demand is continuous and decreasing. The equilibrium price P_i is the unique solution to

$$S_i(P_i, Z_i) = Q_i(P_i, Z_i).$$

Thus Z_i is the IV, P_i is treatment, and $Q_i(P_i, Z_i)$ is the outcome Y_i.

$$Z_i \longrightarrow S_i \longrightarrow P_i \longrightarrow Q_i(P_i).$$

Conditions:

- Exclusion: $Q_i(p, z) = Q_i(p)$ for every p, z.
- Exogeneity: $\{Q_i(p), S_i(p, z)\} \perp Z_i$.
- Relevance: $\mathrm{Cov}(P_i, Z_i) \neq 0$.
- Monotonicity: $S_i(p, 1) \leq S_i(p, 0)$ for all relevant p [the source writes the equilibrium-price symbol in this condition].

If $Q_i(p)$ is differentiable, let $Q_i'(p)$ denote its derivative. The author's parenthetical qualification says that ordinary differentiability is not necessary: because $Q_i(\cdot)$ is monotone decreasing, a derivative in the distributional sense exists.

$$\begin{aligned}\tau_{\text{LATE}} &= \frac{\mathbb{E}[Q_i \mid Z_i = 1] - \mathbb{E}[Q_i \mid Z_i = 0]}{\mathbb{E}[P_i \mid Z_i = 1] - \mathbb{E}[P_i \mid Z_i = 0]} \\ &= \frac{\displaystyle\int \mathbb{E}[Q_i'(p) \mid P_i(0) \leq p \leq P_i(1)] P[P_i(0) \leq p \leq P_i(1)]\, dp}{\displaystyle\int P[P_i(0) \leq p \leq P_i(1)]\, dp}.\end{aligned}$$

The denominator is annotated "total integral."

Qualification / unresolved point (S1-044): Supply-demand existence and the derivative representation

Source: Lecture1-10.pdf, pp. 53

Continuity and opposing strict slopes give at most one intersection; existence also requires that the curves cross. The supply shift must hold at every relevant price to imply $P_i(1) \geq P_i(0)$. The displayed ordinary integral with $Q_i'(p)\, dp$ is valid when demand is absolutely continuous over the affected price intervals and integrable enough to interchange expectation and integration. A monotone continuous demand curve can have a singular continuous component, so a distributional derivative must instead be treated as a signed measure $dQ_i(p)$. Then use $Q_i(P_i(1)) - Q_i(P_i(0)) = \int_{(P_i(0), P_i(1)]} dQ_i$. The author's distributional-derivative idea is valid with this measure interpretation, not as an ordinary density in all cases.

Derivation in the notebook.

$$\begin{aligned}
&\mathbb{E}[Q_i \mid Z_i = 1] - \mathbb{E}[Q_i \mid Z_i = 0] \\
&\quad = \mathbb{E}[Q_i(P_i(1)) \mid Z_i = 1] - \mathbb{E}[Q_i(P_i(0)) \mid Z_i = 0] \\
&\quad = \mathbb{E}[Q_i(P_i(1)) - Q_i(P_i(0))] \\
&\quad = \mathbb{E}\left[\int_{P_i(0)}^{P_i(1)} Q_i'(p)\,dp\right] \\
&\quad = \int \mathbb{E}[Q_i'(p) \mid P_i(0) \le p \le P_i(1)] P[P_i(0) \le p \le P_i(1)]\,dp.
\end{aligned}$$

Fubini: interchange integration order (p. 53) $\longrightarrow$ $E[\int Q_i'(p)\,dp]$ becomes an integral of the expectation.

Threshold crossing and willingness to pay

Scan reference: Lecture1-10.pdf, page 55, with the diagram and decomposition on page 54

Suppose each subject has an unobserved endogenous variable U_i such that

$$W_i = \mathbf{1}\{U_i \ge c(Z_i)\},$$

where $c(z)$ is a cutoff function depending on the instrument. Without loss of generality, take $U_i \sim$ Unif$[0,1]$, giving

$$c(z) = 1 - P(W_i = 1 \mid Z_i = z).$$

Qualification / unresolved point (S1-046): Uniform normalization and cutoff support

Source: Lecture1-10.pdf, pp. 55

The identity $c(z) = 1 - P(W = 1 \mid Z = z)$ uses $U \perp Z$, $U \sim$ Unif$[0,1]$ and an effective cutoff in $[0,1]$. Uniform normalization follows directly from a continuous latent CDF (with a suitable monotone reparameterization); atoms need additional care, such as a randomized latent rank. These are assumptions/normalizations of the threshold model, not a consequence of the indicator rule alone.

The IV setup imposes:

1. Potential outcomes $Y_i(0), Y_i(1)$ with $Y_i = Y_i(W_i)$: the instrument has no direct route to the outcome.
2. $\{Y_i(0), Y_i(1), U_i\} \perp Z_i$: exogeneity.
3. $c(Z_i)$ has a nontrivial distribution: relevance.
4. $c(z)$ is càdlàg (right-continuous with left limits) and nondecreasing. The margin says that W_i is nondecreasing in Z_i.

Original-note correction (S1-045): Increasing cutoffs decrease treatment receipt

Source: Lecture1-10.pdf, pp. 55, 57

For fixed U_i, $W_i(z) = \mathbf{1}\{U_i \ge c(z)\}$ is nonincreasing when c is nondecreasing. The single-jump middle group therefore switches from treated to untreated as z increases. Relative to the earlier increasing-treatment instrument convention it is the opposite switching direction; it can be called a complier group after reversing the instrument orientation. Both IV covariances

change sign together, so the positive-weight treatment-effect interpretation still holds. This convention issue does not invalidate the IV ratio.

Define the marginal treatment effect (MTE)

$$\tau(u) = \mathbb{E}[Y_i(1) - Y_i(0) \mid U_i = u]. \tag{10.6}$$

The IV estimand is a weighted average of $\tau(u)$. In the notes, $Z_i \sim \mathcal{N}(0,1)$, with density φ. Let $S \subset \mathbb{R}$ be the discontinuity set of c, and

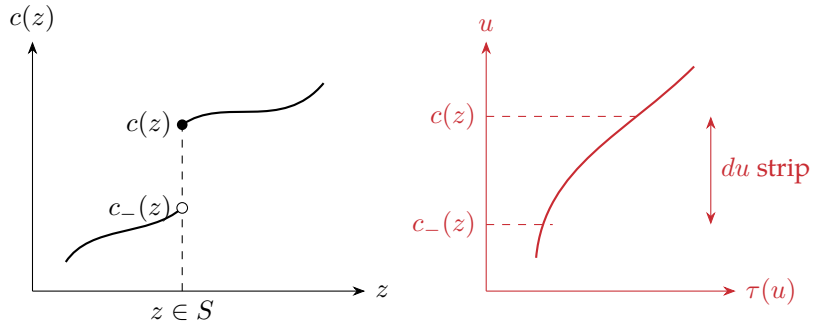
$$c_-(z) = \lim_{a \uparrow z} c(a).$$

The displayed representation is

$$\tau_{\text{LATE}} = \frac{\displaystyle\sum_{z \in S} \left\{ \int_{c_-(z)}^{c(z)} \tau(u)\, du \right\} \varphi(z) + \int_{\mathbb{R}\setminus S} \tau(c(z)) c'(z) \varphi(z)\, dz}{\displaystyle\sum_{z \in S} (c(z) - c_-(z)) \varphi(z) + \int_{\mathbb{R}\setminus S} c'(z) \varphi(z)\, dz}.$$

This displays the IV estimand as a convex average of marginal treatment effects.

My left-page explanation of the weights. The drawing plots $c(z)$ against z with an open dot at the left limit and a filled dot at the value after a jump. The jump location belongs to S.



At a jump, integrate $\tau(u)$ over the strip from $c_-(z)$ to $c(z)$ and multiply by $\varphi(z)$. On the continuous part, evaluate the treatment effect at $u = c(z)$ and weight it by $c'(z)\varphi(z)$.

Probability weight (source wording, p. 54) $\longrightarrow$ $\varphi(z)$.

Treatment effect at $u = c(z)$ (p. 54) $\longrightarrow$ $\tau(c(z))$.

Weight (p. 54) $\longrightarrow$ $c'(z)\varphi(z)$.

The denominator adds the total weight.

Original-note correction (S1-047): The Gaussian factor is a density, not point probability

Source: Lecture1-10.pdf, pp. 54

For continuous Gaussian Z, $P(Z = z) = 0$ at every individual jump location. The factor $\varphi(z)$ is the Gaussian density evaluated there, arising from integration by parts; it is not a point mass or $P(Z \in S)$. The jump's weight is $(c(z) - c_-(z))\varphi(z)$, normalized by the total denominator.

A second sketch shades the interval in u corresponding to the jump, making the du contribution explicit.

Bottom qualification in the source. The note says that a monotone $c(z)$ must have bounded variation, then writes

$$c(z) = c_0 + \int_{-\infty}^{z} c'(a)\, da,$$

where c' is a nonnegative Lebesgue-integrable function.

Original-note correction (S1-048): Monotone cutoffs may have singular continuous variation

Source: Lecture1-10.pdf, pp. 54, 55, 56, 57

A bounded monotone cutoff has a Stieltjes measure dc, not necessarily a density $c'(z)\, dz$. It may have an absolutely continuous part, atoms at jumps, and a singular continuous part (the Cantor function supplies a counterexample). Thus the jump-plus-ordinary-integral formula also needs absence of a singular continuous part; "no jumps" alone is insufficient for the later continuous formula.

Take c to be right-continuous, nondecreasing, $[0, 1]$-valued and nonconstant, and assume the outcome/covariance moments exist. Then $\int \varphi\, dc > 0$. A formulation that retains all parts is obtained by defining $F(u) = \int_0^u \tau(v)\, dv$ (for bounded or suitably integrable τ):

$$\tau_{\text{IV}} = \frac{\int_{\mathbb{R}} \varphi(z)\, d[F(c(z))]}{\int_{\mathbb{R}} \varphi(z)\, dc(z)}.$$

Indeed $E[W \mid Z = z] = 1 - c(z)$ and $E[Y \mid Z = z] = E[Y(0)] + F(1) - F(c(z))$; Gaussian integration by parts gives a minus sign in both covariances, which cancels. At a jump, $d[F(c)]$ has mass $\int_{c_-}^{c} \tau(u)\, du$, explaining the author's strip intuition. On an absolutely continuous part it has density $\tau(c(z))c'(z)$. The source's p. 56 distributional derivative should be read as this measure, rather than as a piecewise ordinary function. A Dirac delta is a measure/distribution, not an ordinary function with the displayed pointwise properties. This corrects an error also present in lecture footnote 4 and qualifies (10.7)–(10.11).

Single jumps, multiple jumps, and continuous cutoffs

Scan reference: Lecture1-10.pdf, page 57, with the derivative supplement on page 56

Example 1: one jump. For a cutoff otherwise constant with a single jump,

$$c(z) = c_0 + \delta_1 \mathbf{1}\{z \geq z_1\},$$

the source lists

$$\begin{array}{ll} \text{Never-taker:} & U_i < c_0, \\ \text{Complier:} & c_0 \leq U_i < c_0 + \delta_1, \\ \text{Always-taker:} & U_i \geq c_0 + \delta_1. \end{array}$$

For K jumps,

$$c(z) = c_0 + \sum_{k=1}^{K} \delta_k \mathbf{1}\{z \geq z_k\},$$

and the note gives

$$\tau_{\text{LATE}} = \frac{\displaystyle\sum_{k=1}^{K} \mathbb{E}[\tau(U_i) \mid c_-(z_k) \leq U_i < c(z_k)](z_k - c_-(z_k))\varphi(z_k)}{\displaystyle\sum_{k=1}^{K} (c(z_k) - c_-(z_k))\varphi(z_k)}.$$

It is a convex combination of the average effects in the switching strata. The weights depend on the size of each stratum in U space and on $\varphi(z_k)$.

Original-note correction (S1-049): The jump width is a cutoff difference

Source: Lecture1-10.pdf, pp. 57

Replace the numerator's factor $z_k - c_-(z_k)$ by $c(z_k) - c_-(z_k)$. The stratum is $[c_-(z_k), c(z_k))$ in U-space; its probability under uniform U is the difference of these cutoff values, not a difference between instrument location and cutoff. With that correction the formula is the stated convex combination.

Example 2: a continuous cutoff. If $c(z)$ has no jumps, the displayed formula becomes

$$\tau_{\text{LATE}} = \frac{\displaystyle\int_{\mathbb{R}} \tau(c(z))c'(z)\varphi(z)\,dz}{\displaystyle\int_{\mathbb{R}} c'(z)\varphi(z)\,dz}.$$

The covariance derivation is

$$\begin{aligned} \operatorname{Cov}(Y_i, Z_i) &= \operatorname{Cov}(Y_i(0) + (Y_i(1) - Y_i(0))W_i, Z_i) \\ &= \operatorname{Cov}((Y_i(1) - Y_i(0))W_i, Z_i) \\ &= \operatorname{Cov}((Y_i(1) - Y_i(0))\mathbf{1}\{U_i \geq c(Z_i)\}, Z_i) \\ &= \operatorname{Cov}(\tau(U_i)\mathbf{1}\{U_i \geq c(Z_i)\}, Z_i). \end{aligned}$$

Define

$$H(z) = \mathbb{E}[\tau(U_i)\mathbf{1}\{U_i \geq c(z)\}].$$

For standard Gaussian Z_i, the note uses

$$\mathrm{Cov}(H(Z_i), Z_i) = \mathbb{E}[H'(Z_i)].$$

Left-page question: distributional derivatives. The starred supplement asks: "Does $H'(z)$ denote the distributional derivative of $H(\cdot)$?" It writes, in a mixed ordinary/distributional notation,

$$-H'(z) = \begin{cases} \left\{ \int_{c_-(z)}^{c(z)} \tau(u)\, du \right\} \delta_z, & z \in S, \\ \tau(c(z))c'(z), & \text{otherwise.} \end{cases}$$

Here δ_z is a Dirac delta at z. The accompanying reminder gives the informal defining properties of a delta centered at zero: $\delta(x) = 0$ for $x \neq 0$ and $\int_{-\infty}^{\infty} \delta(x)\, dx = 1$. The question mark and the distributional nature of this supplement are retained.

Estimating the marginal treatment effect

Scan reference: Lecture1-10.pdf, page 58

At a smooth point of $c(z)$,

$$\tau(c(z)) = \frac{\frac{d}{dz}\mathbb{E}[Y_i \mid Z_i = z]}{\frac{d}{dz}P(W_i = 1 \mid Z_i = z)}. \tag{10.12}$$

Assume the derivatives are well behaved, the numerator and denominator can be estimated, and the denominator is nonzero.

Qualification / unresolved point (S1-050): Where the local-IV derivative identifies an MTE

Source: Lecture1-10.pdf, pp. 58

The chain-rule proof is correct at points where c and the conditional mean are differentiable, $c'(z) \neq 0$, and $F'(c(z)) = \tau(c(z))$. For merely integrable τ, the last equality is an almost-everywhere statement; continuity of τ at the target is a convenient sufficient condition. Identification is restricted to latent ranks reached by the instrument propensity support. Smoothness of c alone does not identify MTEs outside that range.

Proof in the notebook.

$$\begin{aligned} \mathbb{E}[Y_i \mid c(Z_i) = c] &= \mathbb{E}[Y_i(0) + \mathbf{1}\{U_i \geq c\}(Y_i(1) - Y_i(0)) \mid c(Z_i) = c] \\ &= \mathbb{E}[Y_i(0) + \mathbf{1}\{U_i \geq c\}(Y_i(1) - Y_i(0))] \\ &= \mathbb{E}[Y_i(0)] + \int_c^1 \tau(u)\, du. \end{aligned}$$

The last step is marked "Fubini." Therefore

$$\tau(c) = -\frac{d}{dc}\mathbb{E}[Y_i \mid c(Z_i) = c].$$

By the chain rule,

$$\frac{d}{dz}\mathbb{E}[Y_i \mid Z_i = z] = \frac{d}{dc}\mathbb{E}[Y_i \mid c(Z_i) = c]\Big|_{c=c(z)} c'(z).$$

Since $U_i \sim \text{Unif}(0,1)$ independently of Z_i,

$$c'(z) = -\frac{d}{dz}P(W_i = 1 \mid Z_i = z).$$

Combining these identities gives (10.12), as checked in the original.

Lecture 11: Policy learning

Scan reference: Lecture11-14.pdf, page 1

We study how to allocate treatment so as to maximize benefit. The decision is simply which group a person should enter: treatment or control.

A treatment policy is a mapping

$$\pi : \mathcal{X} \longrightarrow \{0, 1\}.$$

An individual with characteristics $X_i = x$ is treated if and only if $\pi(x) = 1$. The goal is to find a policy that maximizes expected utility:

$$V(\pi) = \mathbb{E}[Y_i(\pi(X_i))].$$

(The note takes $Y_i = 1$ to be a good outcome.) For a policy class Π, define

$$\pi^* = \arg\max_{\pi' \in \Pi} V(\pi').$$

The regret of another policy is

$$R(\pi) = \sup_{\pi' \in \Pi} V(\pi') - V(\pi).$$

Emphasis in the notebook: the loss is regret, rather than squared-error loss. The objective is to learn a policy with a guaranteed upper bound on worst-case regret: a minimax regret criterion.

Assume IID observations $(X_i, Y_i, W_i) \in \mathcal{X} \times \mathbb{R} \times \{0, 1\}$, $i = 1, \ldots, n$, with

$$\{Y_i(0), Y_i(1)\} \perp W_i \mid X_i, \qquad Y_i = Y_i(W_i),$$

$$0 < \eta \leq e(X_i) \leq 1 - \eta < 1, \qquad e(x) = \mathbb{P}(W_i = 1 \mid X_i = x).$$

After learning $\widehat{\pi}$, set $W_i = \widehat{\pi}(X_i)$ and expect $Y_i = Y_i(\widehat{\pi}(X_i))$ to be large. **My observation:** once treatment is assigned this way, assignment has no randomness; we can no longer learn the causal effect from it in the same way.

Qualification / unresolved point (S2-001): Deterministic deployment removes overlap

Source: Lecture11-14.pdf, page 1

A deterministic policy gives $e(x) \in \{0, 1\}$, so both potential-outcome means cannot generally be recovered nonparametrically at the same x from deployment data alone. Earlier randomized data, renewed exploration, or additional structural assumptions can still support causal learning.

Policy learning by empirical maximization

Scan reference: Lecture11-14.pdf, page 3, with the analogy on page 2

In general,

$$\widehat{\pi} = \arg\max_{\pi \in \Pi} \widehat{V}(\pi).$$

If $e(x)$ is known, an unbiased IPW choice is

$$\widehat{V}_{\mathrm{IPW}}(\pi) = \frac{1}{n} \sum_{i=1}^{n} \frac{\mathbf{1}\{W_i = \pi(X_i)\} Y_i}{\mathbb{P}[W_i = \pi(X_i) \mid X_i]}, \qquad \widehat{\pi}_{\mathrm{IPW}} = \arg\max_{\pi \in \Pi} \widehat{V}_{\mathrm{IPW}}(\pi).$$

In words, retain observations whose treatment W_i agrees with the policy $\pi(X_i)$, and weight their observed outcomes using the inverse assignment probability. Unconfoundedness gives

$$\begin{aligned}\mathbb{E}[\widehat{V}_{\text{IPW}}(\pi)] &= \mathbb{E}\left[\frac{\mathbf{1}\{W_i = \pi(X_i)\}Y_i(\pi(X_i))}{\mathbb{P}[W_i = \pi(X_i) \mid X_i]}\right] \\ &= \mathbb{E}\left[\mathbb{E}\left[\frac{\mathbf{1}\{W_i = \pi(X_i)\}}{\mathbb{P}[W_i = \pi(X_i) \mid X_i]}\middle| X_i\right]\mathbb{E}[Y_i(\pi(X_i)) \mid X_i]\right] \\ &= \mathbb{E}[Y_i(\pi(X_i))] = V(\pi).\end{aligned}$$

The source begins this calculation with

$$E[\widehat{V}_{\text{IPW}}(\pi)] = E\left[\frac{\mathbf{1}\{W_i = \pi(X_i)\}}{P[W_i = \pi(X_i) \mid X_i]}\right].$$

The conditional expectation of the indicator divided by its probability is 1.

$$\boxed{E\left[\frac{\mathbf{1}\{W_i = \pi(X_i)\}}{P[W_i = \pi(X_i) \mid X_i]}\middle| X_i\right]} = 1.$$

Original-note correction (S2-003): The first IPW expectation drops the outcome

Source: Lecture11-14.pdf, page 3

The handwritten right side equals 1, not $V(\pi)$ in general. Insert Y_i in the numerator and use consistency to replace the selected observed outcome by $Y_i(\pi(X_i))$. The preceding full derivation gives the valid steps.

Left-page analogy: unbiasedness of the ATE IPW estimator.

$$\begin{aligned}\mathbb{E}[\widehat{\tau}^*_{\text{IPW}}] &= \mathbb{E}\left[\frac{W_iY_i}{e(X_i)} - \frac{(1-W_i)Y_i}{1-e(X_i)}\right] \\ &= \mathbb{E}\left[\frac{W_iY_i(1)}{e(X_i)} - \frac{(1-W_i)Y_i(0)}{1-e(X_i)}\right] \\ &= \mathbb{E}\left[\mathbb{E}\left[\frac{W_iY_i(1)}{e(X_i)}\middle| e(X_i)\right] - \mathbb{E}\left[\frac{(1-W_i)Y_i(0)}{1-e(X_i)}\middle| e(X_i)\right]\right] \\ &= \mathbb{E}[Y_i(1) - Y_i(0)].\end{aligned}$$

Thus it is unbiased for τ.

Policy learning as weighted classification

Scan reference: Lecture11-14.pdf, page 5, with the regret bound on page 4

Decompose the value function:

$$V(\pi) = \underbrace{\mathbb{E}[Y_i(0)]}_{E_0} + \mathbb{E}[(Y_i(1) - Y_i(0))\pi(X_i)].$$

Interpretation in the margin: this displays the baseline outcome E_0 and the contribution from the treatment effects of people treated by $\pi(\cdot)$.

With $\tau(x)$ denoting the CATE, the policy advantage is

$$\begin{aligned} A(\pi) &= 2\mathbb{E}[Y_i(\pi(X_i))] - \mathbb{E}[Y_i(0) + Y_i(1)] \\ &= \mathbb{E}[(2\pi(X_i) - 1)\tau(X_i)]. \end{aligned}$$

$A(\pi)$ is the advantage of $\pi(\cdot)$. The same π^* maximizes $A(\pi)$ over Π.

Rewrite IPW: $\widehat{\pi}_{\text{IPW}}$ maximizes

$$\begin{aligned} \widehat{A}_{\text{IPW}}(\pi) &= 2\widehat{V}_{\text{IPW}}(\pi) - \frac{1}{n}\sum_{i=1}^{n}\left\{\frac{W_i Y_i}{e(X_i)} + \frac{(1-W_i)Y_i}{1-e(X_i)}\right\} \\ &= \frac{1}{n}\sum_{i=1}^{n}(2\pi(X_i) - 1)\left\{\frac{W_i Y_i}{e(X_i)} - \frac{(1-W_i)Y_i}{1-e(X_i)}\right\}. \end{aligned} \tag{11.10}$$

This estimates the advantage without bias.

Qualification / unresolved point (S2-002): Fixed versus learned policy

Source: Lecture11-14.pdf, pages 3, 5

The IPW value and advantage are unbiased for a fixed π, or conditional on a policy learned from independent data. In general $\widehat{V}(\widehat{\pi})$ evaluated on the training sample is optimistic: maximizing noisy unbiased values does not preserve unbiasedness.

Write

$$\widehat{A}_{\text{IPW}}(\pi) = \frac{1}{n}\sum_{i=1}^{n}(2\pi(X_i) - 1)\Gamma_i^{\text{IPW}}, \qquad \Gamma_i^{\text{IPW}} = \frac{W_i Y_i}{e(X_i)} - \frac{(1-W_i)Y_i}{1-e(X_i)}. \tag{11.11}$$

Here Γ_i^{IPW} is the IPW score used in ATE estimation: $\widehat{\tau}_{\text{IPW}} = n^{-1}\sum_i \Gamma_i^{\text{IPW}}$. **My interpretation:** $\widehat{A}_{\text{IPW}}(\pi)$ resembles the ATE IPW estimator. When $\pi(X_i) = 1$, we "earn" that individual's treatment effect; when $\pi(X_i) = 0$, we "pay" that individual's treatment effect.

For optimization, write

$$\widehat{A}_{\text{IPW}}(\pi) = \frac{1}{n}\sum_{i=1}^{n}\underbrace{(2\pi(X_i) - 1)\,\text{sign}(\Gamma_i)}_{\text{classification target}}\underbrace{|\Gamma_i|}_{\text{weight }\star}. \tag{11.12}$$

The sign defines a classification target and $|\Gamma_i|$ is the weight (starred in the scan). Maximizing the advantage is equivalent to optimizing a weighted classification objective. In practice one often minimizes a weighted surrogate loss instead. A marginal reference reads "Wager [2020]" [year somewhat unclear]; the author's added warning is: **"But the estimator may be wrong."**

Qualification / unresolved point (S2-038): Why a surrogate can change the policy target

Source: Lecture11-14.pdf, page 5

The author's blue warning is substantive. The exact objective satisfies

$$(2\pi(X_i) - 1)\Gamma_i = |\Gamma_i|\big[1 - 2\mathbf{1}\{\pi(X_i) \neq \mathbf{1}\{\Gamma_i > 0\}\}\big].$$

Thus maximizing the displayed score exactly minimizes weighted classification error. Replacing the indicator loss by an arbitrary smooth surrogate, especially within a restricted policy class,

need not preserve its optimizer. An approximation/optimization guarantee or a suitable surrogate-consistency argument is needed; the warning does not invalidate the exact weighted-classification identity.

Supplement: complexity and a regret rate. The VC dimension measures the complexity of Π; the note says to look up its precise definition. If $|Y_i| \le M$, $\eta \le e(X_i) \le 1-\eta$ (so the scores are bounded), and Π has finite Vapnik–Chervonenkis dimension, then

$$R(\widehat{\pi}_{\text{IPW}}) = O_P\left(\frac{M}{\eta}\sqrt{\frac{\text{VC}(\Pi)}{n}}\right), \quad \widehat{\pi}_{\text{IPW}} = \arg\max_{\pi\in\Pi} \widehat{A}_{\text{IPW}}(\pi).$$

The author underlines this as an optimal convergence rate, then adds a superefficiency caveat: for a fixed policy $\pi(x)$, the convergence rate with n observations may be faster than $\sqrt{\text{VC}(\Pi)/n}$.

Qualification / unresolved point (S2-004): Minimax regret and pointwise rates

Source: Lecture11-14.pdf, page 4

The rate is a worst-case policy-learning claim over a model and policy class. Faster regret at particular distributions does not contradict it. A fixed policy has no learning regret rate by itself; estimating its value is a different problem. Thus the red remark is a useful warning about pointwise versus uniform statements, but not a fixed-policy superefficiency theorem. See Athey and Wager.

Efficient scoring rules for policy learning

Scan reference: Lecture11-14.pdf, page 7, then page 6

Is there a better learning method? Observe that

$$\begin{aligned} A(\pi) &= 2\mathbb{E}[Y_i(\pi(X_i))] - \mathbb{E}[Y_i(0) + Y_i(1)] \\ &= \mathbb{E}[Y_i(\pi(X_i))] - \mathbb{E}[Y_i(1-\pi(X_i))]. \end{aligned}$$

Thus $A(\pi)$ is the ATE in an experiment comparing deployment of $\pi(\cdot)$ with deployment of its opposite policy.

Oracle IPW is feasible but inefficient. The oracle AIPW estimator uses an efficient score and reaches the semiparametric efficiency bound:

$$\widehat{A}^*_{\text{AIPW}}(\pi) = \frac{1}{n}\sum_{i=1}^{n}(2\pi(X_i)-1)\Gamma^*_i,$$

$$\Gamma^*_i = \mu_1(X_i) - \mu_0(X_i) + W_i\frac{Y_i - \mu_1(X_i)}{e(X_i)} - (1-W_i)\frac{Y_i - \mu_0(X_i)}{1-e(X_i)}.$$

Assume further that there are $o_P(n^{-1/4})$-consistent regression adjustments $\widehat{\mu}_w(x)$ and $\widehat{e}(x)$. The red note emphasizes that, as $n \to \infty$, both nuisance-estimation errors are $o_P(n^{-1/4})$. Construct a doubly robust estimator close to the efficient oracle:

$$\widehat{A}_{\text{AIPW}}(\pi) = \frac{1}{n}\sum_{i=1}^{n}(2\pi(X_i)-1)\widehat{\Gamma}_i,$$

$$\widehat{\Gamma}_i = \widehat{\mu}_1^{(-k(i))}(X_i) - \widehat{\mu}_0^{(-k(i))}(X_i) + W_i \frac{Y_i - \widehat{\mu}_1^{(-k(i))}(X_i)}{\widehat{e}^{(-k(i))}(X_i)} - (1 - W_i) \frac{Y_i - \widehat{\mu}_0^{(-k(i))}(X_i)}{1 - \widehat{e}^{(-k(i))}(X_i)}.$$

This too is a weighted classification objective. If Π is a VC class, the feasible and oracle objectives are uniformly asymptotically equivalent faster than $1/\sqrt{n}$:

$$\sqrt{n} \sup_{\pi \in \Pi} \left| \widehat{A}_{\text{AIPW}}(\pi) - \widehat{A}^*_{\text{AIPW}}(\pi) \right| \xrightarrow{P} 0. \tag{11.17}$$

Qualification / unresolved point (S2-005): Conditions for uniform oracle equivalence

Source: Lecture11-14.pdf, page 7

Interpret nuisance rates in $L_2(P_X)$ and use the displayed cross-fitting. A standard sufficient regime has fixed finite VC dimension, suitable boundedness or moment envelopes, $\widehat{e} \in [\eta', 1-\eta']$, nuisance consistency and

$$\|\widehat{e} - e\|_2 \max_w \|\widehat{\mu}_w - \mu_w\|_2 = o_P(n^{-1/2}).$$

The two $o_P(n^{-1/4})$ rates suffice for this product. Arbitrarily growing VC classes require additional restrictions. Athey and Wager supply a regularity framework.

Left-page continuation: regret and efficiency. Combining the preceding result with empirical-process concentration gives

$$R(\widehat{\pi}_{\text{AIPW}}) = O_P\left(\sqrt{\frac{V^* \, \text{VC}(\Pi)}{n}}\right), \quad \widehat{\pi}_{\text{AIPW}} = \arg\max_{\pi \in \Pi} \widehat{A}_{\text{AIPW}}(\pi),$$

$$V^* = \mathbb{E}[\tau^2(X_i)] + \mathbb{E}\left[\frac{\text{Var}(Y_i(0) \mid X_i)}{1 - e(X_i)}\right] + \mathbb{E}\left[\frac{\text{Var}(Y_i(1) \mid X_i)}{e(X_i)}\right].$$

The reference written in the notebook is "Wager 2017."

The displayed V^* decomposition. $\longrightarrow$ This bound is optimal when the treatment effect just emerges from the noise.

Author's red annotation: "Not quite right..."

Qualification / unresolved point (S2-006): Interpret the efficient-score second moment

Source: Lecture11-14.pdf, page 6

The display satisfies $V^* = E[(\Gamma_i^*)^2]$. Since $(2\pi(X) - 1)^2 = 1$,

$$\text{Var}((2\pi(X) - 1)\Gamma_i^*) = V^* - A(\pi)^2.$$

A noise-dominated local model helps explain a minimax lower bound; it does not establish pointwise optimality at every distribution. The author's reservation asks for the precise lower-bound quantifiers. See Athey and Wager.

The role of the policy class

Scan reference: Lecture11-14.pdf, page 8

Start from a nonparametric model: $\mu_w(x)$ and $e(x)$ may be arbitrary. Then the Bayes-optimal treatment assignment rule is

$$\pi_{\text{Bayes}}(x) = \mathbf{1}\{\tau(x) > 0\}.$$

Our goal, however, is to work with a specified policy class Π, not necessarily to approximate this Bayes rule. Examples include linear decision rules, k-sparse decision rules, and decision trees of depth ℓ. We do not require $\pi_{\text{Bayes}} \in \Pi$.

Reasons:

1. We use X_i to make unconfoundedness plausible: $\{Y_i(0), Y_i(1)\} \perp W_i \mid X_i$. Generally, access to more pretreatment variables makes this assumption more plausible. For reliable modeling, it is preferable to use richer features and flexible nonparametric models for $e(x)$ and $\mu_w(x)$.
2. A policy has a concrete form and practical restrictions: it may not use certain features; resources or treatment budgets may be limited; and the functional form of $\pi(\cdot)$ may be constrained.

Qualification / unresolved point (S2-007): More covariates do not automatically remove confounding

Source: Lecture11-14.pdf, page 8

Unconfoundedness is not preserved under arbitrary additions to a conditioning set: even a pretreatment collider can open a noncausal path. Flexible nuisance models help after an appropriate adjustment set is chosen; they do not establish identification.

Lecture 12: Evaluating dynamic policies

Scan reference: Lecture11-14.pdf, page 10, with the setup and second assumption on page 9

The heading connects this topic with reinforcement learning. We evaluate a dynamic policy: a policy that changes over time.

There are n IID individuals $i = 1, \ldots, n$ and times $t = 1, \ldots, T$. At each time we observe time-varying covariates X_{it} and a treatment $W_{it} \in \{0, 1\}$. At the final time T, we additionally observe an outcome $Y_i \in \mathbb{R}$.

To reflect the dynamic structure, every time-varying observation may depend on all earlier treatments. For each $X_{it} \in \mathcal{X}_t$, define 2^{t-1} potential values $X_{it}(w_{1:t-1})$, with

$$X_{it} = X_{it}(W_{i,1:t-1}).$$

The final outcome has 2^T potential outcomes $Y_i(w_{1:T})$, and

$$Y_i = Y_i(W_{i,1:T}).$$

Similarly, the note writes $W_{it} = W_{it}(W_{i,1:t-1})$: treatment at time t depends on observed covariate history $X_{i,1:t}$ and past treatment values.

Left-page intuition. Picture one individual observed in successive columns $1, \ldots, T$, with covariates $X_{i1}, X_{i2}, \ldots$ and decisions $W_{i1}, W_{i2}, \ldots$. Make one decision at each time. The decision at time t depends on what happened through $t-1$, and there are 2^T possible final outcomes.

$X_{i1}, W_{i1};\ X_{i2}, W_{i2};\ \ldots;\ X_{iT}, W_{iT}$ $\longrightarrow$ 2^T potential final outcomes $Y_i(w_{1:T})$

There are two types of estimand:

1. Evaluate a fixed treatment sequence $w \in \{0,1\}^T$:
$$V(w) = \mathbb{E}[Y_i(w)].$$
2. Evaluate a policy with decision rules π_t mapping the available history to $\{0,1\}$:
$$V(\pi) = \mathbb{E}\big[Y_i\big(\pi_1(X_{i1}), \pi_2(X_{i1}, \pi_1(X_{i1}), X_{i2}(\pi_1(X_{i1}))), \ldots\big)\big]. \tag{12.2}$$

The second decision depends on covariate history and previous treatment values. $\longrightarrow$ The arguments of π_2 in (12.2).

Both problems use potential treatment sequences to define the estimand, including interventions on the assignment distribution that generated the data.

Two natural unconfoundedness-type assumptions are considered:

1. **Sequential unconfoundedness / sequential ignorability:**
$$\{\text{potential outcomes after time } t\} \perp W_{it} \mid \{\text{history at time } t\}.$$
At each time, treatment is unconfounded given the data collected up to that time.
2. **Complete randomization:**
$$\{\text{all potential outcomes}\} \perp W_{i,1:T}.$$
Author's judgment in the left-page note: this is unreasonable here.

Qualification / unresolved point (S2-008): Sequential identification needs support

Source: Lecture11-14.pdf, pages 9, 10, 12, 13

Besides sequential ignorability and consistency, require
$$P(W_t = \pi_t(\mathcal{F}_t) \mid \mathcal{F}_t) > 0$$
at histories reachable under π. Stable estimation often needs uniform bounds and moments. Complete randomization of a whole sequence is possible as another design, but generally inappropriate for the adaptive design here. The notation $W_{it}(W_{i,1:t-1})$ is shorthand: decisions also use covariates observed by time t.

Treatment–confounder feedback

Scan reference: Lecture11-14.pdf, page 11

An example of what can go wrong. All individuals have $X_{i1} = 0$, and
$$\mathbb{P}(W_{i1} = 1) = 0.5, \qquad \mathbb{P}(W_{i2} = 1 \mid X_{i2} = 1) = 0.8, \qquad \mathbb{P}(W_{i2} = 1 \mid X_{i2} = 0) = 0.4.$$
The target is written $\tau = \mathbb{E}[Y(1) - Y(0)]$. Analysis shows
$$\mathbb{E}[Y_i \mid W_{i1} = 0] = \mathbb{E}[Y_i \mid W_{i1} = 1] = 0,$$

$$\mathbb{E}[Y_i \mid W_{i2} = 0, W_{i1} = w_1, X_{i2} = x] = \mathbb{E}[Y_i \mid W_{i2} = 1, W_{i1} = w_1, X_{i2} = x]$$

for every w_1, x.

The first mistake is to ignore adaptive sampling and use the ordinary difference in means,

$$\widehat{\tau} = \widehat{\mathbb{E}}[Y \mid W = 1] - \widehat{\mathbb{E}}[Y \mid W = 0].$$

The second mistake is to stratify on X_{i2}:

$$\begin{aligned}
\widehat{\tau}_0 &= \mathbb{E}[Y \mid W = 1, X_{i2} = 0] - \mathbb{E}[Y \mid W = 0, X_{i2} = 0], \\
\widehat{\tau}_1 &= \mathbb{E}[Y \mid W = 1, X_{i2} = 1] - \mathbb{E}[Y \mid W = 0, X_{i2} = 1], \\
\widehat{\tau} &= \frac{n_0 \widehat{\tau}_0 + n_1 \widehat{\tau}_1}{n_0 + n_1}.
\end{aligned}$$

Why might one try this? We know a statement of the form $Y_i(\cdots) \perp W_{i2} \mid X_{i2}$. But the stratification would actually need

$$Y_i(\cdots) \perp (W_{i1}, W_{i2}) \mid X_{i2}.$$

The note illustrates three types:

Type	X_{i2} if $W_{i1} = 0$	X_{i2} if $W_{i1} = 1$
Stable (1)	0	0
Responder (2)	1	0
Acute (3)	1	1

Under the consistency comparison drawn in the notebook, the group with $W = 1, X_{i2} = 0$ contains types (1) and (2), whereas $W = 0, X_{i2} = 0$ contains only type (1). The group with $W = 1, X_{i2} = 1$ contains type (3), whereas $W = 0, X_{i2} = 1$ contains types (2) and (3). Thus, even in a sequentially randomized experiment, this stratification does not control confounding.

Qualification / unresolved point (S2-009): Feedback example: missing data and ambiguous indexing

Source: Lecture11-14.pdf, page 11

The handwritten 0 is clear, so this is not OCR. In the corresponding Lecture 12 example, both uncentered conditional means are 60. They become 0 if all outcomes are centered at 60, but no centering or outcome table is stated in the scan. The intention therefore remains uncertain. Also, $Y(1) - Y(0)$ and unsubscripted W need treatment-sequence definitions. The stable/responder/acute composition argument correctly illustrates selection on a post-treatment variable and compares W_{i1}.

Sequential inference under sequential ignorability

Scan reference: Lecture11-14.pdf, page 12

Since simple stratification does not work, focus directly on estimating $V(\pi)$. Define the filtration

$$\mathcal{F}_t = \sigma(X_1, W_1, X_2, W_2, \ldots, W_{t-1}, X_t),$$

which contains all information available before selecting treatment at time t. Let $\mathbb{E}_\pi$ denote expectation when treatment is assigned according to π. Then (12.2) becomes $V(\pi) = \mathbb{E}_\pi[Y]$.

Define the value function

$$V_{\pi,t}(X_1, W_1, \ldots, W_{t-1}, X_t) = \mathbb{E}_\pi[Y \mid \mathcal{F}_t].$$

It measures the expected return from starting in the state represented by $\mathcal{F}_t$ and following π thereafter. The red note calls the overall policy value $V_{\pi,0}$.

By the tower property,

$$\begin{aligned}
&\mathbb{E}_\pi[V_{\pi,t+1}(X_1, W_1, \ldots, W_t, X_{t+1}) \mid \mathcal{F}_t] \\
&\quad = \mathbb{E}_\pi[\mathbb{E}_\pi[Y \mid \mathcal{F}_{t+1}] \mid \mathcal{F}_t] \\
&\quad = \mathbb{E}_\pi[Y \mid \mathcal{F}_t] = V_{\pi,t}(X_1, W_1, \ldots, W_{t-1}, X_t). \qquad (12.6)
\end{aligned}$$

My interpretation: given $\mathcal{F}_t$, the expected future return remains the same when we express it through time $t+1$. The annotation connects this with the Bellman equation,

$$V(s) = R(s) + \gamma \sum_{s'} P(s' \mid s, a) V(s').$$

Qualification / unresolved point (S2-010): Bellman analogy and the policy argument

Source: Lecture11-14.pdf, page 12

The history-based tower identity needs no Markov assumption. In a discounted Markov reward model the fixed deterministic-policy analogue is

$$V^\pi(s) = r(s, \pi(s)) + \gamma \sum_{s'} P(s' \mid s, \pi(s)) V^\pi(s').$$

A stochastic policy also averages over actions. The final-outcome problem here has no separately specified reward sequence or discount factor.

If we have a good estimate of $V_{\pi,t+1}$, seek a good estimate of $V_{\pi,t}$ and continue backward until we recover $V(\pi)$.

Write $X_{T+1} = Y$. The joint distribution factors as

$$P_\pi(X_1, W_1, \ldots, X_T, W_T, Y) = P_\pi(X_1) \prod_{t=1}^{T} P_\pi(W_t \mid \mathcal{F}_t) P_\pi(X_{t+1} \mid \mathcal{F}_t, W_t).$$

Unconfoundedness means that the non-treatment factors do not depend on the policy:

$$P_\pi(X_1) = P(X_1), \qquad P_\pi(X_{t+1} \mid \mathcal{F}_t, W_t) = P(X_{t+1} \mid \mathcal{F}_t, W_t), \quad \text{for all } \pi.$$

The effect of treatment on the subsequent covariates does not depend on the selecting policy. $\longrightarrow$ The invariant transition factor $P(X_{t+1} \mid \mathcal{F}_t, W_t)$.

Inverse-propensity weighting

Scan reference: Lecture11-14.pdf, page 13

To make (12.6) useful with off-policy data, change measure: the actual data were not generated by $\pi(\cdot)$.

$$\begin{aligned} V_{\pi,t} &= \mathbb{E}_\pi[V_{\pi,t+1} \mid \mathcal{F}_t] \\ &= \mathbb{E}\left[\frac{P_\pi(W_t, X_{t+1} \mid \mathcal{F}_t)}{P(W_t, X_{t+1} \mid \mathcal{F}_t)} V_{\pi,t+1} \middle| \mathcal{F}_t\right] \\ &= \mathbb{E}\left[\frac{P_\pi(W_t \mid \mathcal{F}_t) P_\pi(X_t \mid \mathcal{F}_t, W_t)}{P(W_t \mid \mathcal{F}_t) P(X_t \mid \mathcal{F}_t, W_t)} V_{\pi,t+1} \middle| \mathcal{F}_t\right] \\ &= \mathbb{E}\left[\frac{\mathbf{1}\{W_t = \pi_t(\ldots, X_t)\}}{P[W_t = \pi_t(\ldots, X_t) \mid \mathcal{F}_t]} V_{\pi,t+1} \middle| \mathcal{F}_t\right]. \end{aligned}$$

Change from the observed policy to π; cancel the transition factors. $\longrightarrow$ The two transition factors cancel (with the index correction below).

Original-note correction (S2-011): Transition factors need the next covariate

Source: Lecture11-14.pdf, page 13

The factors must be $P_\pi(X_{t+1} \mid \mathcal{F}_t, W_t)$ and $P(X_{t+1} \mid \mathcal{F}_t, W_t)$. Current X_t is already in $\mathcal{F}_t$ and cannot factor the preceding joint law of (W_t, X_{t+1}). The official lecture repeats this typo. With the corrected index, invariance cancels the transition factors.

Apply the change of measure at every $t = 1, \ldots, T$. Since

$$V(\pi) = \mathbb{E}[V_{\pi,1}(X_1)],$$

$$V_{\pi,1}(X_1) = \mathbb{E}\left[\frac{\mathbf{1}\{W_1 = \pi_1(X_1)\}}{P[W_1 = \pi_1(X_1)]} V_{\pi,2}(X_1, W_1, X_2) \middle| \mathcal{F}_1\right],$$

Original-note correction (S2-037): History arguments in the one-step recursion

Source: Lecture11-14.pdf, page 13

The handwritten one-step value is $V_{\pi,2}(X_1, W_2, X_2)$, but the time-2 history contains W_1, not the as-yet-unselected W_2. Similarly the last argument of the general next value must be X_{t+1}, not the source's W_{t+1}. The displayed compact formulas use the corrected history arguments.

---

repeated substitution eventually leaves $V_{\pi,T+1} = \mathbb{E}_\pi[Y \mid \mathcal{F}_{T+1}] = Y$ and gives

$$V(\pi) = \mathbb{E}\left[\left\{\prod_{t=1}^{T} \frac{\mathbf{1}\{W_t = \pi_t(\ldots, X_t)\}}{P[W_t = \pi_t(\ldots, X_t) \mid \mathcal{F}_t]}\right\} Y\right].$$

$| \mathcal{F}_1$? $\longrightarrow$ The denominator $P[W_1 = \pi_1(X_1)]$

Original-note correction (S2-012): Condition the one-step probability on history

Source: Lecture11-14.pdf, page 13

The author's question correctly identifies the omission. The denominator must be $P[W_1 = \pi_1(X_1) \mid \mathcal{F}_1]$. It can be replaced by an unconditional probability only when this conditional matching probability is constant in the history. Independence of W_1 and X_1 alone is insufficient for a policy that varies with X_1: if $P(W_1 = 1) = 0.2$, the matching probability is either 0.2 or 0.8. The general product already uses the proper conditional probabilities.

An IPW estimator is

$$\widehat{V}_{\text{IPW}}(\pi) = \frac{1}{n}\sum_{i=1}^{n} \gamma_{iT}(\pi) Y_i, \qquad \gamma_{it}(\pi) = \gamma_{i,t-1}(\pi)\frac{\mathbf{1}\{W_{it} = \pi_t(\ldots, X_{it})\}}{P[W_{it} = \pi_t(\ldots, X_{it}) \mid \mathcal{F}_{it}]}.$$

Source transcription: the initial value at the bottom of the page is written $\gamma_{i0}(\pi) = 0$.

Original-note correction (S2-013): Initialize the IPW recursion at one

Source: Lecture11-14.pdf, pages 13, 15

Set $\gamma_{i0}(\pi) = 1$. A zero initial value forces every subsequent weight and the IPW estimator to be zero. This correction is also essential to the AIPW telescoping identity. The official lecture contains the same typo; the product algebra determines the answer.

My interpretation: average outcomes matching π, and use IPW to remove the confounding effects of observed time-varying variables. If the weights, that is, each conditional assignment probability, are known, $\widehat{V}_{\text{IPW}}$ is unbiased.

Backward regression adjustment

Scan reference: Lecture11-14.pdf, page 14

Use (12.6) and sequential unconfoundedness, iterating backward. At $t = T$,

$$\begin{aligned} V_{\pi,T}(X_1, W_1, \ldots, X_T) &= \mathbb{E}_\pi[Y \mid \mathcal{F}_T] \\ &= \mathbb{E}_\pi[Y \mid \mathcal{F}_T, W_T = \pi_T(X_1, W_1, \ldots, X_T)] \\ &= \mathbb{E}[Y \mid \mathcal{F}_T, W_T = \pi_T(X_1, W_1, \ldots, X_T)]. \end{aligned}$$

Learn $\widehat{V}_{\pi,T}$ directly by nonparametric regression. Given a reasonable estimate of the next value function,

$$V_{\pi,t}(X_1, W_1, \ldots, X_t) \approx \mathbb{E}[\widehat{V}_{\pi,t+1}(\ldots, X_{t+1}) \mid \mathcal{F}_t, W_t = \pi_t(\ldots, X_t)]. \tag{12.12}$$

Continue backward until an estimate of $V(\pi)$ is recovered.

Unlike IPW, quantify how regression errors propagate through backward iteration. $\longrightarrow$ Q-learning

Qualification / unresolved point (S2-014): Backward evaluation versus optimization

Source: Lecture11-14.pdf, page 14

The displayed recursion is backward regression or fitted evaluation of a fixed π. Optimal Q-learning additionally selects actions by maximizing fitted action values. The author's error-propagation concern applies to both.

A doubly robust estimator

Scan reference: Lecture11-14.pdf, page 15

We now have both IPW and regression. Equation (12.12) yields a value estimate $\widehat{V}_{\pi,1}(X_1)$, and hence

$$\widehat{V}_{\mathrm{REG}}(\pi) = \frac{1}{n}\sum_{i=1}^{n} \widehat{V}_{\pi,1}(X_{i1}).$$

Backward regression (12.12) $\longrightarrow$ $\widehat{V}_{\pi,1}$, then $n^{-1}\sum_i \widehat{V}_{\pi,1}(X_{i1})$

If we trust $\widehat{V}_{\pi,2}$ more than $\widehat{V}_{\pi,1}$, use it to debias the earlier estimate:

$$\widehat{V}(\pi) = \frac{1}{n}\sum_{i=1}^{n}\left[\widehat{V}_{\pi,1}(X_{i1}) + \gamma_{i1}(\pi)\{\widehat{V}_{\pi,2}(X_{i1}, W_{i1}, X_{i2}) - \widehat{V}_{\pi,1}(X_{i1})\}\right].$$

Weight γ_{i1}; correction term. $\longrightarrow$ $\gamma_{i1}(\widehat{V}_{\pi,2} - \widehat{V}_{\pi,1})$

$$\widehat{V}_{\mathrm{AIPW}}(\pi) = \frac{1}{n}\sum_{i=1}^{n}\left[\gamma_{iT}(\pi)Y_i + \sum_{t=1}^{T}\{\gamma_{i,t-1}(\pi) - \gamma_{it}(\pi)\}\widehat{V}_{\pi,t}(X_{i1},\ldots,X_{it})\right].$$

This generalizes AIPW from the nondynamic case.

Continue the correction until $V_{\pi,T+1} = Y$. $\longrightarrow$ The full sequential AIPW estimator above.

The underlined final statement in the scan reads: For any unbiased $\widehat{V}_{\pi,t}$, $\widehat{V}_{\mathrm{AIPW}}$ is unbiased.

Qualification / unresolved point (S2-015): Conditions for sequential double robustness

Source: Lecture11-14.pdf, page 15

Let q_t be fixed or independently trained value functions, $q_{T+1} = Y$, and $\gamma_0 = 1$. The summand equals

$$q_1 + \sum_{t=1}^{T}\gamma_t(q_{t+1} - q_t).$$

If every treatment probability is correct, weighted expectations telescope to $E_\pi[Y]$ for arbitrary q_t; unbiased regressions are unnecessary. If every $q_t = V_{\pi,t}$, each residual has conditional mean zero under the target action, allowing positive predictable propensity models. Marginally unbiased fitted regressions alone do not suffice, especially when the same observations are reused. Estimated

nuisances generally require consistency/product-rate conditions for asymptotic claims. See Jiang and Li (2016).

Lecture 13: Structural equation modeling

Scan reference: Lecture11-14.pdf, page 16

SEM means structural equation model. Recall the earlier equation $Y_i = \alpha + W_i\tau + \varepsilon_i$: set $W_i = w$ and observe $Y_i(W_i = w) = \alpha + w\tau + \varepsilon_i$.

Nonparametric SEMs

Let $(X_1, \ldots, X_p)$ be p random variables with joint distribution P. Suppose P can be represented by a directed acyclic graph (DAG) G. This means that

$$P(X_1, \ldots, X_p) = \prod_{j=1}^{p} P(X_j \mid \text{pa}_j), \tag{13.2}$$

where pa_j is the set of parents of X_j in G. Equivalently,

$$\text{pa}_j = \{X_i : E_{ij} = 1\},$$

with E_{ij} indicating an edge $X_i \to X_j$. Assume further that deterministic functions f_j, $j = 1, \ldots, p$, yield

$$X_j = f_j(\text{pa}_j, \varepsilon_j), \qquad \varepsilon_j \sim F_j, \quad \text{mutually independent.} \tag{13.3}$$

An SEM/DAG specifies each node through its parents and noise.

Even when upstream parent values change, $f_j(\cdot)$ does not change. $\longrightarrow$ $X_j = f_j(\text{pa}_j, \varepsilon_j)$

This gives a concrete meaning: f_j can be viewed as the distributional index for X_j conditional on its parents.

A causal query externally sets the values of some nodes and examines how other nodes' distributions change. For disjoint node sets $W, Y \subset X$, write the effect of setting $W = w$ on Y as

$$P[Y \mid \text{do}(W = w)].$$

In (13.3), delete the equations associated with W and replace W by w in the other equations. For an intervention on X_j, the source writes the truncated factorization as

$$P[X \mid \text{do}(X_j = x_j)] = \begin{cases} \dfrac{P[X]}{P[X_j = x_j \mid \text{pa}_j]}, & X_j = w_j, \\ 0, & \text{otherwise.} \end{cases}$$

1? $\longrightarrow$ The denominator $P[X_j = x_j \mid \text{pa}_j]$

Original-note correction (S2-016): Match the imposed intervention value

Source: Lecture11-14.pdf, page 16

The nonzero branch must use $X_j = x_j$, matching the intervention. For discrete variables,

$$p^{\mathrm{do}(X_j=x_j)}(x) = \mathbf{1}\{x_j \text{ equals the imposed value}\} \prod_{\ell \neq j} p(x_\ell \mid \mathrm{pa}_\ell).$$

The blue "1?" refers to the forced node's new mass, which is 1 at its imposed value. It does not justify replacing the displayed denominator by 1: that denominator removes the old observational mechanism.

Qualification / unresolved point (S2-017): Continuous interventions require a point mass

Source: Lecture11-14.pdf, page 16

For a continuous coordinate, deterministic intervention creates a point mass; the resulting joint law need not have a density with respect to the original full-dimensional Lebesgue measure. The truncated product of the other conditional kernels is the general construction. Ratio notation also requires the relevant observational support.

The do-calculus: separation and graph modifications

Scan reference: Lecture11-14.pdf, page 17

For disjoint node sets X, Y, Z, consider an undirected path p between a node in X and a node in Y. The path is blocked by Z if it contains a node W such that either:

1. W is a collider on p, and neither W nor any descendant of W is in Z; or
2. W is not a collider on p, and $W \in Z$.

The two path sketches (solid arrows are causal; dashed arrows link annotations):



The scan calls this d-separation.

Qualification / unresolved point (S2-018): Quantify over every path

Source: Lecture11-14.pdf, page 17

Blocking one path does not suffice. Read the initial "arbitrary path" as requiring the criterion for every path between the node sets. In the sketches, arrows from the conditioning-set labels are annotations, not extra causal edges.

If Z d-separates X and Y, (13.2) implies $X \perp Y \mid Z$, written

$$(X \perp Y \mid Z)_G.$$

Graph notation:

- $G_{\overline{X}}$ deletes edges pointing into X.
- $G_{\underline{X}}$ deletes edges leaving X.
- $G_{\underline{X}\overline{Z}}$ deletes edges leaving X and entering Z.

The three rules

Scan reference: Lecture11-14.pdf, page 19, with the red mnemonic on page 18

The margin reiterates: $\text{do}(X = x)$ removes incoming edges, hence the overbar on X. For disjoint node sets X, Y, Z, W:

1. **Insertion/deletion of observations.** If

   $$(Y \perp Z \mid X, W)_{G_{\overline{X}}},$$

   then

   $$P[Y \mid \text{do}(X = x), Z = z, W = w] = P[Y \mid \text{do}(X = x), W = w].$$

2. **Action/observation exchange.** If

   $$(Y \perp Z \mid X, W)_{G_{\overline{X}\underline{Z}}},$$

   then

   $$P[Y \mid \text{do}(X = x), \text{do}(Z = z), W = w] = P[Y \mid \text{do}(X = x), Z = z, W = w].$$

3. **Insertion/deletion of actions.** If

   $$(Y \perp Z \mid X, W)_{G_{\overline{X}\overline{Z(W)}}},$$

   where $Z(W)$ consists of nodes in Z that are not ancestors of any node in W in $G_{\overline{X}}$, then

   $$P[Y \mid \text{do}(X = x), \text{do}(Z = z), W = w] = P[Y \mid \text{do}(X = x), W = w].$$

My left-page mnemonic.

The common $\text{do}(X = x)$ and the extra bars/underlines. $\longrightarrow$ The three do-calculus rules above.

All three rules contain $\text{do}(X = x)$; the bar over X supplies that intervention. Then, when the appropriate separation condition holds, deleting outgoing edges from Z lets us exchange "do Z" with "control for Z." Deleting incoming edges to $Z(W)$ lets us remove "do Z" entirely. $Z(W)$ is the part of Z with no ancestry relation to W of the prohibited kind.

Repeatedly apply do-calculus to reduce a causal query to an observable query about P: an expression without the do-operator, depending only on observed random variables. Do-calculus is complete: if it cannot reduce the causal query to such an observational expression, the query is not nonparametrically identifiable in the stated SEM framework.

Qualification / unresolved point (S2-019): Completeness is relative to a causal model

Source: Lecture11-14.pdf, page 19

Completeness concerns the absence of any valid reduction in the given causal model and observed-variable set, not the failure of one informal search. Conditional probabilities used in an adjustment formula also require relevant support. The three rules and $Z(W)$ definition agree with the official notes.

Example 1: the back-door criterion

Scan reference: Lecture11-14.pdf, page 20

Take three disjoint node sets X, Y, W and query $P[Y \mid \text{do}(W = w)]$. Suppose X contains no descendants of W and X d-separates W and Y after the outgoing edges of W are removed:

$$(Y \perp W \mid X)_{G_{\underline{W}}}. \tag{13.8}$$

Then

$$\begin{aligned} P[Y \mid \text{do}(W = w)] &= \sum_x P[X = x \mid \text{do}(W = w)] P[Y \mid X = x, \text{do}(W = w)] \\ &= \sum_x P[X = x] P[Y \mid X = x, \text{do}(W = w)] \\ &= \sum_x P[X = x] P[Y \mid X = x, W = w]. \end{aligned} \tag{13.9}$$

The second equality uses the fact that X is not downstream of W; the last is rule 2. The red margin links the first replacement to $(X \perp W)_{G_{\overline{W}}}$.

My interpretation of the graphical condition. Think of W and Y as single nodes. Removing the outgoing arrows from W removes its downstream causal influence on Y; what remains is the dependence that the adjustment set must block. The top annotation says that X must separate the back-door connection between Y and W so that X can be used when evaluating the effect of intervening on W [part of the topmost red wording is unclear in the scan].

Suppose W and Y are single nodes and W has no descendants other than Y in G. Cutting the outgoing arrows of W removes its downstream influence. Then (13.8) becomes

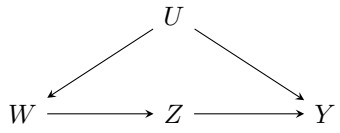
$$F_Y(w) \perp W \mid X, \qquad F_Y(w) = f_Y(w, X, \varepsilon_Y). \tag{13.10}$$

Infer this graphically; a concrete conditional-independence argument. $\longrightarrow$ $F_Y(w) \perp W \mid X$ in (13.10)

All factors other than w remain unspecified in $X_j = f_j(\text{pa}_j, \varepsilon_j)$.

Example 2: the front-door criterion

Scan reference: Lecture11-14.pdf, page 21



We want $P[Y \mid \text{do}(W = w)]$. Since U is unobserved, we cannot apply back-door adjustment using $X = U$. But suppose Z fully mediates the effect of W on Y and is not affected by U.

$$\begin{aligned} P[Y \mid \text{do}(W = w)] &= \sum_z P[Z = z \mid \text{do}(W = w)] P[Y \mid Z = z, \text{do}(W = w)] \\ &= \sum_z P[Z = z \mid W = w] P[Y \mid Z = z, \text{do}(W = w)]. \end{aligned}$$

No back-door path connects W to Z. $\longrightarrow$ The replacement of $\text{do}(W = w)$ by $W = w$ in the Z factor.

Next,

$$\begin{aligned} P[Y \mid Z = z, \text{do}(W = w)] &= P[Y \mid \text{do}(Z = z), \text{do}(W = w)] && \text{(rule 2)} \\ &= P[Y \mid \text{do}(Z = z)] && \text{(rule 3)} \\ &= \sum_{w'} P[W = w'] P[Y \mid Z = z, W = w'] && \text{(back-door)}. \end{aligned}$$

Combining the steps,

$$P[Y \mid \text{do}(W = w)] = \sum_z P[Z = z \mid W = w] \sum_{w'} P[W = w'] P[Y \mid Z = z, W = w']. \qquad (13.11)$$

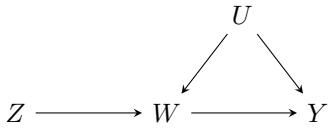
Overall $W \to Y$ effect. $\longrightarrow$ $W \to Z$: $P(Z = z \mid W = w)$; $Z \to Y$: $\sum_{w\prime} P(w\prime) P(Y \mid z, w\prime)$

My observation: although this directly describes an intervention $\text{do}(W = w)$, it still has to integrate over the observational distribution $P[W = w']$.

Even an intervention on W integrates an observational distribution. $\longrightarrow$ The $P[W = w']$ factor in (13.11).

Example 3: instrumental variables

Scan reference: Lecture11-14.pdf, page 22, upper part



We would like to estimate $P[Y \mid \text{do}(W = w)]$. Do-calculus is not useful here: adding the IV node does not provide the required separation; it makes the relevant graphical conditions more complicated.

For IV analysis, the assumptions cannot be specified only as an SEM graph; for example, monotonicity is also needed. Even then, we may identify only a nonstandard causal quantity such as LATE. Encoding monotonicity in an SEM is itself complicated.

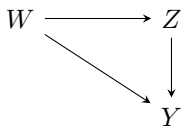
SEMs and potential outcomes

Scan reference: Lecture11-14.pdf, page 22, lower part

For a two-node graph $W \to Y$, the two formulations coincide: the potential outcome $Y_i(w)$ and the SEM expression $f_Y(w, \varepsilon_Y)$ can be mapped one to one.

More generally, the notebook contrasts the formulations. Potential-outcome notation asks what Y would be if specified variables were set to particular values. The SEM formulation is flexible enough to express nested operations. For the graph $W \to Z \to Y$ together with $W \to Y$, define

$$Z(w) := f_Z(w, \varepsilon_Z).$$



The source then writes

$$P[Y \mid \text{do}(W = 0), \text{do}(Z = Z(w = 1))] = f_Y\big(w = 0, z = f_Z(w = 1, \varepsilon_Z), \varepsilon_Y\big).$$

The margin highlights combining different counterfactual settings and asks whether this is possible in potential-outcome notation without drawing a graph.

Original-note correction (S2-022): A counterfactual variable is not its distribution

Source: Lecture11-14.pdf, page 22

A well-typed equality is

$$Y(0, Z(1)) = f_Y(0, f_Z(1, \varepsilon_Z), \varepsilon_Y).$$

The distribution is $\mathcal{L}(Y(0, Z(1)))$, not the random variable itself. The mismatch occurs in the handwritten equation and the corresponding official example.

Views recorded in the notebook: some regard the ability of an SEM to identify effects of this kind of operation as a defect, and the inability of the potential-outcome formulation to do so as a limitation.

Identifiable? $\longrightarrow$ The claim that an SEM identifies this nested operation.

These are questions and viewpoints in the notes, not assertions resolved by this translation.

Original-note correction (S2-023): Potential outcomes can express nested counterfactuals

Source: Lecture11-14.pdf, page 22

The notational limitation is incorrect: $Y(0, Z(1))$ expresses exactly the nested quantity. Neither notation alone proves identification. See Imai, Keele, and Tingley (2010) for nested potential outcomes in mediation.

Qualification / unresolved point (S2-024): Answer to the margin question: which identification assumptions?

Source: Lecture11-14.pdf, page 22

For the drawn fully observed graph with mutually independent structural errors and adequate support,

$$E[Y(0, Z(1))] = \int E[Y \mid W = 0, Z = z]\, dP(Z = z \mid W = 1).$$

The SEM supplies the relevant cross-world independence. Latent mediator–outcome confounding or weaker single-world assumptions can invalidate this formula. Thus the red "Identifiable?" is a substantive model question. Imai et al. distinguish definition from identification.

Lecture 14: Adaptive experiments

Scan reference: Lecture11-14.pdf, page 24, with the explanatory note on page 23

Adaptive experiments dynamically adjust the experimental design. Earlier fixed experiments worked in an IID setting, an assumption that is sometimes too restrictive.

Setting and notation

There are actions $k = 1, \ldots, K$ and subjects $t = 1, \ldots, T$. Across subjects, potential outcomes are IID. Write

$$Y_t(k) \sim F_k, \qquad Y_t = Y_t(W_t), \qquad \mu_k = \mathbb{E}[Y_t(k)].$$

W_t is the action selected for subject t, F_k is the potential-outcome distribution for arm k, and μ_k is its mean. Define regret by

$$R_T = \sum_{t=1}^{T} (\mu^* - \mu_{W_t}), \qquad \mu^* = \sup_{1 \le k \le K} \mu_k.$$

μ^* is the largest expected reward among the available actions. Regret is the expected reward lost by choosing a sequence of actions instead of always choosing the best action.

Define

$$n_{kt} = \sum_{j=1}^{t} \mathbf{1}\{W_j = k\}, \qquad \widehat{\mu}_{kt} = \frac{1}{n_{kt}} \sum_{j=1}^{t} \mathbf{1}\{W_j = k\} Y_j.$$

n_{kt} $\longrightarrow$ Number of pulls of arm k through time t.

$\widehat{\mu}_{kt}$ $\longrightarrow$ Average reward of arm k.

These use the information collected from time 1 through t.

My left-page restatement. Think of T similar, independent individuals. Each has K treatment options and hence K potential outcomes, with the same arm-specific distributions F_k. For each individual we observe only the outcome of the chosen treatment, $Y_t = Y_t(W_t)$. The vector μ collects expected rewards. Total regret adds, across the T individuals, the difference between the chosen treatment and the best treatment; the best arm is the same arm throughout this setup.

In an ordinary randomized experiment, W_t is uniform and nonadaptive over $1, \ldots, K$, so regret grows linearly with T:

$$R_T \sim T \sum_{k=1}^{K} \frac{\mu^* - \mu_k}{K}.$$

The annotation identifies the summand average as the expected loss per individual. The first objective of an adaptive experiment is sublinear regret.

Regret should not grow linearly with the number of experiments or time; choices get closer to the optimum. $\longrightarrow$ Sublinear regret.

This requires balancing *exploration* (trying things) with *exploitation* (using what is known).

Optimism in the face of uncertainty

Scan reference: Lecture11-14.pdf, page 25

The upper confidence bound (UCB) method balances exploration and exploitation:

1. Initialize by sampling each arm t_0 times: choose it and observe it t_0 times, for Kt_0 initial samples.
2. For each $t = Kt_0 + 1, Kt_0 + 2, \ldots$, use data collected through $t - 1$ to construct a confidence interval $\widehat{U}_{kt}$ for each μ_k.
3. Select the action W_t whose interval has the largest upper endpoint and observe $Y_t = Y_t(W_t)$.

Each new time point adds only one observation.

To understand how UCB controls regret, simplify to Gaussian rewards:

$$Y_t(k) \sim \mathcal{N}(\mu_k, b^2), \tag{14.4}$$

with known b^2.

Known variance simplifies the calculation. $\longrightarrow$ b^2 in the Gaussian UCB radius.

Use

$$\widehat{U}_{kt} = \widehat{\mu}_{k,t-1} \pm b\sqrt{\frac{4 \log T}{n_{k,t-1}}}. \tag{14.5}$$

For $t_0 = 1$, the stated regret expression is

$$R_T = \sum_{k \neq k^*} \left\{ \frac{16 b^2 \log T}{\mu_{k^*} - \mu_k} + (\mu_{k^*} - \mu_k) \right\}, \tag{14.6}$$

with probability at least $1 - K/T$, where k^* denotes the best arm.

Original-note correction (S2-025): The UCB regret statement is an upper bound

Source: Lecture11-14.pdf, pages 25, 27, 28

Replace $=$ in (14.6), its repeated statement on page 27, and the final display on page 28 by $\leq$. For $\Delta_k = \mu^* - \mu_k > 0$ the proof gives $n_{kT} \leq 16 b^2 \log T / \Delta_k^2 + 1$, not equality. Sum over strictly suboptimal arms; tied optimal arms have zero regret and cannot appear with zero denominators. The official notes repeat the equality typo.

My intuition about this bound.

The gap $\mu_{k^*} - \mu_k$ in the denominator. $\longrightarrow$ Closer arms take longer to distinguish; far worse arms can be discarded quickly.

UCB regret grows only logarithmically. An arm closer to the optimum takes longer to distinguish from it; a much worse arm can be discarded quickly. A small gap $\mu_{k^*} - \mu_k$ appears to send the displayed bound to infinity. In practice there are two competing bounds: for a strong effect, UCB discards an arm when its interval is separated; for a weak effect, total regret is at most $T(\mu_{k^*} - \mu_k)$. Combining these gives a finite bound, and even in the worst case over the means, the notebook gives an order $K\sqrt{T \log T}$ upper bound. The two sketches on the continuation page compare the growth rates (reconstructed below).

Original-note correction (S2-026): A uniform worst case cannot allow unbounded mean gaps

Source: Lecture11-14.pdf, page 25

The claimed uniform bound over all real means is false: the initial pull of every arm already costs $\sum_k \Delta_k$, which is unbounded. On the confidence event,

$$R_T \le \sum_{k:\Delta_k>0} \min\left\{T\Delta_k, \frac{16b^2 \log T}{\Delta_k} + \Delta_k\right\}.$$

For uniformly bounded gaps and fixed b, the strong/weak-effect comparison yields the stated $O(K\sqrt{T \log T})$ scale plus initialization. The intuition is valid after this restriction.

A regret bound for UCB

Scan reference: Lecture11-14.pdf, page 27, with the attempted concentration calculation on page 26, then page 28

Rewrite regret as

$$R_T = \sum_{k \ne k^*} n_{kT}(\mu_{k^*} - \mu_k).$$

The main task is to bound n_{kT}, the number of times UCB pulls a suboptimal arm.

This is also the design principle behind UCB. $\longrightarrow$ Bound n_{kT}, the number of pulls of a suboptimal arm.

Let ζ_{kj} be the time of the jth pull of arm k. For $t = K+1, \dots, T$, the scan develops the following probability bound:

$$\begin{aligned}
&P\left[\sup_{K<t\le T}\left\{\mu_k - \widehat{\mu}_{k,t-1} - b\sqrt{\frac{4\log T}{n_{k,t-1}}}\right\} \ge 0\right] \\
&\le P\left[\sup_{1\le j\le n_{kT}}\left\{\mu_k - \widehat{\mu}_{k,\zeta_{kj}} - b\sqrt{\frac{4\log T}{j}}\right\} \ge 0\right] \\
&= P\left[\sup_{1\le j\le n_{kT}}\left\{\mu_k - \frac{1}{j}\sum_{\ell=1}^{j} Y'_\ell(k) - b\sqrt{\frac{4\log T}{j}}\right\} \ge 0\right] \\
&\le P\left[\sup_{1\le j\le T}\left\{\mu_k - \frac{1}{j}\sum_{\ell=1}^{j} Y'_\ell(k) - b\sqrt{\frac{4\log T}{j}}\right\} \ge 0\right] \le \frac{1}{T}.
\end{aligned}$$

Here $Y'_\ell(k)$ are independent draws from $\mathcal{N}(\mu_k, b^2)$.

Enlarge the index set: the supremum and its event can only grow. The reward stream has the same distribution. $\longrightarrow$ $K < t \le T \rightsquigarrow 1 \le j \le n_{kT} \rightsquigarrow 1 \le j \le T$

The two corresponding handwritten lines use $Y'_\ell(0)$.

Original-note correction (S2-027): Time-changed observations belong to arm k

Source: Lecture11-14.pdf, page 27

Both occurrences must be $Y'_\ell(k)$, as in the displayed derivation. Pre-generate an IID stream for each arm and reveal its next observation when that arm is selected. A union bound over deterministic counts $j \le T$ avoids incorrectly treating the stopped sample mean as conditionally IID given its random count.

Consequently, for each arm k, with probability at least $1 - 1/T$,

$$\mu_k \le \widehat{\mu}_{k,t-1} + b\sqrt{\frac{4\log T}{n_{k,t-1}}} \quad \text{for all } t = K+1, \dots, T. \tag{14.8}$$

A union bound over the K arms gives probability at least $1 - K/T$.

Left-page work: Hoeffding's inequality, and my unresolved calculation. For independent bounded random variables $Z_1, \dots, Z_n$ with $Z_i \in [a, b]$ almost surely, and $s > 0$,

$$P\left(\frac{1}{n}\sum_{i=1}^{n}(Z_i - \mathbb{E}[Z_i]) \ge s\right) \le \exp\left(-\frac{2ns^2}{(b-a)^2}\right).$$

The same bound holds for a lower-tail deviation $\le -s$; the two-sided absolute-deviation bound has a factor 2.

Almost surely bounded. $\longrightarrow$ $Z_i \in [a, b]$

For events $A_1, \dots, A_n$, the union bound is

$$P(A_1 \cup \cdots \cup A_n) \le \sum_{i=1}^{n} P(A_i).$$

The note connects this to the $1 - K/T$ guarantee.

My attempted calculation (with the questions left open): perhaps it is enough to show

$$P\left(\mu_k - \frac{1}{j}\sum_{\ell=1}^{j} Y'_\ell(k) - b\sqrt{\frac{4\log T}{j}} \ge 0\right) \le \frac{1}{T}.$$

The left side is

$$P\left(\frac{1}{j}\sum_{\ell=1}^{j} Y'_\ell(k) - \mu_k \le -b\sqrt{\frac{4\log T}{j}}\right).$$

Applying the displayed bounded-variable inequality would suggest

$$\exp\left(-\frac{2j\, b^2(4\log T)/j}{(b-a)^2}\right) = \exp\left(-\frac{8b^2}{(b-a)^2}\log T\right) \le \frac{1}{T}.$$

Author's red doubts: "$Y'_\ell(k)$ is unbounded; is k written incorrectly?" and "I do not know how this is simplified; perhaps there is an unstated assumption."

Qualification / unresolved point (S2-028): Resolve the unfinished Gaussian tail calculation

Source: Lecture11-14.pdf, pages 26, 27

The author's red objection is correct: bounded Hoeffding does not apply. Instead,

$$\sqrt{j}(\overline{Y}'_{kj} - \mu_k)/b \sim N(0,1),$$

$$P\left(|\overline{Y}'_{kj} - \mu_k| > b\sqrt{4\log T/j}\right) = 2\Phi(-2\sqrt{\log T}) \le T^{-2} \quad (T \ge 2).$$

The last inequality follows from the normal Mills bound $2\Phi(-x) \le 2e^{-x^2/2}/(x\sqrt{2\pi})$. A union bound over at most KT arm/count pairs gives simultaneous two-sided coverage at least $1 - K/T$. A per-count $1/T$ bound would not suffice after a T-count union. These are added proof details, not statements present in the scan. The two uses of the letter b also should not be confused.

Continuation of the arm-count argument. If (14.8) holds for all arms and $W_t = k$, the UCB selection rule gives

$$\begin{aligned}\widehat{\mu}_{k,t-1} + b\sqrt{\frac{4\log T}{n_{k,t-1}}} &\ge \widehat{\mu}_{k^*,t-1} + b\sqrt{\frac{4\log T}{n_{k^*,t-1}}} \\ &\ge \mu_{k^*}.\end{aligned}$$

The notebook then writes

$$\mu_k + 2b\sqrt{\frac{4\log T}{n_{k,t-1}}} \ge \mu_{k^*}, \qquad n_{k,t-1} \le \frac{16b^2\log T}{(\mu_{k^*} - \mu_k)^2}.$$

Confidence statement (1), then UCB selects k. $\longrightarrow$ Count bound (2): $n_{k,t-1} \le 16b^2\log T/\Delta_k^2$.

Original-note correction (S2-029): The count argument needs the other deviation direction

Source: Lecture11-14.pdf, pages 27, 28

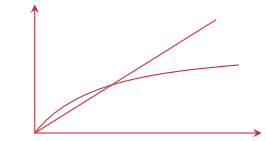
Equation (14.8) alone does not justify the implication. Put $r_{kt} = b\sqrt{4\log T/n_{k,t-1}}$. On the preceding two-sided event, selection of k implies

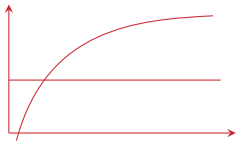
$$\mu^* \le \widehat{\mu}_{k^*,t-1} + r_{k^*t} \le \widehat{\mu}_{k,t-1} + r_{kt} \le \mu_k + 2r_{kt}.$$

The last step needs $\widehat{\mu}_{k,t-1} - \mu_k \le r_{kt}$, the missing direction. Squaring yields the count threshold. The theorem is recoverable with the same Gaussian confidence constant; the handwritten proof is incomplete.

My interpretation: pulling arm k means its upper confidence endpoint must still exceed that of the best arm. Once its pull count passes the threshold, while the threshold grows only with $\log T$, it can no longer be selected in this argument.

Substitute the arm-count bound in the regret definition:


qualitative growth comparison


crossing a threshold

$$R_T = \sum_{k \neq k^*} (\mu_{k^*} - \mu_k) n_{kT} = \sum_{k \neq k^*} \left\{ \frac{16 b^2 \log T}{\mu_{k^*} - \mu_k} + (\mu_{k^*} - \mu_k) \right\}.$$

The final additive gap is the regret from the initial $t_0 = 1$ pull of the arm; the first term covers $t = K+1, \ldots, T$. The equality is as written in the scan; the purple upper-bound correction applies here too.

Adaptive randomization schemes

Scan reference: Lecture11-14.pdf, page 29

UCB has limitations. It is difficult to combine with methods developed for IID sampling and to generalize to more complicated sampling designs. For example, uncertainty may differ across arms, so constructing comparable intervals $\widehat{U}_{kt}$ in more general designs can be awkward [part of the handwritten parenthesis is unclear]. A red note contrasts UCB's dependence on past observations with methods based on randomized IID data [a short parenthetical aside at the far right is unclear].

Thompson sampling. This is an alternative to UCB: a heuristic algorithm based on Bayesian updating.

1. Choose a prior $\Pi_{k,0}$ for each potential-outcome distribution F_k.
2. At time t, compute the posterior probability that arm k is the best arm:
$$e_{k,t-1} = P_{\Pi_{t-1}}(\mu_k = \mu^*).$$
3. Randomly select an action:
$$W_t \sim \text{Multinomial}(e_{\cdot,t-1}).$$
4. Observe $Y_t = Y_t(W_t)$ and update the posterior to Π_t.

Thompson sampling and UCB have similar statistical behavior: they sample each arm periodically until they are essentially convinced that it is inferior.

Posterior probability drops below $1/T$. $\longrightarrow$ Essentially convinced that the arm is inferior.

Qualification / unresolved point (S2-031): Best-arm ties and the Thompson heuristic

Source: Lecture11-14.pdf, page 29

Best-arm probabilities sum to one if the posterior best arm is almost surely unique; otherwise specify a tie-breaking rule. Here Multinomial means one categorical draw. Falling below $1/T$ is an intuition about negligible remaining selection probability, not a cutoff imposed by the algorithm. Similar UCB/Thompson regret behavior requires model, prior and regularity conditions.

In Thompson sampling the selected action is random, with adaptive randomization probabilities depending on past data. This connects naturally with causal inference.

Thompson sampling chooses a random action using past-data-dependent probabilities. $\longrightarrow$ Combine with causal inference.

The notebook observes that tuning the prior Π_0 can be more reasonable than tuning the UCB interval $\widehat{U}_{kt}$.

Inference in adaptive experiments

Scan reference: Lecture11-14.pdf, page 31, with the derivation on page 30

Analyze data from UCB or Thompson sampling and provide confidence statements about the underlying problem parameters. This is much harder than in the IID setting.

Two natural estimators of μ_k are the sample average,

$$\widehat{\mu}_k^{\text{AVG}} = \widehat{\mu}_{kT} = \frac{1}{n_{kT}} \sum_{t=1}^{T} \mathbf{1}\{W_t = k\} Y_t, \tag{14.11}$$

and, under Thompson sampling, the IPW estimator,

Qualification / unresolved point (S2-033): Use the assignment probability before the current outcome

Source: Lecture11-14.pdf, pages 29, 31

Define $e_{tk} = P(W_t = k \mid \mathcal{H}_{t-1})$, computed before subject t's outcome; this is the earlier $e_{k,t-1}$. Require positivity. Ordinary deterministic UCB gives conditional probabilities 0 or 1, so these IPW/AW formulas do not directly apply to its unselected arms. The sample mean requires $n_{kT} > 0$.

$$\widehat{\mu}_k^{\text{IPW}} = \frac{1}{T} \sum_{t=1}^{T} \frac{\mathbf{1}\{W_t = k\} Y_t}{e_{tk}}. \tag{14.12}$$

Because the data collection is adaptive, these estimators do not have asymptotically normal limiting distributions, making confidence intervals difficult.

Original-note correction (S2-032): Adaptivity does not exclude every normal limit

Source: Lecture11-14.pdf, page 31

The correct warning is that these estimators *need not* have a normal limit under unrestricted adaptivity. Fixed positive assignment probabilities give ordinary CLTs; suitably stabilized adaptive designs can also admit CLTs. Positive predictable probabilities still make the unnormalized IPW mean unbiased at a fixed horizon, but unbiasedness does not imply normality. See Hadad et al. (2021).

An adaptively weighted estimator can admit a Gaussian pivot: A Gaussian pivot approaches, or converges in distribution to, $N(0,1)$.

$$\widehat{\mu}_k^{\mathrm{AW}} = \frac{\sum_{t=1}^{T} \frac{\mathbf{1}\{W_t = k\} Y_t}{\sqrt{e_{tk}}}}{\sum_{t=1}^{T} \frac{\mathbf{1}\{W_t = k\}}{\sqrt{e_{tk}}}}. \tag{14.13}$$

Under reasonable regularity conditions,

$$\widehat{V}_k^{-1/2}(\widehat{\mu}_k^{\mathrm{AW}} - \mu_k) \Rightarrow \mathcal{N}(0,1),$$

where

$$\widehat{V}_k = \frac{\sum_{t=1}^{T} \left\{ \frac{\mathbf{1}\{W_t = k\}(Y_t - \widehat{\mu}_k^{\mathrm{AW}})}{\sqrt{e_{tk}}} \right\}^2}{\left\{ \sum_{t=1}^{T} \frac{\mathbf{1}\{W_t = k\}}{\sqrt{e_{tk}}} \right\}^2}. \tag{14.14}$$

My explanation: a CLT can be recovered because the estimator stabilizes variance. Its variance becomes controllable, allowing a central limit theorem.

Left-page derivation: the martingale behind the estimator. A preliminary definition at the top of the page describes a martingale: the variables are measurable with respect to the available history, satisfy $\mathbb{E}[X_i] < \infty$, and satisfy the conditional-mean identity. Another line describes an adapted sequence M_n as depending only on $X_0, \ldots, X_n$, with $\mathbb{E}[M_n \mid X_0, \ldots, X_{n-1}] = M_{n-1}$.

Original-note correction (S2-034): Martingale integrability uses an absolute moment

Source: Lecture11-14.pdf, page 30

Use $E|X_i| < \infty$. The one-sided inequality $E[X_i] < \infty$ alone can allow $-\infty$ and fails to ensure the required conditional expectation. If "finite expectation" is intended to mean a finite real expectation, integrability is implicit; absolute values make it explicit.

From (14.13),

$$\widehat{\mu}_k^{\mathrm{AW}} - \mu_k = \frac{\sum_{t=1}^{T} \frac{\mathbf{1}\{W_t = k\}(Y_t - \mu_k)}{\sqrt{e_{tk}}}}{\sum_{t=1}^{T} \frac{\mathbf{1}\{W_t = k\}}{\sqrt{e_{tk}}}}.$$

Let the partial sum in the numerator be

$$M_t = \sum_{j=1}^{t} \frac{\mathbf{1}\{W_j = k\}(Y_j - \mu_k)}{\sqrt{e_{jk}}}.$$

W_t is randomized using the information available through time t and is independent of $Y_t(k)$ given the past. Therefore M_t is a martingale:

$$\mathbb{E}[M_t \mid M_1, \ldots, M_{t-1}] = M_{t-1}.$$

The red restatement is $\mathbb{E}[M_{t+1} \mid M_1, \ldots, M_t] = M_t$.

The source then states, "by the weighting scheme,"

$$\mathrm{Var}[M_t \mid M_1, \ldots, M_{t-1}] = b_k^2, \qquad b_k^2 = \mathrm{Var}[Y_t(k)]. \qquad \text{(14.8, as labeled in the scan)}$$

Conditioning on the previous partial sum makes this the conditional variance of the new increment as well. The reference number (14.8) is reused in the notebook.

A martingale CLT gives, provided e_{tk} does not decay too quickly,

$$\frac{M_T}{\sqrt{T b_k^2}} \Rightarrow \mathcal{N}(0, 1).$$

Author's red notes: adaptively weighted CLT; finite second moments; the sampling probability e_{tk} should decay more slowly than $1/t$. This excludes the most aggressive strongly efficient exploration regimes that pull an arm only on the order of $\log T$ times.

Qualification / unresolved point (S2-035): Sufficient conditions for the martingale pivot

Source: Lecture11-14.pdf, pages 30, 31

One transparent sufficient regime is $0 < b_k^2 < \infty$, $E|Y_t(k) - \mu_k|^4 < \infty$, predictable $e_{tk} \geq ct^{-\alpha}$ with $c > 0$, $0 \leq \alpha < 1$, and current potential outcomes independent of past data. For $D_t = \mathbf{1}\{W_t = k\}(Y_t - \mu_k)/\sqrt{e_{tk}}$,

$$E[D_t \mid \mathcal{H}_{t-1}] = 0, \quad E[D_t^2 \mid \mathcal{H}_{t-1}] = b_k^2, \quad T^{-2} \sum_{t \leq T} E[D_t^4] = O(T^{\alpha - 1}) \to 0.$$

This gives a martingale Lyapunov condition and concentration of $T^{-1} \sum D_t^2$. The AW denominator has predictable mean scale at least $T^{1-\alpha/2}$ and fluctuations $O_P(\sqrt{T})$, giving consistency and permitting fitted residuals in the variance. These conditions are sufficient, not necessary. The source's $\mathrm{Var}(M_t \mid M_{1:t-1}) = b_k^2$ is correct here: M_{t-1} is known and the new increment has that variance. See Hadad et al. (2021) for broader conditions.

Next,

$$\frac{1}{T b_k^2} \sum_{t=1}^{T} \left\{ \frac{\mathbf{1}\{W_t = k\}(Y_t - \widehat{\mu}_k^{\mathrm{AW}})}{\sqrt{e_{tk}}} \right\}^2 \xrightarrow{P} 1.$$

This is a concentration step; the margin again requires the probabilities not to decay too fast. The denominators in $\widehat{\mu}_k^{\mathrm{AW}} - \mu_k$ and $\widehat{V}_k^{1/2}$ cancel, yielding the Gaussian pivot above.

The denominators of $\widehat{\mu}_k^{\mathrm{AW}} - \mu_k$ and $\widehat{V}_k^{1/2}$ cancel. $\longrightarrow$ The Gaussian pivot in (14.14).